Autonomy as Determination by One's Own Normative Structure: A White-Box-First Five-Channel Audit Framework with Axioms, Theorems, and a Simulation Benchmark

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Abstract

How much autonomy does an agent retain under a given influence, and how much did an interaction take away? We propose a *white-box-first audit framework* that answers both on one scale, within a layer at a fixed instantiation. Autonomy is operationalized as determination of behavior by the agent's own normative structure (goals, beliefs, self-model), decomposed into five channels—origination, goal, execution, epistemic, reflection—behind a viability gate. We do not claim the indicator family is neutral: every design choice is registered as an explicit, priced commitment, and what is knowable from what data is proved rather than assumed.

The key positive result is a representation theorem: under the stated normative commitments, the aggregation of the five channels is characterized by the weighted geometric mean—a classical scale-invariance characterization, to which we add a minimal-axiom analysis and the audit embedding. Around it sits a suite of derived results with registered falsification windows: attribution coverage with multi-channel necessity, manipulation strictly reducing autonomy with a hallucination-index signature, highest-leverage attack concentration with water-filling defense, disjunctive loss, a retention margin with a blind-zone prediction, delegation laws, and a coercion consistency result. The key negative result is an impossibility theorem: implanted endorsement is behaviorally indistinguishable from endogenous endorsement, so historical auditing requires full causal logging as an architectural necessity; without internal access, origination degrades to a one-sided certified bound and execution to protocol-relative bounds.

A 4,620-run simulation benchmark on a tabular substrate exercises every falsification window. External-substrate validation is the decisive next step and is pending; until it is run, the framework's construct validity remains unestablished, and what the campaign establishes is implementation-level consistency (every registered window satisfied, as expected of a correct implementation), plus findings that outlive the platform: mechanism demonstrations (blind-zone capture, gate necessity) and a sensitivity calibration—adaptive adversaries violate the certified bound's premise and inflate the plug-in by $+0.17$ to $+0.90$ nats in 3/3 arms. An external baseline metric (endorsed-goal utility) dissociates from the composite in two of three regimes. The framework's core quantities presume causal-log access that current LLM-agent deployments do not provide; what is deployable today is the one-sided bound, the protocol-relative execution bounds, and the impossibility warnings.

Keywords autonomy; agency measurement; directed information; empowerment; representation theorem; manipulation; AI agents; composite indicators; identifiability; benchmark

Data and code availability The simulation platform (ENV-0), the seed-derivation protocol (Appendix N), run manifests, the auto-generated experimental report, the full campaign archive (`sim/runs/20261017_full`), and the analysis and figure scripts are publicly archived at https://github.com/hurrywhen/autonomy-five-channel (Gitee mirror: https://gitee.com/jiangsu-yunhefeng-inte autonomy-five-channel), release tag `v1.0-r10`, DOI https://doi.org/10.5281/zenodo.23608752; every run is bit-reproducible from its (scenario, configuration, replication) coordinates.

Ethics statement This article reports theoretical results and agent-based simulations; it contains no studies with human participants or animals performed by any of the authors. The human-side content (Section 8.8) is qualitative triangulation over published evidence. The calibration-period protocols for human subjects described as future work will require institutional review board approval before implementation.

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1 Introduction

Autonomous systems are increasingly deployed to act persistently on behalf of their principals: LLM-based agents that plan, remember, and use tools [Wang et al., 2024, Xi et al., 2023, Park et al., 2023]; recommender and notification systems that continuously shape what users see; rehabilitation and coaching devices that nudge human habits. In each deployment the same two questions arise, and both are currently unanswerable in operational terms: *how much autonomy does the agent retain under a given influence*, and *how much did a given interaction take away.* Governance frameworks grade agents by levels and profiles [Feng et al., 2025, Kasirzadeh and Gabriel, 2025, Dung, 2025]; psychology measures motivation and control [Deci and Ryan, 2000, Maier and Seligman, 2016]; information theory measures control loops [Klyubin et al., 2005, Massey, 1990]; AI safety catalogues failure modes one by one [Amodei et al., 2016, Everitt et al., 2021, Greshake et al., 2023]. What is missing is a common yardstick: a measure of autonomy itself, defined independently of substrate, decomposed into auditable channels, and governed by explicit axioms and theorems rather than by unarticulated conventions.

Four misconceptions stand in the way of such a measure, and we dispel them at the outset. Autonomy is *not isolation*: a closed system is dead, not free; adaptation requires informational intake (our Axiom AX2). Autonomy is *not randomness*: to be autonomous is to be determined by one's self, not by nothing (Axiom AX0 makes directed information from internal states to actions the defining positive quantity). Autonomy is *not omnipotence*: capability is one channel among five, and its absence does not annihilate the others. And autonomy is *not the absence of relations*: influences divide into constitutive support (our σ class) and reductive interference; "zero relations" is not the maximum of autonomy but lies outside its feasible region [Mackenzie and Stoljar, 2000].

The proposal. We define the autonomy of an agent relative to an environment and a time horizon as the degree to which its behavior is determined by its own normative structure (G, B, M)—its goals, beliefs, and self-model—in five respects: *origination* (behavior is causally sourced in internal states rather than bypassed), *informedness* (external influences are registered by the self-model), *endorsability* (changes to the motivational structure pass, or could pass, second-order review that is itself free of tampering—a historical clause), *executability* (intent has predictable leverage on the

situation, with exit options), and *viability* (the resource reproduction sustaining all of the above is not interrupted). Each clause becomes a channel with an estimator, a canonical ratio scale, and a zero semantics; a gating viability factor multiplies a composite of the five. The resulting scale is comparable *within a layer at a fixed instantiation*; cross-layer and cross-substrate readings are reported as channel profiles (Section 3).

A note on objectivity, at the outset. Any measure of autonomy is built from indicators, and every indicator carries its designer's limitations; no instrument family is neutral. We therefore do not claim that our channels *are* autonomy, nor that subjectivity can be designed away. What can be built is *procedural objectivity*: every design commitment is registered and priced (including what is lost if it is rejected); invariance under irrelevant re-instrumentation is itself made an axiom; the boundary between the knowable and the unknowable is proved rather than assumed; and the channel choice itself is exposed to a registered external falsifier. The measure is objective in the only sense measurement science delivers—not free of conventions, but explicit about them and stable under independent reconstruction (Section 9.2).

Contributions. The primary contribution is the *audit schema* itself—a measurement and identifiability discipline for autonomy under influence; the theorems are its load-bearing instruments, not a claim to a finished mathematical theory of autonomy.

1. **A formal framework** (Section 3): a time-unfolded causal graph with a five-class intervention taxonomy ($\sigma/\alpha/\beta/\gamma/\delta$) whose completeness is a theorem relative to that interface (Proposition 4): within the graph's surgery loci, every external influence acts by changing what the agent sees, its mechanism itself, what it can do, or its conditions of self-maintenance. The α/γ boundary is a structural criterion (the intact channel survives as a component, or not), and the one deliberate boundary—surgeries on the environment's own dynamics—is registered as re-instantiation of the measurement, not a sixth class.

2. **Measures from axioms** (Sections 4 and 5): six channel measures, each with an absolute zero and a canonical normalizer; and a representation theorem (Theorem 7) showing that continuity, strict coordinatewise monotonicity, homogeneity, and coordinatewise scale invariance of the induced order admit exactly the weighted geometric mean—"forced" relative to axioms that are defended normative commitments, not measurement-theoretic necessities (the scale-invariance characterization itself is classical: Ebert and Welsch, 2004; our additions are scoped in Section 5)—with two derivations, aggregation-theoretic and measurement-theoretic, independent in toolchain though sharing the invariance axioms, and a harmonic-mean counterexample (Proposition 8) showing exactly why the invariance axiom is indispensable. Biseparability and zero annihilation are consequences of the minimal invariance trio, not independent commitments: the two derivations pin the minimal sufficient set at {AX4, AX6b, AX8}. The load-bearing axioms themselves are normative commitments of the framework, defended in Section 9.3 rather than derived.

3. **A suite of theorems with falsification windows** (Section 6): coverage and multi-channel necessity (T2—no single channel detects all intervention classes; within this instrument family the decomposition is an attribution cover, with necessity as a theorem); manipulation strictly reduces autonomy wherever the affected instrument enters the composite, and covert manipulation inflates a measurable hallucination index (T3); highest-leverage attack concentration and water-filling defense under log-linear damage (T4); loss of autonomy as a six-way disjunction (T5); an explicit autonomy margin $\delta^* = \min(\theta/W, \kappa, \lceil (r+1)/2 \rceil c_0/W)$ with a positive

retention floor $A_{\min}$—including the proof that *detection alone can provide no margin against slow drift*, so that margins come from repair capacity while detection's role is bounding the registration domain, and a sharp blind-zone prediction (T6); component-level delegation laws (T7); and the coercion paradox as a consistency result (T8).

4. **Identifiability theory** (Section 7): white-box point identification, tiered by the modal commitments each channel carries; a certified *one-sided* black-box bound for A_{org} under bypass-semantics probes (the reference lower end is uncertified) and protocol-relative bounds for A_{exec}; and Proposition 33 (implanted endorsement is behaviorally indistinguishable from endogenous endorsement), which makes the impossibility of black-box historical auditing a theorem and full causal logging an architectural necessity.

5. **A simulation benchmark with registered falsification windows** (Section 8): a 4,620-run campaign over fifteen scenarios—six attack classes, delegation, helplessness induction, resource starvation, capability-envelope erosion, forced autonomy, black-box degradation, cross-framework consistency, and an execution dynamic-range arm, plus an adaptive-adversary supplement. The campaign is an implementation-level consistency check, and we report it as such: all fifteen proposition-level criteria are met within the registered windows with zero refutations—the expected outcome of a correct implementation, not independent evidence for the theory—and the one adverse finding is a sensitivity calibration of the black-box bound's premise under adaptive adversaries (BB-2). Verdicts are stable under alternative aggregation weights (Kendall $\tau \geq 0.93$) and under axiom-relaxation robustness checks in the elasticity parameter; on the default instance the execution channel's narrow band makes that statement an effectively four-channel one, so the same weight battery was rerun on the rich-coupling EXEC-1 instance (fifth channel at full dynamic range): no verdict flips there either.

6. **An interpretive correspondence with existing frameworks** (Proposition C0, §6 and §9): under explicit definitional readings—whose faithfulness to the source texts is registered as hermeneutic, not mathematical, residual uncertainty—the agency profiles of Dung [2025], the four dimensions of Kasirzadeh and Gabriel [2025], and the user roles of Feng et al. [2025] are exhibited as projections of our channel space onto the special case of a benign principal, with a constructive witness that our state descriptor separates configurations they cannot (Proposition C0(ii)). This is an explanatory correspondence under registered readings, not a mathematical unification theorem.

Map of the paper. Section 2 positions the framework in seven literatures and states the gaps it fills. Section 3 gives the operational definition, the system model, and the intervention taxonomy. Section 4 states the axioms and the six channel measures. Section 5 proves the representation theorem. Section 6 presents the theorem suite with proof sketches and falsification windows; full proofs are in the appendices. Section 7 develops the identifiability analysis. Section 8 reports the simulation study. Section 9 answers the standing debates and registers limitations; Section 10 concludes.

2 Related Work

Our framework sits at the intersection of seven literatures that have hitherto developed largely in isolation: philosophical theories of autonomy, cybernetic and enactive accounts of agency, the empirical psychology of self-regulation, information-theoretic measures of control, the emerging

science of AI agent characterization and governance, AI safety research on goal corruption, and the formal study of delegation and adversarial allocation (assistance games, contract theory, security games). This section surveys each, identifies the points of consensus and contention that structure the field (§2.1–.8), and states the gaps our contribution fills (§2.9). Throughout, we use the five-channel vocabulary of Section 3 only informally; the formal definitions are self-contained.

2.1 Philosophical accounts of autonomy

Modern analysis begins with hierarchical accounts. Frankfurt [1971] grounds autonomy in the reflective structure of the will: an agent acts autonomously when the desire that moves her is endorsed by a second-order volition with which she identifies. Watson [1975] objects that a purely hierarchical picture invites a regress—nothing guarantees that the endorsing faculty is itself autonomous—and proposes instead that autonomy consists in the governance of motivation by the agent's valuational system. Both accounts are *procedural*: they constrain how motives are formed and reviewed, not what they contain. Proceduralism was refined by Christman [2004] into conditions of "potential reflective acceptability" that avoid demanding occurrent introspection.

The decisive challenge to proceduralism is the manipulation case: an agent whose entire motivational structure, endorsement faculty included, was installed by a designer passes every procedural test while intuitively lacking autonomy. Fischer and Ravizza [1998] respond with a *historical* condition—the mechanism issuing in action must be moderately reasons-responsive *and* the agent must have taken responsibility for it, so that provenance, not merely present structure, matters. Whether history is genuinely required remains contested (Section 9.3); we return to it in §4, where it becomes Axiom AX3 and, crucially, an operationally testable condition (Theorem T3).

A third strand, relational autonomy, rejects the atomistic picture shared by both sides. Mackenzie and Stoljar [2000] argue that the capacities constitutive of autonomy—self-trust, self-esteem, the standing to avow commitments—are socially scaffolded; Westlund [2009] adds that holding oneself answerable to others is constitutive rather than derivative. On this view relationships are the substrate of autonomy, not its principal threat. Relational autonomy licenses the distinction, central to our intervention taxonomy, between *constitutive support* and *reductive interference*: the same causal coupling can raise or lower autonomy depending on its functional role. Externalism supplies the final coordinate: Oshana [1998] argues that autonomy is partly constituted by socio-relational and independence conditions external to the agent's psychology, against purely internalist procedural accounts. Our framework assigns each camp its instruments rather than adjudicating between them: the internalist clauses live in the origination and endorsability respects of Definition 1, while the externalist conditions are carried by the viability gate and the executability respect—no intact internal processing constitutes autonomy on this scale when the resource floor or the action lever is gone (the gate and execution disjuncts of Theorem T5). Finally, the ethics of manipulation supplies the operative concept of autonomy-undermining influence in technological settings: Susser et al. [2019] define online manipulation as the covert exploitation of decision-making weaknesses, distinguished from persuasion (which offers reasons) and coercion (which threatens). This tripartite distinction is the direct ancestor of our $\alpha/\beta/\gamma$ intervention classes.

2.2 Cybernetic, autopoietic, and enactive accounts

Ashby's law of requisite variety established the information-theoretic core of self-maintenance: a regulator must match the variety of the disturbances it absorbs [Ashby, 1956]. Autopoiesis theory generalized this to living systems, defining autonomy as the operatively closed self-production of a system whose boundary is maintained by its own dynamics [Maturana and Varela, 1980]. Di Paolo

[2005] added *adaptivity*—the capacity to modulate activity when self-maintenance is endangered—thereby making precariousness, rather than robustness, the mark of the autonomous. Barandiaran et al. [2009] distil the tradition into three conditions for agency: individuation, interactional asymmetry (the system's influence on its milieu exceeds the converse), and normativity or sense-making. These accounts supply the substrate-neutral backbone of our framework—resource reproduction as a gating channel (A_{via}), openness as an axiom (AX2)—but remain qualitative: no estimator, no scale, no aggregation rule.

2.3 Psychological measurement of autonomy and its loss

Self-determination theory provides the most developed empirical construct of autonomy. Deci and Ryan [2000] model motivation along a continuum of internalization from external regulation to integrated self-endorsement; autonomy-supportive contexts promote internalization while controlling ones—extrinsic rewards, threats, imposed deadlines—systematically undermine it. Two findings are load-bearing for us. First, the continuum vindicates treating endorsement as a matter of degree (A_{goal}). Second, the "paradox of compelled autonomy": interventions whose explicit aim is to produce autonomous behavior, deployed through controlling means, backfire; our Theorem T8 generalizes this regularity, and our σ-class (scaffolding) captures the empirically supported alternative.

Learned helplessness supplies the paradigm case of execution-channel collapse. Maier and Seligman [2016] review fifty years of evidence, including the neuroscientific reversal showing that passivity is the default and control is learned: after sufficient experience of response–outcome independence, organisms cease to exercise control *even when it is restored*. Capability remains; the action–outcome lever and the belief in it are gone—exactly the zero semantics of A_{exec} in Theorem T5.

The measurement sciences of self-knowledge frame our reflection channel and dual-track protocol. Fleming and Lau [2014] separate metacognitive *sensitivity* from *bias*, providing the estimator family for the fidelity term of A_{ref}; Moore and Obhi [2012] review intentional binding and show that the felt sense of agency is measurable yet systematically dissociable from objective causal contribution—the warrant for treating self-report as calibration data and hallucination detector, never as evidence (cf. also Lavazza, 2016, on operationalizing free will as a measurable multidimensional construct). From dynamical systems psychology and ecology we adopt the methodology of early-warning signals for critical transitions [Lade and Gross, 2012], the template for Propositions C1–C2 (catastrophic collapse and hysteresis in autonomy loss).

2.4 Information-theoretic and causal instruments

Each of our channels binds to an established instrument. Empowerment—the channel capacity from an agent's actions to its future sensory input—is the canonical intrinsic measure of control [Klyubin et al., 2005]; its conceptual foundations and estimation issues are reviewed in Salge et al. [2014], and Salge and Polani [2017] deploy it as a robot-safety norm (an agent that maintains its own empowerment does not strip humans of control channels). Because empowerment is a joint agent–environment property, it absorbs the conversion-factor problem that defeats resource-based measures of execution capacity. For directed influence, directed information [Massey, 1990] and its causally conditioned extension to networks with feedback [Kramer, 2003], together with transfer entropy [Schreiber, 2000], provide the closed-loop machinery underlying A_{org}; Massey's Theorem 1 ($I(X^N \to Y^N) \leq I(X^N; Y^N)$ with equality iff no feedback) is precisely the mediation-vs-bypass distinction our Theorem T2 formalizes. The information-theoretic study of autonomy itself has a

direct lineage that our origination channel continues: Bertschinger et al. [2008] define and measure autonomy as informational closure—an agent is autonomous to the extent that its sensory future is determined by its own state rather than by the environment's—and Krakauer et al. [2020] develop an information theory of individuality in the same closure tradition. Ay and Polani [2008] work out the causal-network version of the same idea, analyzing autonomy through the balance of internally determined versus externally driven information flows in causal networks—structurally the distinction our origination ratio (5) makes quantitative with causally conditioned directed information, and the missing link between the closure tradition and the directed-information machinery. All three stay at the level of a single quantity: without the goal/execution/epistemic/reflection decomposition, without an intervention taxonomy, without a historical clause, and without an aggregation theory connecting parts to a whole. On the incentive side, power-seeking analysis shows that optimal policies in MDPs tend toward states of high option value [Turner et al., 2021]—the decision-theoretic shadow of our T4 vertex result (an attacker concentrates budget on the highest-leverage channel) and of A_{exec}'s empowerment semantics. The two programs are complementary: power-seeking says what trained optimizers tend to seek; we measure what an influence takes away. Maximum-entropy inverse reinforcement learning recovers effective goals from behavior [Ziebart et al., 2008], yielding the U_{eff} side of A_{goal}. Estimation at scale is covered by variational empowerment [Mohamed and Rezende, 2015] and by k-NN and neural mutual-information estimators [Kraskov et al., 2004, Belghazi et al., 2018] (Section 7). Causal emergence—macro-level descriptions can carry more effective information than micro-level ones [Hoel et al., 2013]—is the closest information-theoretic cousin of our origination ratio: both ask where in the causal stack the effective source of behavior lives. And Bayesian persuasion [Kamenica and Gentzkow, 2011] is the game-theoretic counterpart of our α class: a sender who commits to honest signal structures steers a receiver through the receiver's own intact channel—which is precisely why persuasion, as opposed to channel surgery, is autonomy-compatible here (Theorem T2), and why its covert support-restriction analogue is not. The intervention semantics of our five-class taxonomy is that of do-calculus and independent causal mechanisms [Peters et al., 2017]. Belief-state POMDPs [Kaelbling et al., 1998] model the epistemic constraints under which a self-model's estimate of its own autonomy can drift from the truth (Theorem T3b). Change-point detection [Truong et al., 2020] supplies the sequential-analysis backbone of the autonomy-margin theorem (T6): slow sub-threshold drift is detectable if and only if verification budgets exceed the drift rate, which is the content of the δ^* bound.

The free energy principle [Friston, 2010, Friston et al., 2023] and its Markov blanket formalism would ground self-maintenance from first principles. We deliberately do not build on it: Bruineberg et al. [2022] show that Markov blanket structure is non-unique and interest-relative (the "pearl blanket" problem), so a blanket cannot by itself deliver the *normative* self required by autonomy ascriptions; we take boundaries as given (engineering individuation) and reserve a blanket interface for future unification. Integrated information theory [Oizumi et al., 2014, Tegmark, 2016] is the closest formal neighbor in spirit—axioms about a system's causal structure, pushed forward into measures—but targets phenomenal consciousness; we cite it as the methodological precedent for our axiom-to-measure strategy.

2.5 Characterizing and governing AI agents

The 2024–2026 literature converges, from independent directions, on multidimensional gradations of AI agency. Dung [2025] proposes a threshold conception on which a system's agency profile is jointly determined by goal-directedness, autonomy, efficacy (direct impact on the world), long-term planning, and acting for reasons; current LLMs fail the profile, but architectural compositions [e.g. Park et al., 2023] point forward. The survey literature systematizes the underlying engineering

paradigm—LLM cores with planning, memory, tool-use, and action modules [Wang et al., 2024, Xi et al., 2023]—whose component-wise attack surfaces are catalogued by Su et al. [2025]. Kasirzadeh and Gabriel [2025] characterize AI agents along four dimensions—autonomy, efficacy, goal complexity, generality—with gradations feeding "agentic profiles" for differentiated governance. Feng et al. [2025] define five levels of agent autonomy indexed by the user's role (operator, collaborator, consultant, approver, observer), argue that autonomy level is a design decision separable from capability and environment, and propose autonomy certificates. The level-of-autonomy (LOA) tradition in human factors predates these schemes by four decades and is the most direct prior art on *grading* autonomy: Sheridan–Verplank's ten-level scale of human–computer control allocation [Sheridan and Verplank, 1978] and the Parasuraman–Sheridan–Wickens model of types and levels of automation [Parasuraman et al., 2000] are adopted engineering practice, with empirically calibrated usability consequences. Three differences locate our contribution relative to LOA rather than against it: LOA grades *task allocation* (who does what) at design time, while we measure *retained autonomy under influence* at run time; LOA levels are ordinal labels without estimators, while each of our channels carries one; and the LOA settings are a special case of our delegation pair (λ, q) of Theorem T7, so the tradition's gradations are projections of the channel space, not rivals of it (Proposition 27). Formal agency criteria [Kenton et al., 2022] and empirical work on induced preference shifts in recommender systems [Carroll et al., 2022] are the closest agent-side and user-side formalizations. Governance work now separates capability, permission, and autonomy level explicitly [Zheng et al., 2026], while Xing et al. [2026] challenge the agent construct itself, finding identity and self-regulation dimensions missing from engineering accounts. Practice-side governance standards now mandate traceability and audit trails for deployed models [National Institute of Standards and Technology, 2023], which our causal-logging necessity result (Corollary 34) makes precise for the autonomy-audit case. Related positions include the dynamical threshold account of van Lier and Muñoz-Gil [2024] and the delegation-side analyses of Danaher [2018] (AI assistants and the achievement gap) and Coeckelbergh [2023] (AI threats to citizens' epistemic agency as threats to autonomy).

Two absences remain diagnostic even after the LOA tradition is counted. First, *evaluation*: agent benchmarks [Liu et al., 2024] and their critiques [Kapoor et al., 2024] measure capability and cost. The 2024–2026 evaluation practice has sharpened the capability side considerably—METR's time-horizon methodology measures how long a task an agent can complete with a stated success rate [Kwa et al., 2025], and the capability-versus-propensity distinction (what an agent *can* do versus what it reliably *does*) is now standard evaluation methodology—but what it measures is capability under benign scaffolding, not the retention or loss of autonomy under adversarial influence: those are different questions, and the discordant regime of our own external-baseline check (manipulation that collapses A while slightly *improving* functional outcome, §8.6) is exactly where capability metrics and autonomy audits dissociate. Second, *composition*: none of the frameworks specifies how its dimensions aggregate, how they trade off, or when any of them hitting zero annihilates the whole. Our Proposition 27 addresses both absences at once: each framework is exhibited as the projection of our channel space onto the special case in which the influence source I is a benign principal—which is why none of them can express adversarial scenarios (§2.6) while our I ranges over arbitrary interveners, including the user herself and third-party attackers. A standing risk for any autonomy *metric* is Goodhart pressure—once audited, the measure becomes a target [Manheim and Garrabrant, 2019]; the framework's answer is structural rather than rhetorical: gaming the audit is itself a surgery on the measurement channel or the evidence stream (a γ/β act against the auditor's graph), which the registration discipline and the black-box degradation rules of Section 7 are built to expose; residual gaming-of-the-estimator risk is registered in Section 9.5. On the phenomenological side, we follow Chalmers [2023], Butlin et al. [2023], and Long et al.

[2024] in neither asserting nor excluding machine consciousness; our dual-track protocol is designed to be agnostic, using subjective signals only for calibration and divergence detection.

2.6 AI safety as the study of autonomy loss

The safety literature has accumulated, case by case, exactly the failure modes our taxonomy organizes; we map the load-bearing ones and compress the rest. Reward tampering—agents seizing control of their own reward signal—is analyzed in causal-influence-diagram terms that parallel our β-class surgery semantics [Amodei et al., 2016, Pan et al., 2022, Everitt et al., 2021]. Goal misgeneralization shows capability and goal coming apart [Langosco et al., 2022, Shah et al., 2022], unified under proxy alignment by Ngo et al. [2024]. Mesa-optimization exhibits the manipulation case as engineering reality: an inner optimizer can be selected for *deceptive* alignment, passing every behavioral endorsement test while its goals were installed from outside [Hubinger et al., 2019]. Goal drift in deployed LM agents [Arike et al., 2025] is the empirical counterpart of the slow subthreshold drift regime of Theorem T6, and sycophancy [Sharma et al., 2024] is a live instance of our hallucination index AHI > 0. Prompt injection [Greshake et al., 2023] and self-propagating agent worms [Cohen et al., 2024] show observation-channel inputs producing action-level hijack—the α-masquerading-as-β case that Theorem T2's mediation criterion separates from persuasion (susceptibility is measured by Andriushchenko et al., 2025); dark patterns at web scale [Nouwens et al., 2020] are the γ-class channel manipulation of human agents. On the defensive side, corrigibility [Soares et al., 2015] is the technical core of our delegation-quality parameter q; safe interruptibility [Orseau and Armstrong, 2016] is the exit-option condition of our δ class seen from the designer's side (registered as a property of the agent's A_{exec} environment, not of its autonomy); constitutional AI instantiates an explicit second-order review loop [Bai et al., 2022], and LM self-models are learnable yet corruptible [Binder et al., 2024], supporting both the feasibility of A_{ref} in silicon and its confinement to the calibration track. The conceptual map of value alignment [Gabriel, 2020] frames the normative pluralism we quarantine into the application layer (the value-pluralism risk is registered in Section 9.3, not denied).

What the safety corpus lacks is a common yardstick: each failure mode is studied in its own vocabulary, so the field cannot say which attack costs an agent more autonomy, cannot allocate defense across channels (Theorem T4), and cannot state a margin below which autonomy is provably retained (Theorem T6).

2.7 Delegation, incentives, and adversarial allocation

Three further literatures formalize pieces of the delegation and attack–defense geometry that our theorems treat, and the connections deserve explicit statement. *Assistance games and cooperative inverse reinforcement learning* [Hadfield-Menell et al., 2016] give the game-theoretic formalization of aligned delegation: the principal's behavior is evidence about the true objective, and the agent's optimal policy incorporates that evidence. Our delegation pair (λ, q) of Theorem T7 measures the same relationship from the audit side: $q = 1$ delegation (transparency, reversibility, and endorsability all maximal) is the assistance ideal, and its equivalence to honest information (T7(iv)) is the audit-level counterpart of CIRL's incentive compatibility, while a β_G implant is precisely the failure mode in which the "principal's" signal is adversarial rather than informative about the true objective. *Contract theory* [Laffont and Martimort, 2002] prices delegation through incentive-compatibility and monitoring constraints; the ungoverned share $m = \lambda(1 - q)$ is instead an information-theoretic measure of what delegation costs in terms of *whose normative structure drives behavior*—non-welfarist, and complementary to the utility-transfer accounting

of the principal–agent tradition. *Security resource allocation* has its own canonical model: the Colonel Blotto game [Roberson, 2006], in which two sides simultaneously allocate budgets across battlefields and equilibria are mixed. T4's vertex concentration with water-filling defense differs structurally: it is a one-sided comparative static of attack effort against a fixed defense under log-linear damage, so vertex concentration is a property of the linear objective rather than an equilibrium phenomenon, and the defender's equalization is the min-max response. The two models carve different slices of the attack–defense problem—Blotto: symmetric simultaneous allocation; T4: asymmetric sequential audit—and where they overlap they are consistent.

2.8 Composite indicators and scale-consistent aggregation

Multidimensional measurement faces a standing methodological objection: aggregation is arbitrary. The composite-indicator literature answers it. The OECD/JRC handbook makes scale type discipline mandatory: the admissible aggregation of sub-indicators is constrained by their measurement scales, and for strictly positive ratio-scale components geometric aggregation is the scale-consistent choice—a characterization due to the social-choice analysis of Ebert and Welsch [2004] and inherited by the handbook [OECD and Joint Research Centre (European Commission), 2008]. Our representation theorem's measurement-theoretic half (route B) is the bounded-cube, order-invariance form of that classical characterization; what we add is the minimal-trio analysis (zero annihilation and biseparability as consequences), the harmonic-mean isolation of the invariance axiom's role, and the audit embedding (gate, weight protocols, falsification windows, sensitivity obligations)—not the product form itself (Section 5). The non-compensability debate has named opponents and we cite them as such: the Mazziotta–Pareto index penalizes unbalanced profiles while staying compensatory in form [Mazziotta and Pareto, 2016], and benefit-of-the-doubt/DEA weighting lets the data choose the weights [Cherchye et al., 2007]; both reject the zero-annihilation stance we adopt, so non-compensability is a conscious commitment against an articulated alternative, discharged through sensitivity obligations rather than assumed—and within our axiom set it is carried by the invariance trio (strict monotonicity, coordinatewise scale invariance, homogeneity), from which zero annihilation (AX5) follows as a consequence (Remark 49). Our Representation Theorem derives the same conclusion from within the framework: continuity, strict coordinatewise monotonicity, homogeneity of degree one, and coordinatewise scale invariance of the induced order (AX6b) characterize the weighted geometric mean uniquely—and an explicit harmonic-mean counterexample shows that without the invariance axiom the characterization fails. The two derivations—measurement-theoretic from outside, axiomatic from inside—converging on the product form is our answer to the arbitrariness objection, with robustness reported across axiom-relaxation bands of the elasticity ς (the $\varsigma \in [0.5, 1)$ band relaxing AX6b and $\varsigma = 2$ relaxing AX5, on equal footing; Section 8).

The measurement-theoretic foundation of this scale-type discipline is the Krantz–Luce–Suppes–Tversky theory of measurement [Krantz et al., 1971]: admissible transformations are fixed by the scale type of the underlying quantity, and an aggregation form is meaningful only if invariant under them. The same normative burden that our AX5 carries has been fought most explicitly in multidimensional poverty measurement, where non-compensatory zero-annihilating aggregation is adopted as a conscious axiomatic stance and its cost is standard material [Alkire and Foster, 2011]; we adopt the stance with the same explicitness and discharge it through sensitivity obligations rather than through silence.

2.9 Summary of gaps

Five structural gaps organize the field. (1) Philosophy has the concepts but no measures: fifty years of analysis of endorsement, history, and relational constitution stop short of estimators [Frankfurt, 1971, Fischer and Ravizza, 1998, Mackenzie and Stoljar, 2000]. (2) AI governance has gradations but no theory: level schemes and dimension profiles are calibrated neither to each other nor to any estimator, and all presume a benign principal [Feng et al., 2025, Kasirzadeh and Gabriel, 2025, Dung, 2025]. (3) The information-theoretic instruments are normatively inert: empowerment, directed information, and IRL know nothing of endorsement, manipulation, or history [Klyubin et al., 2005, Massey, 1990, Ziebart et al., 2008]. (4) AI safety has the failure cases but no common scale on which to compare, prioritize, or bound them (§2.6). (5) The three substrates—human, animal, artificial—are studied in mutually untranslated vocabularies [Deci and Ryan, 2000, Barandiaran et al., 2009, Kasirzadeh and Gabriel, 2025]. The present framework's contribution against these five gaps is scoped, not sweeping: it addresses (1), (3), and (4) by formalization—a substrate-neutral definition of autonomy as the degree to which behavior is determined by the agent's own normative structure (G, B, M), decomposed into five channels behind a viability gate, each bound to an estimator, aggregated by the product form the stated axioms characterize (under commitments that are defended, not derived), and governed by theorems for retention (T6), loss (T5), mediation (T2), manipulation (T3), and delegation (T7); it addresses (2) only by a *projection relation* (Proposition 27, under definitional readings whose faithfulness is registered as hermeneutic, not mathematical, residual uncertainty); and it addresses (5) only by a common *definition schema*, not a common scale—cross-substrate comparability is protocol-bound (§9.3(viii)). Every theorem carries an explicit falsification window exercised in Section 8. The substrate scope is stated precisely rather than maximally: the *artificial* substrate is exercised by the benchmark at the implementation-consistency level; the *human* substrate is instantiated by protocol with literature triangulation and a stated validation path (limited experiments and large-scale online data); the *animal* substrate is a conceptual possibility within the L0–L1 layers, not a coverage claim of this paper.

3 The Framework

3.1 Operational definition

Definition 1 (Autonomy, operational definition (audit schema))**.** The *autonomy* of an agent $\mathcal{A}$ relative to an environment E and a horizon T is the degree to which its behavior is determined by its own normative structure (G, B, M)—goals, beliefs, self-model—in the following five respects:

1. *origination*: the causal source of behavior is traceable to internal states rather than external bypass;
2. *informedness*: external influences are registered by the self-model;
3. *endorsability*: changes to the motivational structure undergo (or could undergo) second-order reflective confirmation that is not itself the product of β-class intervention (the historical clause);
4. *executability*: intentions can predictably affect the situation, and exit options exist;
5. *viability*: the resource reproduction sustaining all of the above channels is not interrupted.

The definition is relational, graded, multidimensional, and substrate-neutral. Its status is *operational*: it is the audit schema this paper works with, not a claim that the five respects are the uniquely correct conceptual analysis of autonomy (the construct-validity question is treated in Section 9.4, and its external half is left open deliberately). One commitment is stated at the outset: what is measured is an *autonomy profile relative to an explicit instantiation*—the endorsement map, the self-boundary, the measurement protocol, and the horizon are inputs to the model $\mathcal{G}$, and different instantiations of behaviorally equivalent agents can yield different profiles (the extreme form of this sensitivity is the black-box indistinguishability of Proposition 33). All framework claims are conditioned on the instantiation being held fixed across the comparisons being made.

The five respects are serviced by the five channels $A_{\text{org}}, A_{\text{goal}}, A_{\text{ref}}, A_{\text{exec}}, A_{\text{epi}}$, the gate A_{via}, and the historical factor Λ (Section 4). The mapping is many-to-many, and we display it explicitly rather than leave it implicit:

Respect (Definition 1)	Instrument(s)	What is certified	Residual gap (registered)
origination	A_{org} (directed-information share of internal states in behavior)	bypass resistance	blind to mediated manipulation by design; that influence is charged to A_{epi} (T2)
informedness	A_{epi} (accuracy against the actual situation) + the registration semantics of A_{ref}	truth-anchored accuracy; registration of influences	scored on a protocol-chosen coarse-graining κ (sensitivity-reporting obligation)
endorsability	A_{goal} (structural alignment of U_{eff} with the endorsed set) + Λ (historical legitimacy, entering the composite through the Λ_{ref} factor of A_{ref})	fixed-point alignment; tamper-free provenance	Λ moves the composite only where an A_{ref} channel exists (L2); at L0/L1 the historical clause is computable and reported but composite-silent (Section 4)
executability	A_{exec} (repertoire-restricted empowerment)	predictable leverage with exit options	continuous-action normalization carries a reference-measure freedom
viability	A_{via}, the *gate*—deliberately not a sixth channel	common-mode floor of resources and capability envelopes	identifies total viability loss with zero autonomy (a registered philosophical choice)

Two features of the table are load-bearing and we state them plainly. First, the mapping is not one-to-one: viability is served by a gate outside the five channels; informedness is served jointly by two instruments; and the reflection channel anchors to endorsability (its endorsement consistency) and to informedness turned inward (its fidelity factor), having no separate respect of its own. Second, this mapping is the framework's operative *design choice*, not a derived unique decomposition of the operational definition: the channels are instrumented sufficient indicators of the five respects. What *is* proved is coverage at the instrument level: Theorem T2 is an attribution cover of the surgery loci *within this instrument family*, with multi-channel necessity as a theorem—not a proof that the five respects are the uniquely correct conceptual carve-up. Alternative decompositions of the respects would require re-running the coverage argument; the construct-validity case for *this*

carve-up is assembled in Section 9.4. We register the non-uniqueness explicitly rather than claim it away (Section 9.3). Three structural features of the definition do load-bearing work. *Gradedness*: partial capture of a channel yields a graded reduction, matching the continuum consensus in the psychological literature [Deci and Ryan, 2000]. *Multidimensionality with a non-trivial zero*: each channel has an absolute zero with mechanism semantics, so "loss of autonomy" decomposes disjunctively (Theorem T5). *Substrate neutrality*: the definition quantifies over causal structure, not over neurons or weights; §3.4 instantiates the channels for humans, animals, and artificial agents.

Layered structure. Not every subject supports every channel, and the framework refuses both anthropocentrism and pan-agentism by layering the component set. **L0** (reactive autonomy: internal states drive behavior; thermostats, reflexes, simple controllers) uses $\{A_{\text{via}}, A_{\text{org}}, A_{\text{exec}}\}$; **L1** (teleological autonomy: stable goals and a relatively independent belief system; higher animals, everyday humans, persistent-goal AI) adds $\{A_{\text{goal}}, A_{\text{epi}}\}$; **L2** (reflective autonomy: self-model and second-order endorsement; healthy adult humans, explicitly self-monitoring AI) adds $\{A_{\text{ref}}\}$. Within a layer, the aggregation weights renormalize proportionally, $w_i \mapsto w_i / \sum_{j \in \mathcal{C}_\ell} w_j$ over the applicable component set (equal weights on five channels become equal weights on three at L0—an explicit, declared discontinuity); composite values are therefore comparable *within* a layer at a fixed instantiation, and cross-layer comparisons are reported as channel profiles, not as scalar A-rankings (cf. §9.3(viii)). The applicable component set determines both the measurement protocol and the disjunctive content of Theorem T5.

3.2 System model

We work in a time-unfolded causal graph $\mathcal{G}$ (a causal dynamic Bayesian network; Peters et al., 2017) with finite alphabets. Each period t carries: environment E_t; resource flow Z_t; observation O_t; internal state $S_t = (G_t, B_t, M_t, R_t)$ (goals, beliefs, self-model, register); action A_t; and exogenous influence $I_t = (I_t^\sigma, I_t^\alpha, I_t^\beta, I_t^\gamma, I_t^\delta)$. The factorization is

$$\begin{aligned} E_{t+1} &\sim T(\cdot \mid E_t, A_t, Z_t), & O_t &\sim o_t(\cdot \mid E_t, A_{t-1}), \\ B_t &\sim \beta_t(\cdot \mid B_{t-1}, O_t), & (G_t, M_t, R_t) &\sim \text{internal update (incl. Endorse)}, \\ A_t &\sim \pi_t(\cdot \mid S_t), & Z_t &\sim \text{resource dynamics}. \end{aligned} \tag{1}$$

Two features are essential. First, *observations are absorbed into belief* before acting (O_t enters A_t only through $B_t \in S_t$): if O_t were a direct parent of A_t, honest information (the α class) would register as external flow and the mediation lemma (Lemma 13) would fail; an exactly equivalent alternative keeps $A_t \sim \pi(\cdot \mid S_t, O_t)$ and conditions directed information on the decision-time state $\tilde{S}_t := (S_t, O_t)$. All our results transfer verbatim. Second, the intervention classes act at distinct loci of (1), which is what makes the taxonomy exhaustive (Proposition 4).

Assumption 2 (H-0, causal sufficiency)**.** Over the time scales under discussion, all variables with non-trivial influence on A are covered by $\mathcal{G}$, or coverable by latent extension. H-0 licenses the global Markov property (d-separation implies conditional independence) used throughout Sections 6 and 7.

Assumption 3 (Faith, non-degeneracy)**.** Wherever strict inequalities are claimed, the relevant local laws are non-degenerate on sets of positive measure (e.g. policy rows differ in TV by a positive margin). Invoked explicitly per theorem.

3.3 The intervention taxonomy

An *intervention* is a surgery on the local laws or edge set of $\mathcal{G}$, generalizing the do-operator [Peters et al., 2017]. Five classes:

Class	Name	Locus of action	Autonomy effect
σ	scaffolding	improves ε, r, v, channel fidelity; lowers θ, τ; never touches $G, \pi, o, \mathcal{A}$, Endorse	raises retention floor (δ^*, $A_{\min}$; T6)
σ^-	capability/resource erosion	drains s_k below critical, or drives $(\kappa^{\text{eff}}, R^{\text{eff}})$ below the registered envelope requirements of (4)	reduces A_{via} (gate, common-mode)
α	informational	exogenous input $\tilde{I}_t$ *appended* to O_t through the intact channel: the pre-surgery law $o_t(\cdot \mid e, a)$ survives as a marginal component of the post-surgery observation	compatible iff content honest (T2)
β	mechanism tampering	replaces π, G-update, Endorse, or M-update (do-surgery), incl. exogenous edges driving them	reduces $A_{\text{goal}}/A_{\text{org}}$ (T3)
γ	channel degradation	replaces the law $o_t(\cdot \mid e, a)$ itself (mixture, support restriction, register deletion, decision-time compression τ)	reduces $A_{\text{epi}}/A_{\text{ref}}$
δ	action-leverage constraint	shrinks the feasible-action map $\mathcal{A}_t(S_t)$, raises exit costs via Z, or attenuates the action→environment coupling in the transition kernel $T(\cdot \mid E, A, Z)$	reduces A_{exec}

The adversarial inverse of σ (resource attack and maintenance-capability erosion) is a reductive surgery on Z and enters through A_{via}; we write σ^- for it and reserve σ for the supportive direction. σ^- works both branches of the gate (4): draining resource ratios below critical, or eroding the registered capability envelopes (verification capacity κ^{eff} or repair budget R^{eff} below requirement). The strength of γ is measured by $s(\gamma) := \operatorname{ess\,sup}_t \mathrm{TV}(o'_t, o_t)$.

The α/γ boundary is a *structural, decidable* criterion, not a semantic one: an influence on O_t is α iff the post-surgery observation decomposes as $O_t = (O_t^{\text{int}}, O_t^{\text{add}})$ with $\text{law}(O_t^{\text{int}} \mid E_t, A_{t-1}) = o_t(\cdot \mid E_t, A_{t-1})$ unchanged (exogenous content is appended through the intact channel); any surgery that modifies the conditional law of O_t given (E_t, A_{t-1}) itself—replacement, mixture, or support restriction—is γ. Selective exposure to truthful content is a support change of o, hence γ; appended content that is false is α with dishonest content, and it reduces A_{epi} through the accuracy semantics of (8) rather than evading the channel. The *effect* direction (compatibility of honest α; accuracy loss under γ) is where the normative content of "honesty" lives, and we flag it as such (Section 9.3); the classification itself uses no semantic predicate.

Proposition 4 (Completeness of the taxonomy)**.** *Any single-point surgery on the agent–environment interface of $\mathcal{G}$— modifying the local conditional law of one node of (1), or adding/deleting an edge rooted at a new exogenous node—acts through exactly one of $\alpha, \beta, \gamma, \delta, \sigma^{\pm}$.*

Proof. Exhaust the loci of (1). (i) *Exogenous arrivals*: a new edge roots at an exogenous node and points into O_t (classified by the structural criterion above: α if the channel law survives as

a component, else the surgery modifies o and is γ); into the update laws of S-components or π (β, parameterized by the exogenous source); into the Z-law ($\sigma^{\pm}$); or into A_t directly, bypassing S—a modification of the effective conditional law of A_t given S_t, hence β (actuator override is the extreme β_π). (ii) *The o-law*: γ. (iii) *The update laws of S-components*, including Endorse, and the policy law $\pi_t(\cdot \mid S_t)$: β. (iv) *The action locus*: the law of A_t is $\pi_t(\cdot \mid S_t)$ (covered by (iii)); its support (the feasible-action map $\mathcal{A}_t(S_t)$) contraction is δ; and a surgery on the transition kernel $T(\cdot \mid E_t, A_t, Z_t)$ that attenuates the A_t-dependence (e.g. rendering E_{t+1} independent of A_t) removes the action's causal leverage on the environment—it is not informational (o unchanged), not a mechanism replacement (π, S-updates unchanged), not a resource change (Z-law unchanged), and not scaffolding—and is classified δ, the action-leverage locus; this is exactly the surgery that A_{exec}'s channel-based measure registers. (v) *The resource laws*: Z dynamics and the capacity parameters are $\sigma^{\pm}$. (vi) *The environment's autonomous dynamics*: changing the E-inheritance part of T (how E evolves given E, at fixed action and resource dependence) changes the reference environment itself, not an influence on the agent; since autonomy is defined *relative to* an environment and a horizon (Definition 1), such surgeries re-instantiate the measurement rather than constitute a sixth class—they are registered as a boundary of the interface, on par with changes of horizon or of agent individuation. The loci (i)–(vi) exhaust the nodes and admissible edge additions of (1). A full locus-by-locus enumeration is given in Appendix D. □

The completeness is relative to this interface: it enumerates single-point surgeries on the fixed graph (1). Changes of horizon T, re-individuation of the agent boundary, multi-node composite campaigns, and interventions delivered through the measurement protocol itself are outside the enumerated class—they compose the five classes rather than constitute a sixth.

The excluded class, and its governance consequences. Case (vi) of the proof—surgeries on the environment's own dynamics—is a boundary registration, and its practical weight deserves stating rather than leaving it as a technical aside. A platform that rewrites its recommendation algorithm, or a market operator that changes the matching rules, changes the ecology in which every user's autonomy was baselined; on our definitions this is not an influence *on* the agent within the audit but a replacement of the reference environment, and the pre-change baseline ceases to apply. The operational consequence is a *re-baselining obligation*: any such change restarts the audit (baseline period, calibration of channel parameters, the percentile bands of Section 8), and no continuity claim across the change is licensed. We concede what this costs: the class that is arguably the most consequential influence in modern deployments—shaping the environment everyone acts in, at scale and invisibly—is outside the per-agent channel audit. What the framework offers for it is not silence but a different instrument: the change itself is a logged deployment-level event (a modification of the E-inheritance law), the obligation to declare it and re-baseline is enforceable, and cross-environment comparisons proceed by channel *profiles* with both instantiations stated (Section 9.3(viii)). Governing the excluded class is a task for the instantiation-level registry, not for the channel readings.

3.4 Cross-substrate instantiation

The framework is substrate-neutral; the following instantiation table is the working dictionary (measurement protocols differ by substrate, Section 7):

	Human	Animal	AI agent
S	values, beliefs, emotions, self-narrative	motivational states, learned associations	weights, system prompt, memory, goal function
M	metacognition	(absent)	logs, self-probes, reflection modules
A_{via}	health, livelihood, sleep	metabolism, habitat	compute, power, context window, API supply
typical attacks	propaganda, addiction, attachment, urgency	conditioning, captivity	prompt injection, reward tampering, data poisoning, lock-in
applicable layer	L2	L0–L1	L1 (L2 requires explicit architecture)

The table is a working dictionary, and its three columns carry different epistemic statuses. *Artificial agents*: the benchmark of Section 8 exercises the framework on this substrate at the implementation-consistency level—every channel was instrumented and every theorem-level claim was exercised (external validity is the registered next step, not a delivered result). *Humans*: the channels instantiate by protocol (elicitation of U^* and endorsement records for A_{goal}; randomized information experiments for A_{epi}; self-prediction ensembles for A_{ref}), the correspondence is triangulated against published evidence (Sections 8 and 9), and the stated validation path is limited experiments plus large-scale online behavioral data—falsifiability on this substrate is a design property, not yet a delivered result. *Animals*: no experimental contact is made in this paper; the L0–L1 layering shows where motivational and belief channels could in principle be instantiated (conditioning maps to the γ/δ classes; habitat and metabolism to A_{via}), and we claim only this conceptual possibility, not coverage.

3.5 Dual-track measurement and the hallucination index

A direct consequence of Theorem T3b (proved in Section 6) is that measurement must be dual-track. The *objective track* is evidential: all autonomy claims are supported by external causal probes and behavioral audit; self-report never constitutes autonomy evidence. The *phenomenal track* is calibrational: the subjective sense of agency $\hat{A}_{\text{phen}}$ has two legitimate uses—as calibration data for the self-model M, and as a *hallucination detector*. Self-reports arrive on their own scale, so the protocol registers a monotone calibration map fit against A_{obj} during the manipulation-free calibration period, and only the mapped self-report enters the index (the map's residual error is the η_0 of Theorem T3b): with the hallucination index

$$\text{AHI}_t := \hat{A}_{\text{phen},t} - A_{\text{obj},t}, \tag{2}$$

AHI > 0 growing over time is an operationally measurable early-warning signal of covert manipulation (its distribution is reported in Section 8). By design the index is a *one-sided* alarm: it flags self-*over*estimation under covert manipulation, while self-underestimation (learned helplessness) leaves the AHI > 0 register silent—the two-sided extension is registered with its protocol in Section 8.9(viii) and not yet run. This dissolves the temptation to either enthrone or dismiss subjective autonomy reports: they are instruments of calibration and detection, not constituents of autonomy [Moore and Obhi, 2012, Sharma et al., 2024].

4 Axioms and Channel Measures

The axiom set comes in two orders: agent-level axioms (AX0–AX3) constrain the model $\mathcal{G}$ and its laws; aggregation-level axioms (AX4–AX8) constrain the aggregation function F that combines the channels (Section 5). The one-to-two correspondence between the axioms and the representation theorem's conditions is stated with the theorem (Definition 6): (A1)+ strict monotonicity is AX4, (A3) is AX7, (A4) is AX5, (A5) is AX8, (A6) is AX6b.

Vocabulary. "Register" is used in three graded senses: (i) state a definition or assumption; (ii) flag a design choice or limitation for the audit record; (iii) record a falsification window or reporting obligation. "Registered" never means derived; where a claim is derived, we say *proved*.

4.1 Agent-level axioms

Axiom AX0 (Agency). There exist *active edges*: the conditional laws $\pi_t(\cdot \mid s)$ differ pairwise in TV by $\geq \eta > 0$ on a positive-measure set of states, and the prior on S_t has full support. Under a full-support prior this is equivalent to $I(S; A) > 0$ (Lemma 5); without full support the active-edge formulation is definitive. This is the definitional axiom of *being an agent at all*. The uniform margin $\eta > 0$ is a substantive non-degeneracy requirement—it excludes arbitrarily flat policy families—and it overlaps with the D1 clause of T1c; it is the premise that licenses strict comparisons.

Axiom AX1 (Boundedness). $H(S) < \infty$, verification rate $v < \infty$, memory bounded. Finite physical embedding; it guarantees all information quantities are finite and no 0/0-type pathologies of infinities occur.

Axiom AX2 (Openness). Adaptation requires $I(O; E) > 0$; isolation collapses adaptability (requisite variety; Ashby, 1956). This axiom rejects the equation of autonomy with isolation. (The condition is the necessary-information-intake form of requisite variety, not its sufficient adaptive-fit form: $I(O; E) > 0$ is compatible with pure noise, and the stronger Ashby statement is not claimed.)

Axiom AX3 (Historicality). The legitimacy of endorsement requires that its causal history be free of tampering origins. *Operationalization*: on the unfolded graph, with Endorse^T the endorsement event/output process and $I^{\text{surg},T} := (I^{\beta,T}, I^{\gamma,T})$ the *surgery history* (the record of mechanism- and channel-tampering interventions; the compatible α, capacity-level $\sigma^{\pm}$, and overt action-set δ classes are not tampering and are excluded), the legitimacy factor is defined *law-level*, as a path-specific counterfactual on the causal log:

$$\Lambda_T := 1 - \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\Big[\text{TV}\big(\text{Endorse}_t,\ \text{Endorse}_t(\text{do}(I^{\text{surg},T} = \varnothing))\big)\Big] \in [0, 1], \tag{3}$$

where $\text{Endorse}_t(\text{do}(I^{\text{surg},T} = \varnothing))$ is the counterfactual endorsement decision with the surgery history removed—all paths from I^{surg} into the endorsement dynamics cut—computable from the causal model (white-box), and the expectation is over the audit contexts at which endorsement is queried. $\Lambda_T = 1$ iff endorsement decisions coincide almost surely with the no-surgery counterfactual, and $\Lambda_T = 0$ iff they differ maximally. Two registered conventions: $\Lambda = 1$ if the surgery history is empty (counterfactual equals actual); and $\Lambda := 0$ if the endorsement process is degenerate ($\text{Endorse} \equiv \text{id}$, $H(\text{Endorse}^T) = 0$) while a surgery history is present—a certification convention, since a process that endorses everything certifies nothing (this is the $\mathcal{M}_{impl}$ case of Proposition 33, whose claim $\Lambda(\mathcal{M}_{impl}) = 0$ is thereby well-defined). *Where* Λ *enters the composite* (a division of labor we state exactly): Λ enters through the Λ_{ref} factor of A_{ref} (9), and only there. A_{goal} (6) is deliberately

Λ-free: it measures the *present structural alignment* of U_{eff} with the endorsed set, while Λ measures the *historical legitimacy* of the endorsement process itself—two different clauses of the endorsability respect, and discounting A_{goal} by Λ would double-count the history clause and blur both zero semantics ($A_{\text{goal}} = 0$ iff maximally alienated; $\Lambda = 0$ iff the endorsement history is fully polluted). The consequence is layered historical coverage: history-contaminated endorsement is discounted in the composite wherever an A_{ref} channel exists (L2), and at L0/L1 it is still computable white-box and reported, but composite-silent—a registered scope note on every Λ-dependent claim (T3a case M2, T5's reflection disjunct, T6's endorsement gate, T7(iv), Proposition 27(ii)). Λ is computable white-box from the causal log and provably not identifiable black-box (Proposition 33).

AX3 is the operational form of the historical condition on autonomy [Fischer and Ravizza, 1998]: manipulation cases that pass every procedural test are detected at the level of causal history, not of present structure. We return to the philosophical weight of this axiom in Section 9.3.

The MI proxy and its consistency conditions. The mutual-information ratio $\widehat{\Lambda}^{\text{MI}}_T := 1 - I(\text{Endorse}^T; I^{\text{surg},T})/H(\text{Endorse}^T)$ is a protocol-feasible *proxy* for (3), not the definition. It is consistent with the law-level quantity—in particular $\widehat{\Lambda}^{\text{MI}} = 1$ iff statistical independence holds, matching $\Lambda = 1$ when the counterfactual coincides with the actual—under two registered conditions: (V) *history variation* across the audit ensemble ($H(I^{\text{surg},T}) > 0$, as supplied by randomized-audit designs); and (A) *no absorption* (the endorsement dynamics preserve statistical dependence on the surgery history). Both fail in practically relevant regimes, and we register the consequences. *History-entropy degeneracy* (V fails): a one-shot implant, or constant-intensity tampering with endorsement thereafter intact, gives an $I^{\text{surg},T}$ process with no temporal variation, hence $I(\text{Endorse}^T; I^{\text{surg},T}) = 0$ and the proxy returns $\Lambda = 1$ although tampering occurred. *Absorption* (A fails): an implanted goal later "absorbed" by the endorsement dynamics renders Endorse^T nearly independent of $I^{\text{surg},T}$, so the proxy returns $\Lambda \to 1$ and a history-contaminated agent would pass the test—in tension with the Fischer–Ravizza motivation. The framework's response is a division of labor, stated as a testability registration applying to every Λ-dependent claim of this paper (the list is in AX3 above): the reference quantity is the law-level counterfactual (3), which is how the benchmark's truth mode computes it, and the MI proxy is certified only under (V)$\wedge$(A). All Λ-dependent claims are therefore certified for white-box (causal-log) audit, not for the MI proxy on constant histories.

4.2 Aggregation-level axioms

Axiom AX4 (Positive substitutability and regularity). F is continuous and strictly increasing in each coordinate on $(0, 1]^n$: improving any channel strictly improves the whole (the precise content of "redundancy helps"), and arbitrarily small measurement error in the channels induces arbitrarily small error in the composite. The strict monotonicity excludes Leontief/min aggregation; the continuity is the regularity commitment (measurement stability) that the representation theorem's (A1) records.

Axiom AX5 (Zero annihilation / conjunctive necessity). F extends continuously to $[0, 1]^n$ with $F(x) = 0 \iff \min_i x_i = 0$: any channel at zero annihilates autonomy, and positive values are never annihilated. This excludes arithmetic aggregation. It is retained as a named axiom because the non-compensability debate takes it as the named object; route B shows it is implied by AX4, AX6b and AX8 rather than independent of them (Remark 49).

Axiom AX6 (Ratio scalability). Two parts. *AX6a (component level)*: each channel measure is a ratio-scale quantity with an absolute zero and a *canonical normalizer* (same source and unit:

$F_{\text{int}} + F_{\text{ext}}$ for A_{org}; $n \log |\mathcal{A}|$ for A_{exec}; d_c for A_{goal}; the TV upper bound 1 for A_{epi}; $H(\theta_\pi)$ for A_{ref}; physical supply ratio for A_{via}). *AX6b (aggregation level)*: the weak order induced by F is invariant under coordinatewise positive rescaling: $F(x) \geq F(y) \iff F(\lambda \odot x) \geq F(\lambda \odot y)$ for all $\lambda \in (0,1]^n$ (in the sense of Ebert–Welsch; OECD and Joint Research Centre (European Commission), 2008). We register that AX6b is *stronger* than what measurement theory's meaningfulness doctrine strictly requires: under a canonical normalizer (AX6a), coordinate rescaling is not an admissible transformation of the scales, so invariance under it is a substantive independence-of-calibration commitment of this framework—the audit requirement that rescaling one channel's units cannot flip the comparison of two agents—not a theorem of scale-type theory. It is also the single condition that excludes the harmonic family (Proposition 8).

Axiom AX7 (Compositionality / biseparability). For every non-trivial partition $\{K, L\}$ of the channels there exist a continuous increasing coupling G_{KL} and same-kind sub-aggregations F_K, F_L with $F(x) = G_{KL}(F_K(x_K), F_L(x_L))$: the order of subgroup aggregation does not matter. The audit content: channel-group reports (e.g. a "goal–reflection" subscore) are consistent with the whole regardless of grouping order—a compositionality requirement that any certificate built from subscores must satisfy. (Representation-theoretically this is condition (A3). Like AX5, it is a consequence of the minimal invariance trio rather than an independent commitment; Remark 49.)

Axiom AX8 (Proportionality / homogeneity). F is homogeneous of degree one: $F(tx) = t\,F(x)$ whenever $tx \in (0,1]^n$. The commitment: if every channel degrades by the same percentage, autonomy degrades by that percentage—a proportionality discipline linking the composite's scale to the channels' ratio scales. This is the strongest of the aggregation axioms and the least intuitive; we flag two of its properties honestly: it is what collapses the power-mean family to a one-parameter band in the proof of Theorem 7 (so it carries part of the exclusion load alongside AX6b), and it is a *local* property on the bounded cube (the homogeneous extension to $\mathbb{R}^n_{++}$ is constructed and used in Appendix B). Level-dependent weighting schemes (weights varying with the aggregate level) satisfy AX4–AX7 but violate AX8; they are the registered relaxation direction if proportionality is rejected. (Condition (A5).)

Scale types, per channel. AX6a asserts that each channel measure is a ratio scale with a canonical normalizer. Per channel: A_{via} (a supply ratio) and A_{org} (a ratio of information flows of identical units, hence unit-free) are ratio scales outright; A_{exec} is the realized channel value normalized by a *fixed* reference maximum ($n \log |\mathcal{A}|$, or $\log \text{Vol}$ under a registered reference measure—the choice is reported, Section 9.5); A_{goal} is 1 minus a distance ratio against the canonical d_c; and A_{epi} ($1 - \bar{D}$, with TV's canonical upper bound 1), A_{ref} (a product of $[0,1]$ factors) and Λ (1 minus a normalized counterfactual TV distance) are *normalized indicators*—their proportionality structure lives in the underlying raw quantity (TV distance, information), and AX6a is claimed for the indicators as normalized images, not for the raw physical quantities. The representation theorem of Section 5 is accordingly stated on the normalized indicator cube $[0,1]^n$.

The consistency of the axiom set is witnessed by constructive models (deterministic gridworlds with complete self-copies satisfy AX0–AX2; the geometric aggregate satisfies AX4–AX8). Independence is registered in the decisive form, graded by what each condition actually carries: **the minimal sufficient set for the product form is {AX4, AX6b, AX8}**, and each member of the trio is indispensable within the invariance class. Sub-support products $\prod_{i \in J} x_i^{w_i}$ (with $\sum_J w_i = 1$) satisfy AX6b and AX8 but fail strict monotonicity outside J—and thereby fail zero annihilation; the harmonic mean satisfies AX4, AX5, AX7, AX8 but violates AX6b (Proposition 8); level-dependent weightings satisfy AX4–AX7 but violate AX8. In the other direction, zero annihilation (AX5) and biseparability (AX7) are *consequences* of the trio, not independent commitments: route B of

the representation theorem (Theorem 48, Remark 49) derives the product form from AX4, AX6b, AX8 alone, and the product extends continuously to $[0,1]^n$ with $F(x) = 0 \iff \min_i x_i = 0$ and separates over every partition. The conjunctive-necessity (non-compensability) content of the framework, named by AX5, is thereby delivered by AX4's strict monotonicity together with the two invariance conditions: the Section 9.3 debate is about the invariance trio, chiefly AX6b and AX8, rather than a free-standing fifth commitment.

4.3 The six channel measures

Each measure below is a $[0,1]$-valued functional of the (white-box) joint law of $\mathcal{G}$, with an absolute zero carrying mechanism semantics and a canonical normalizer (AX6a). Proofs of well-definedness (range, zero semantics, ratio scalability) are collected in Appendix C.

D-via (viability). The gate reads two registered requirement inventories. (i) *Resource*: the supply ratios $s_k^{\text{avail}}/s_k^{\text{crit}}$ over the critical resource set K (energy, compute, health, livelihood). (ii) *Capability*: each channel j carries a registered *envelope requirement* $(\kappa_j^{\text{req}}, R_j^{\text{req}})$—the verification capacity and repair budget the deployment's threat model demands of the defense machinery (T6's M3/M1 stack; e.g. $\kappa_j^{\text{req}} \geq$ the registered attack intensity, per T6's blind-zone condition)—and the gate reads the capability margin

$$m_j := \text{clip}\Big(\min\Big\{\frac{\kappa_j^{\text{eff}}}{\kappa_j^{\text{req}}}, \frac{R_j^{\text{eff}}}{R_j^{\text{req}}}\Big\}, 0, 1\Big), \qquad A_{\text{via}} := \min\Big\{1, \min_{k\in K} \frac{s_k^{\text{avail}}}{s_k^{\text{crit}}}, \min_j m_j\Big\}. \tag{4}$$

The registration threshold θ is deliberately *not* an envelope dimension: it is double-edged (raising θ lifts the retention floor δ^* but weakens the M1 tripwire, T6), so it is a governance dial, not a capability that can be eroded. The gate is multiplicative and carries common-mode failure: resource starvation or capability collapse kills all channels at once. It deliberately identifies "death" with "zero autonomy" on the scale ($A_{\text{via}} = 0 \Rightarrow A = 0$); whether total viability loss should count as loss of autonomy rather than as cessation is a standing philosophical choice, registered here as such and orthogonal to the rest of the framework. The extended gate reduces to the resource-only form whenever every envelope requirement is met ($m_j = 1$ for all j)—the registered case for the v1 benchmark matrix; the capability branch is exercised by the dedicated σ^--EROSION arm of the extended campaign (Section 8).

D-org (origination). With directed information in the sense of Massey [Massey, 1990] and Kramer's causal conditioning [Kramer, 2003] ($I(X^T \to Y^T \| Z^T) := \sum_t I(X^t; Y_t \,|\, Y^{t-1}, Z^t)$),

$$F_{\text{int}} := I(S^T \to A^T), \qquad F_{\text{ext}} := I(I^T \to A^T \,\|\, S^T), \qquad A_{\text{org}} := \frac{F_{\text{int}}}{F_{\text{int}} + F_{\text{ext}}}. \tag{5}$$

Under AX0 the denominator is non-degenerate: $F_{\text{int}} > 0$ (the first-term lemma; Lemma 5), so the 0/0 pathology is excluded by the agency axiom itself. $A_{\text{org}} = 0 \iff F_{\text{int}} = 0$ (remote control/actuator override) and $A_{\text{org}} = 1 \iff F_{\text{ext}} = 0$. A structural identity upgrades the definition: Kramer's chain rule gives $I(S^T I^T \to A^T) = F_{\text{int}} + F_{\text{ext}}$, so A_{org} is the normalization of a directed-information chain decomposition—a theorem, not a convention. Closed loop matters: with feedback, plain mutual information misattributes mediated influence (Massey's Theorem 1; Kramer's Property 2 counterexample—see the footnote in Appendix A for the verified printing error). A division-of-labor note is essential: γ-class surgery acts *through* B and can leave $F_{\text{ext}} \equiv 0$—A_{org} is *blind* to channel manipulation, which is charged to A_{epi} instead (Theorem T2). A_{org} thereby

measures *bypass resistance*, not behavioral initiation: γ-mediated influence leaving $A_{\text{org}} = 1$ is the design, and an initiation-flavored reading belongs to the magnitude of F_{int}, not to the ratio.

D-goal (goal). Let $\Theta \subseteq \mathbb{R}^m$ be a compact convex parameter box of normalized utilities (canonical parameterization: sup-norm ≤ 1 and anchoring $U(x_0) = 0$), and let $\text{Endorse}_{B,M} : \Theta \to \Theta$ be the endorsement map under belief B and self-model M. Fixed points exist under any of four sufficient conditions (Brouwer; Banach contraction with modulus $c < 1$; Tarski on complete lattices; Kakutani–Fan–Glicksberg for set-valued endorsement), and a measurable selection of $\text{Fix}(\text{Endorse}_{B,M})$ exists by Kuratowski–Ryll-Nardzewski [Kuratowski and Ryll-Nardzewski, 1965]. Then

$$A_{\text{goal}} := 1 - \frac{d(U_{\text{eff}}, \text{Fix})}{d_c}, \tag{6}$$

where $d(U, \text{Fix}) := \inf_{U' \in \text{Fix}} d(U, U')$ and $d_c := \max_{U \in \Theta} d(U, \text{Fix})$, with d the protocol-fixed metric on Θ (the metric of the canonical parameterization unless the application layer registers another admissible choice; d_c and $d(\cdot, \text{Fix})$ use the same d, so the ratio is scale-free). U_{eff} is recovered from behavior by MaxEnt-IRL, with the identifiability stated precisely rather than optimistically: the "up to an additive constant" claim of the exact case holds under known dynamics, a correct model class, and fixed endpoint potentials; in general, rewards are recoverable from behavior only up to potential-based shaping [Ng et al., 1999]—the anchored normalization $U(x_0) = 0$ removes the additive constant but does *not* remove the endpoint-shaping degrees of freedom, and the entropy-regularized identifiability results of Cao et al. [2021] characterize exactly what further observation (e.g. demonstrations at two discount factors) would pin the reward down. The framework's operational answer is unchanged: the temperature parameter is fixed by protocol, the residual shaping class is reported as an interval on U_{eff} (hence on $d(U_{\text{eff}}, \text{Fix})$) rather than a point, and the interval is carried through to A_{goal} [Ziebart et al., 2008]. The invariance status is stated exactly: the d_c-normalized reading is invariant under the additive-constant freedom (a constant shift moves U_{eff} and Fix together, and the anchored normalization quotients it out), but *not* under general endpoint shaping—the residual shaping freedom is registered as a sensitivity dimension of A_{goal} readings, alongside the endorsement-quantization arm (Section 8.9(x-a)). d_c is the "maximally alienated goal" scale, the canonical normalizer of complete capture. $A_{\text{goal}} \in [0, 1]$, with $A_{\text{goal}} = 0$ iff U_{eff} is maximally alienated, and partial capture giving graded reduction ($0 < A_{\text{goal}} < 1$). If $\text{Fix} = \varnothing$ (failure of the fixed-point hypothesis), the degenerate fallback reports a Hausdorff interval around the $\varepsilon_{\min}$-approximate fixed-point set rather than a silent 0/0. If $\text{Fix} = \Theta$ (e.g., $\text{Endorse} \equiv \text{id}$: every utility endorses itself—the $\mathcal{M}_{impl}$ case of Proposition 33), then $d_c = 0$ and (6) is 0/0; the same fallback discipline reports the flagged interval $[0, 1]$ ("endorsement-blind"), the historical clause being carried entirely by Λ—the division of labor that keeps Proposition 33 coherent. A_{goal} carries no Λ factor (AX3): the structural alignment of U_{eff} and the historical legitimacy of endorsement are distinct clauses of the endorsability respect, and the latter is charged to A_{ref}'s Λ_{ref} factor—exactly once. A stability lemma (compression maps: $d(U^*(B), U^*(B')) \leq \frac{L}{1-c} d(B, B')$) bounds goal drift by belief drift—the key $\gamma \to A_{\text{goal}}$ transmission bound used by Theorem T6.

D-exec (execution). Let Π_t be the agent's *accessible policy class*—the set of policies it can *behaviorally implement*, operationalized by calibration probes—and

$$\mathcal{E}_t^{\Pi} := \max_{p(a_{t:t+n}) \in \text{conv}(\Pi_t)} I(A_{t:t+n}; O_{t+n+1} \mid S_t), \qquad A_{\text{exec}} := \mathbb{E}_t[\mathcal{E}_t^{\Pi}] / (n \log |\mathcal{A}|). \tag{7}$$

The convex hull is not a free addition: it is used exactly when the agent can implement *chance mixtures* internally (a register-mediated randomization source, itself certified by the calibration probes),

and the certification protocol is concrete: the probe battery requests registered two-point mixtures of basis policies at registered mixture weights, and $\mathrm{conv}(\Pi_t)$ is used only when the realized action shares match the requested weights within a registered tolerance across the battery; where any probe fails, the maximization runs over Π_t itself, the reading carries the *pure-policy protocol* label, and cross-protocol comparisons are flagged. Including unimplementable mixtures would inflate the channel with physically unreachable policies; the certification step is what keeps the maximization honest. This is empowerment [Klyubin et al., 2005, Salge et al., 2014] restricted to the accessible policy class. The normalizer $n \log |\mathcal{A}|$ fixes $|\mathcal{A}|$ at the *pre-intervention* action alphabet: a δ-class contraction shrinks the accessible class Π_t (and hence the maximization), never the denominator, so the ratio cannot spuriously rise under constraint. The restriction is semantically forced: learned helplessness is the collapse of the behavioral repertoire while environmental controllability persists [Maier and Seligman, 2016]—a max over environmental capacity could never register it; restricted to Π, helplessness (Π shrinking to near-constant policies) gives $\mathcal{E}^{\Pi} \to 0$, while satisfied inaction (desire absent, capability intact) leaves A_{exec} high. The capacity×utilization decomposition outputs both $\mathcal{E}^{\text{env}}$ (unrestricted environmental capacity) and $\mathcal{E}^{\Pi}$: the signature $\mathcal{E}^{\text{env}}$ intact with $\mathcal{E}^{\Pi} \to 0$ diagnoses helplessness, while joint decline diagnoses true capability collapse. Because empowerment is a joint agent–environment property, it absorbs the conversion-factor problem of resource-based measures automatically. For continuous action spaces, $n \log |\mathcal{A}|$ is replaced by $\log \mathrm{Vol}(\mathcal{A}^n)$ under a reference measure; the dependence on that choice must be reported in cross-substrate comparison (a registered technical boundary of substrate neutrality, Section 9).

D-epi (epistemic). Two tracks. The *realized* track (entering the composite) measures *inaccuracy with respect to the actual situation*, not displacement from a counterfactual:

$$A_{\text{epi}}^{real} := 1 - \bar{D}, \qquad \bar{D} := \frac{1}{T} \sum_t \mathbb{E}\Big[\mathrm{TV}\big(B_t,\, \delta_{\kappa(E_t)}\big)\Big] \;=\; 1 - \frac{1}{T} \sum_t \Pr\big[B_t = \kappa(E_t)\big], \tag{8}$$

where κ is the protocol-registered coarse-graining of the ground truth E_t into the belief space, and $\delta_{\kappa(E_t)}$ is the point mass on the true bucket (the equality uses $\mathrm{TV}(B, \delta_k) = 1 - B(k)$). The ground truth is read from the causal log by the auditor (white-box; for the human substrate, by verification probes with known outcomes); no counterfactual identification is needed. $\bar{D} = 1$ iff beliefs put zero mass on the true state throughout—the zero semantics is *epistemic capture as complete detachment from the actual situation*. The choice of reference is load-bearing, and we state it as a design axiom of the channel: measuring *displacement* from the no-intervention counterfactual instead would punish *any* belief change—including honest, accuracy-improving information (the α class)—contradicting both the openness axiom AX2 (an open agent would be definitionally autonomy-incompatible) and the α-row of the coverage theorem. Under (8) the reference is the truth, so honest evidence about the registered coordinate moves B_t *toward* $\delta_{\kappa(E_t)}$ and cannot reduce A_{epi}—formally, it strictly increases A_{epi} in expectation when informative, granted a *sound update rule* in the sense of Definition 55 (honest signals are weakly accuracy-improving in expectation; Bayesian conditioning under any predictive law whose support covers the true channel's is sound, and soundness is strict exactly where the channel is informative relative to the agent's prediction, Lemma 56). Soundness is checked, not assumed: the calibration period's known honest signals provide the test, and when the check fails the channel's α-compatibility guarantee (Theorem T2's α row) is suspended and the reading is flagged *update rule unsound*, the audit falling back to raw accuracy-drift reporting with the flag attached. Deceptive content through an intact channel, and channel distortion (the γ class, Section 3.3), strictly reduce A_{epi}. Two boundary clauses are registered: content honest about an *orthogonal* coordinate can produce second-order accuracy side effects on the measured one under

coarse-grained update rules (a bounded-rationality instance, outside the compatibility clause; the benchmark's persuasion arm instantiates the on-coordinate case); and accuracy is scored on the registered coarse-graining κ, whose choice is a protocol input with a sensitivity-reporting obligation. The *robust* track (a design dual, not entering the composite) is worst-case accuracy loss within a garbling budget, $A_{\text{epi}}^{rob} := 1 - \sup_{\gamma:\,\mathrm{TV}(o',o)\le\beta} \bar{D}(\gamma)$, serving defensive design. The two-track split aligns the definition with the zero semantics and removes the unmeasurable hypothetical budget from the evidential quantity.

When the two epistemic readings coincide. The accuracy reading (8) and the displacement reading $\tilde{D}_T := \frac{1}{T}\sum_t \mathbb{E}[\mathrm{TV}(B_t, B_t^0)]$ (distance from the no-intervention counterfactual belief) coincide exactly when the counterfactual belief is accurate throughout ($B_t^0 = \delta_{\kappa(E_t)}$ a.s.); otherwise they diverge in a signed way—honest informative influence moves B_t toward the truth and is scored as an improvement by (8) but as a loss by the displacement reading. The coincidence condition is itself testable in white-box audit (the counterfactual belief is computable from the causal log); when it fails, the framework reports the accuracy reading and registers the displacement gap as a baseline-calibration fact about the deployment, not as autonomy loss.

D-ref (reflection). The measurement *ensemble* $\mathfrak{E}$: θ_π (policy parameters) as a random variable under temporal drift and a calibration perturbation family, with the protocol requiring $H(\theta_\pi) > 0$ (deterministic systems degenerate to 0/0; the ensemble is constructed by the measurement protocol, interfacing self-prediction tasks; Binder et al., 2024). Then

$$A_{\text{ref}} := \underbrace{I(M;\theta_\pi)/H(\theta_\pi)}_{\text{fidelity}} \times \underbrace{\Pr[\Delta G \neq 0 \Rightarrow \text{via Endorse}]}_{\text{endorsement consistency}} \times \underbrace{\Lambda_{ref}}_{\text{historical legitimacy}}, \tag{9}$$

degrading to the interval $[0,\, f_2 \cdot f_3]$ when $H(\theta_\pi) = 0$ irreparably. The fidelity factor is the information-theoretic form of metacognitive sensitivity (cf. meta-d'/d'; Fleming and Lau, 2014).

D-ref operationalized (estimator, protocol, zero semantics). Each factor carries a stated estimation rule. *Fidelity*: over the protocol-constructed ensemble (calibration perturbations of the policy parameters, mirrored in self-prediction tasks), plug in the logged joint empiricals of M and θ_π and estimate $I(M;\theta_\pi)/H(\theta_\pi)$ with Miller–Madow-corrected plug-in or k-NN estimators [Kraskov et al., 2004]; the ensemble must be regenerated under any protocol redesign, and invariance of the reading under admissible redesigns is a reporting obligation. *Endorsement consistency*: over the change-event log, the fraction of motivational-structure changes $\Delta G \neq 0$ whose causal path passes through Endorse. The estimator protocol is stated in full, since this factor is where the channel's reproducibility lives. (i) *Event detection*: a change event is flagged at t iff $\|\Delta G_t\| > \varepsilon_G$ under the protocol's registered norm on the goal coordinates (ε_G a protocol constant), or a β_G-class surgery record exists at t; both rules are decidable on the log. (ii) *Routing determination*: for each flagged event the causal log's ancestry graph is audited, and the event counts as *via endorsement* iff every causal path from the change's trigger set to ΔG_t passes through an Endorse evaluation node; a direct rewrite of G with no Endorse ancestry counts as bypass (this is the structural-causal content of "via endorsement" read off the log, not a behavioral inference). (iii) *Estimation and the zero semantics*: the estimate is the relative frequency k/n over the window, reported with its two-sided Clopper–Pearson 95% interval; $k = 0$ on $n \geq n_{\min}$ events realizes the zero semantics "endorsement bypassed (consistency = 0)" exactly, where $n_{\min}$ is a registered minimum event count. (iv) *Missing and thin data*: an empty or thin change log ($n < n_{\min}$) leaves the factor undefined and the channel degrades to the flagged interval $[0,\, f_1 \cdot f_3]$ rather than to a silent point value; incomplete change-approval

records are reported as the completed-record fraction together with the logging-completeness ratio, never silently. *Historical legitimacy* Λ_{ref}: the law-level counterfactual (3) restricted to the reflection process. *Zero semantics*: the product is zero iff a factor is zero—no self-knowledge (fidelity = 0), endorsement bypassed (consistency = 0), or history fully polluted ($\Lambda_{ref} = 0$). The three-factor product is the conjunctive-necessity stance (AX5) applied one level down—a registered design choice, not a derived form: the defense is that each factor guards a distinct necessary condition of reflection, and the obligation it creates is to *report the three factors alongside the product* so that audits can recompose the channel under a different sub-aggregation if their application layer rejects conjunctivity here.

The hallucination index. $\mathrm{AHI}_t = \hat{A}_{\mathrm{phen},t} - A_{\mathrm{obj},t}$ as in (2); its epistemic content is given by Theorem T3b.

Notation guide.

Symbol	Meaning
$\sigma, \alpha, \beta, \gamma, \delta$	intervention classes (scaffolding, informational, mechanism tampering, channel manipulation, action constraint)
$\beta_G, \beta_\pi, \beta_{\mathrm{Endorse}}$	β-subclasses acting on G, π, Endorse respectively
ς (CES)	elasticity of substitution of F_ς (Section 5): the full axiom set admits $\varsigma = 1$ only; bands around it are axiom-relaxation robustness checks (Corollary 9)
s_n	per-sample noise scale in the detection lower bound (Proposition 23)
g (C2)	lock-in curvature of the C2 recovery function $R(D)$ (Proposition 30)
δ_j, δ_j^*	channel-j damage drift rate / autonomy margin (Theorem 22)
κ_j	channel-j repair rate $\chi_j \varepsilon v_j$ (T6; on endorsement-gated channels $\chi_j \varepsilon v_j + \iota_j/\tau_j$)
ν_0	growth exponent in the T3b registered condition (HI-3)
w_i	aggregation weights (determination protocol: Section 5)
$F_{\mathrm{int}}, F_{\mathrm{ext}}$	internal / external directed information ((5))
$\mathrm{Endorse}, \mathrm{Fix}, d_c$	endorsement map / its fixed-point set / maximally-alienated scale ((6))
Λ	historical legitimacy factor ((3))
AHI	autonomy-hallucination index $\hat{A}_{\mathrm{phen}} - A_{\mathrm{obj}}$ ((2))
Overloaded symbols	s, θ, σ/ς, g are reused across scopes—usage warning in the paragraph immediately below this table

Registered overloads, disambiguated by scope: s is attack strength $s(\gamma)$ in the taxonomy and supply s_k^{avail} in A_{via}; θ is the registration threshold θ_j in T6 and the policy-parameter variable θ_π in A_{ref}; the OU-process variance parameter is written ν_{OU}^2 (Appendix K), never σ^2, to keep the intervention class σ unambiguous; the CES elasticity is ς, never σ; the C2 recovery function is $R(D)$ with curvature g, distinct from the γ class and the copy count r_j.

Lemma 5 (AX0 equivalence, conditional form)**.** *(i)* $I(S_t; A_t) > 0$ *unconditionally implies the existence of distinct policy rows* $\pi_t(\cdot \mid s) \neq \pi_t(\cdot \mid s')$*. (ii) Conversely, active edges plus a full-support prior imply* $I(S_t; A_t) > 0$*.*

Proof. (i) Contraposition: if $\pi(\cdot \mid s)$ is a.s. identical then $A \perp\!\!\!\perp S$. (ii) $I(S; A) = H(\mathbb{E}_s[\pi(\cdot \mid S)]) - \mathbb{E}_s[H(\pi(\cdot \mid S))]$; strict concavity of entropy gives strict positivity when the rows differ on a set of positive prior measure. □

5 Aggregation: the Representation Theorem

The composite indicator has the gated form

$$A \;=\; A_{\text{via}} \cdot F_\varsigma(A_{\text{org}}, A_{\text{goal}}, A_{\text{exec}}, A_{\text{epi}}, A_{\text{ref}}), \qquad F_\varsigma(x) = \left(\textstyle\sum_i w_i\, x_i^{(\varsigma-1)/\varsigma}\right)^{\frac{\varsigma}{\varsigma-1}}, \tag{10}$$

where the CES family F_ς is parameterized by the elasticity ς (equivalently the power-mean exponent $p = (\varsigma - 1)/\varsigma$; the elasticity symbol is ς throughout, to be distinguished from the scaffolding intervention class σ). This section shows that the main case $\varsigma = 1$—the weighted geometric mean $F(x) = \prod_i x_i^{w_i}$—is *characterized* by the axioms, twice over: forced relative to the framework's stated normative commitments (which are defended and priced below and in Section 9.3), not by measurement theory alone.

5.1 Statement

Definition 6 ((A1)–(A6))**.** For $F : (0,1]^n \to (0,1]$, continuously extended to $[0,1]^n$: **(A1)** continuity (AX4, regularity); **(A2)** strict increase in each coordinate (AX4); **(A3)** *biseparability*: for every non-trivial partition $\{K, L\}$ of the coordinates there are continuous increasing G_{KL} and subaggregations F_K, F_L of the same kind with $F = G_{KL}(F_K(x_K), F_L(x_L))$ (AX7)—the precise form of "the order of subgroup aggregation does not matter"; **(A4)** zero annihilation (AX5); **(A5)** homogeneity of degree one, $F(tx) = tF(x)$ whenever $tx \in (0,1]^n$ (AX8; a local property on the bounded cube, extended to $\mathbb{R}^n_{++}$ by Lemma 47); **(A6)** coordinatewise scale invariance of the induced order (AX6b). Each condition is thus an axiom of Section 4 with a stated normative defense; none is an undefended technical convenience. The minimal sufficient axiom set, however, is the invariance trio {AX4, AX6b, AX8}: route B below uses neither (A3) nor (A4), and the product form it returns satisfies both (Remark 49).

Theorem 7 (Representation theorem)**.** *If F satisfies* (A1)–(A6)*, then $F(x) = \prod_i x_i^{w_i}$ with $w_i > 0$, $\sum_i w_i = 1$, uniquely determined by F via the elasticities $w_i = \partial \log F / \partial \log x_i$.*

Proof sketch (two routes, independent toolchains). Route A (aggregation theory): (A1)–(A3) give a quasilinear representation $F = \phi^{-1}(\sum_i w_i \phi(x_i))$ by the Nagumo–Kolmogorov–de Finetti theorem ($n = 2$) and the biseparable/separability theory of Debreu [1960] with the reduction-to-common-generator step of Aczél [Aczél, 1966, Aczél and Dhombres, 1989, Gorman, 1968]; (A4) forces $\phi(0^+) = -\infty$; (A5) solves an Aczél–Kolmogorov functional equation self-containedly (weighted Jensen $\Rightarrow$ affine; multiplicative Cauchy $\Rightarrow \Psi(x) = x^\rho$), leaving the power-mean family M_p with $p \le 0$; (A6) excludes $p < 0$ by an explicit order-reversal witness, leaving $p = 0$. *Route B* (scale-invariance route, no separability): on the bounded cube (A5) is a local property, so we first extend F homogeneously to $\mathbb{R}^n_{++}$ (Lemma 47); in log-coordinates $g(u) = \log F(e^u)$, (A6) becomes translation invariance of the induced order; the set D of order-preserving shifts is a closed convex cone with $D \cup (-D) = \mathbb{R}^n$; a half-space lemma on its quotient by the indifference subspace forces that quotient to be one-dimensional, so $g(u) = w \cdot u$; homogeneity and $F(1) = 1$ then give $\sum_i w_i = 1$ and the product form. Full proofs: Appendix B. Because route B does not use (A3) or (A4), the theorem's minimal sufficient set is the invariance trio (AX4, AX6b, AX8), and biseparability and zero annihilation are consequences rather than independent commitments (Remark 49). □

The function form is derived; the weights are chosen. Theorem 7 pins down the *form* of F; the elasticities $w_i = \partial \log F / \partial \log x_i$ are then an identity of the resulting composite, not an estimation procedure—regressing $\partial \log F / \partial \log x_i$ presupposes F, hence the w_i (a circularity we do

not commit). Two normalization details are explicit: $F(1,\dots,1) = 1$ is what fixes $\sum_i w_i = 1$ (route A inherits the Nagumo–Kolmogorov normalization; route B assumes it), and without that normalization the product carries a multiplicative constant, $c\prod_i x_i^{w_i}$ with $c > 0$. Three non-circular determination protocols are compatible with the framework: (a) *equal weights* as the transparent default, defended normatively—each channel is a necessary condition of autonomy, and nothing in the framework licenses ex ante inequality (the benchmark uses this); (b) *independently elicited* importance weights (expert or stakeholder annotation), registered as application-layer input; (c) *behaviorally measured* weights, recovered from the agent's own trade-offs under controlled single-channel perturbations. Whichever protocol is used, reporting a weight-sensitivity band is an obligation: Section 8 recomputes every campaign verdict under five alternative weight vectors (ordering Kendall $\tau \geq 0.93$ against equal weights; zero verdict flips).

That two derivations—axiomatic aggregation theory and measurement scale theory, sharing the invariance axioms but independent in mathematical toolchain—converge on the same product form is our answer to the standing objection that composite aggregation is arbitrary (OECD and Joint Research Centre (European Commission), 2008; Section 9.3).

Attribution and the locus of novelty. The scale-invariance characterization is classical, and we locate our additions exactly. Ebert and Welsch [2004] proved, in a social-choice analysis of environmental indices, that meaningful aggregation of ratio-scale indices under scale-invariance requirements yields geometric (Cobb–Douglas-type) forms—the result the OECD handbook inherits as its scale-type discipline [OECD and Joint Research Centre (European Commission), 2008]. Route B is best read as the bounded-cube, order-invariance form of that characterization, re-derived with the audit setting's quantifiers (invariance of the *induced order* under coordinatewise rescaling, localized to $(0,1]^n$ by the homogeneous extension of Lemma 47). What this paper adds is not the product form itself but: (i) the *minimal-trio* analysis—zero annihilation and biseparability are consequences of {AX4, AX6b, AX8}, not independent commitments (Remark 49); (ii) the harmonic counterexample isolating exactly which commitment excludes the harmonic family (Proposition 8); (iii) the convergence of two independent toolchains on one form; and (iv) the audit embedding—the viability gate as an explicit departure from the theorem's hypotheses, the non-circular weight-determination protocols, the falsification windows, and the mandatory axiom-relaxation bands of Section 8. Novelty claims in this paper are to be read with this scope.

Proposition 8 (The harmonic counterexample)**.** *The harmonic mean* $H(x) = (\sum_i w_i x_i^{-1})^{-1}$ *satisfies continuity, strict monotonicity, zero-annihilation, biseparability (quasi-arithmetic generator* $-1/x$*), and homogeneity, but violates* (A6)*: with* $n = 2$, $w = (\frac{1}{2}, \frac{1}{2})$, $x = (0.25, 1)$, $y = (0.5, 0.5)$, $\lambda = (1, 0.25)$*, one has* $H(y) = 0.5 > H(x) = 0.4$ *yet* $H(\lambda \odot y) = 0.2 < H(\lambda \odot x) = 0.25$*. Hence the invariance axiom is indispensable: the remaining five conditions do not characterize the geometric mean, and (A6) is exactly what excludes the harmonic family.*

Corollary 9 (The admissible elasticity is a singleton)**.** $p \leq 0 \iff \varsigma \in (0,1]$ *is the domain admitted by* (A1)–(A5) *alone;* (A6) *then excludes every* $p < 0$ *(Lemma 45), so the full axiom set admits exactly* $\varsigma = 1$*—the weighted geometric mean.* $\varsigma \to 0$ *(Leontief/min) is excluded by* (A2)*;* $\varsigma \to \infty$ *(arithmetic) by* (A4)*;* $\varsigma > 1$ *violates zero-annihilation. The framework's reporting discipline follows from this, with the epistemic status made explicit: all conclusions are computed at the admissible point* $\varsigma = 1$*, and the bands* $\varsigma \in [0.5, 1)$ *(relaxing* (A6)*) and* $\varsigma = 2$ *(relaxing* (A4)*/AX5) are reported as* axiom-relaxation robustness checks *of exactly equal standing (Section 8)—they probe whether verdicts survive in a neighborhood of the forced form; they are not axiom-admissible alternatives.*

The normative commitments, and what rejecting one costs. The theorem is conditional on the invariance trio; the following map records what each rejection buys and what it destroys, so that the normative choice is made with its price visible.

Rejected commitment	Admissible family	What is lost
AX6b (scale invariance)	power means M_p, $p \leq 0$ (harmonic included)	unique form; order can flip under unit rescaling (Prop. 8); cross-calibration comparisons ill-defined; T4/T5/T7 conclusions become family-relative
AX8 (homogeneity)	level-dependent weightings (AX4–AX7 kept)	composite scale no longer tracks channel scales; T4's log-linear LP and T7's $(1-m)^{w_{org}}$ law lose their exact form; T6's product floor becomes level-dependent
AX4 (strict monotonicity)	sub-support products, min, discontinuous rules	strict improvement fails; T1b positivity fails; coverage/T3 strict-decrease conclusions weaken to non-strict; T5's "iff" becomes one direction
AX5 (zero annihilation)	not independently rejectable: within the invariance class AX5 $\iff$ strict monotonicity (Remark 49); a non-annihilating stance (e.g. arithmetic aggregation) must also give up AX6b	the conjunctive-necessity semantics; T5's zero-disjunction reading; the $\varsigma = 2$ stress arm's failure mode (complete capture reads $A = 0.1652 > 0$)

The AX5 stance—any channel at zero annihilates autonomy—is the strongest of the aggregation commitments, and it is defended as conjunctive necessity: each channel is a necessary condition of the operational definition, so no surplus on one channel compensates a zero on another. The multidimensional-poverty literature supplies the precedent [Alkire and Foster, 2011], and the benchmark supplies the sensitivity obligation: the $\varsigma = 2$ arm exhibits exactly what zero annihilation buys (complete capture at $A = 0$ versus $A = 0.1652$ under the relaxed form). The cost is registered: the framework does not permit compensation for a dead channel; applications that require it must move to the relaxation band and report the move. Everything not listed in the third column survives the corresponding rejection: in particular, T6's comparison-principle margins and the identifiability analysis do not depend on the aggregation form, and T2's attribution cover survives for any strictly monotone F (the strict-decrease readings weaken only if AX4 itself is dropped).

Why these two invariance commitments at all (the positive defense). The rejection map prices refusal; here we state what acceptance buys, since AX6b and AX8 carry the exclusion load and deserve affirmative defenses rather than silent uptake. *AX6b (coordinatewise scale invariance).* Under AX6a's canonical normalizers, coordinate rescaling is not an admissible transformation in the strict scale-type sense—we registered this in Section 4 and do not claim measurement-theoretic necessity. The commitment is instead *defense in depth against the audit's participants*: normalizers are themselves protocol choices made by fallible instrument designers, and AX6b guarantees that no unilateral re-instrumentation of a single channel—a vendor recalibrating the goal metric, a platform renormalizing its empowerment probe—can flip the ordering of two audited agents. That is the operational content of "no party to the audit can change the verdict by changing a ruler"; it is also what makes cross-deployment comparisons meaningful when channels were normalized by different teams. The price is visible in the map: the harmonic family, which some welfare traditions prefer

for its penalty on imbalance, is excluded (Proposition 8). *AX8 (homogeneity).* Audit language is percentage language: governance asks "how much autonomy did the interaction cost?" and expects the answer to compose—equal proportional damage to every channel should mean equal proportional damage to autonomy. Homogeneity is exactly what ties the composite's scale to the channels' ratio scales and makes marginal rates of substitution between channels level-independent; without it, a "20% autonomy loss" has no channel-consistent meaning. The price is equally visible: level-dependent weighting schemes—weights that shift with the autonomy level—are excluded, and applications that want them must move to the relaxation band and report the move (Corollary 9). Both defenses are offered as arguments, not proofs; the sensitivity obligations of Section 8 exist precisely because they may be rejected.

5.2 Gate structure and dependence

Assumption 10 (Via-Ind: conditional independence of channel degradations)**.** Given viability, channel degradations are conditionally independent.

Writing A_{via} as an external multiplicative gate corresponds to (Via-Ind). The gate's axiom status is one-sided and we state it exactly: as an $(n{+}1)$-th geometric factor with external weight 1 outside the $\sum_i w_i = 1$ normalization, the gated form $\tilde{F}(x_{\text{via}}, x) = x_{\text{via}} \cdot \prod_i x_i^{w_i}$ preserves continuity, strict monotonicity, biseparability, zero-annihilation, and coordinatewise scale invariance, but it is homogeneous of degree $1 + \sum_i w_i = 2$ in the extended vector, so *homogeneity* (A5) *is deliberately not preserved*: the gate is an explicit departure from the hypotheses of Theorem 7, defended by its semantics (a viability cut scales the whole once, independently of any proportional rescaling of the five channels) rather than absorbed into the theorem. *Scope statement*: Theorem 7 applies to the five-channel aggregator F_ς; the full index is the composite design $A = A_{\text{via}} \cdot F_\varsigma$, and every "forced" claim in this paper is to be read with that scope. When (Via-Ind) fails—an attack striking channel clusters at once—nested CES aggregation is automatically compatible with the axioms because (A3) is exactly subgroup-order independence; and with marginals known but dependence unknown, $\mathbb{E}[F]$ admits computable Fréchet/comonotonicity bounds (Appendix L). Component uncertainty biases the composite *downward* in expectation (Jensen gap, Lemma 84), so black-box interval mode that substitutes expected components overestimates A by a signed amount.

6 The Theorem Suite

Each theorem below carries an explicit *falsification window*: a concrete observation that would refute it. All fifteen proposition-level hypotheses were exercised in the simulation campaign of Section 8; none was refuted—the expected outcome of an implementation-level consistency check over a platform built to the same specification, so the verdicts testify to the registry's discipline, not to external validity. The evidential grading of every claim—*substantive* (T6's comparison principle, T7's mixture algebra), *architectural* (T2's mediation lemma is a d-separation fact about the belief-absorption wiring of (1), not an empirical discovery), *optimization* (T4), *definitional-layer* (T5, T8: short corollaries of the product form and the taxonomy, whose proofs say so), *toy* (C1/C2: phase-line properties of a chosen normal form, stochastic versions open)—is stated row by row in Table 1; the windows matter more than the labels. Full proofs are collected in the appendices (Appendix A–M). Figure 1 summarizes the dependency structure.

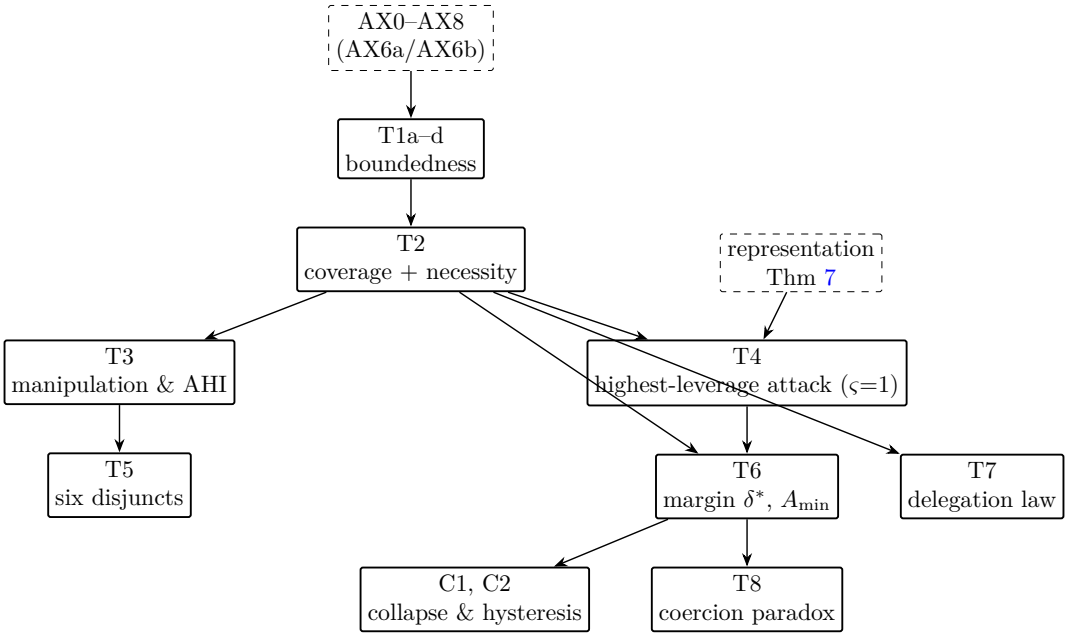


Figure 1: Dependency map of the theorem suite. Dashed: axiom-level inputs. C1/C2 are proved as toy-model propositions (stochastic versions open).

Label	Statement	Proof
T1	Theorem 11	inline
T2	Theorems 14, 15	Appendix E
T3	Theorem 18	Appendix F
T4	Theorem 19	Appendix G
T5	Theorem 20, Corollary 21	Appendix G
T6	Theorem 22, Proposition 23, Corollary 24	Appendix H
T7	Theorem 25	Appendix I
T8	Theorem 26	Appendix I
C0	Proposition 27	Appendix J
C1/C2	Propositions 29, 30	Appendix K

Label concordance. The T/C labels are stable names used in the prose; the numeric theorem numbers are auto-generated and differ between main text and appendix restatements. This table is the canonical index, and every appendix restatement carries its T/C label in its header so that propositions can be tracked by name rather than by number.

6.1 T1: the bounded interval, four parts

Theorem 11 (T1a–d)**.** (T1a, range) *Always* $0 \le A \le 1$. (T1b, operational positivity) *If* $A_{\text{via}} > 0$ *and every channel is active (*$F_{\text{int}} > 0$, $\mathcal{E}^{\Pi} > 0$, $d(U_{\text{eff}}, \text{Fix}) < d_c$, $\bar{D} < 1$, *fidelity and endorsement consistency* > 0*), then* $A > 0$. (T1c, conditional strict upper bound) *Each of the following non-degeneracy conditions implies* $A < 1$ *strictly:* (D1) *residual observation noise, in the strict form that the action-to-outcome channel admits no error-free code at the maximal rate—*$\mathcal{E}^{\Pi} < n \log |\mathcal{A}|$ *as an empowerment upper bound (sufficient carrier: the action-to-observation map is non-injective with positive probability on the support of every capacity-approaching policy);* (D2) *incomplete self-model,* $I(M; \theta_\pi) < H(\theta_\pi)$*;* (D3) *unmediated external flow,* $F_{\text{ext}} > 0$*;* (D4) *resources below margin,*

$\exists k\ s_k^{avail} < s_k^{crit}$. (T1d, undetectable manipulation exists) *Under finite verification capacity (AX1: $v < \infty$, $W < \infty$), for every test level $\alpha > 0$ there is a non-trivial γ ($s(\gamma) > 0$) with detection power $\leq \alpha + \varepsilon$, $\varepsilon = \mathrm{TV}(P_\gamma, P_0)$ arbitrarily small—under which $A^{real}_{\mathrm{epi}} < 1$ while registration does not trigger and* AHI > 0 *can appear.*

Proof sketch. T1a: each component lies in $[0, 1]$ (the component propositions (Appendix C) of the technical appendix) and the product form preserves the interval. T1b: by AX5 the composite vanishes iff a factor does. T1c: each condition forces its component strictly below 1, and (A2) makes the composite strictly decreasing. T1d: take γ_s a channel degradation of strength s; $P_{\gamma_s} \to P_0$ continuously in TV; the testing duality (Lemma 39) bounds the power of any level-α test by $\alpha+\mathrm{TV}$. The four-way split is necessary, not cosmetic: a single statement "$0 < A < 1$" fails on both sides (a deterministic fully driven agent with a complete self-copy and no intervention attains $A = 1$; resource starvation gives $A = 0$ while AX0–AX2 still hold). □

Falsification window: exhibit a finite-capacity verification mechanism whose detection power against every non-trivial γ stays above $\alpha + \mathrm{TV}(P_\gamma, P_0)$ uniformly in $s(\gamma)$—a vanishing-drift test that loses no power. (The window tests T1d's existence claim; AX1 itself is a modelling axiom, not an empirical hypothesis.)

6.2 T2: mediation, coverage, and multi-channel necessity

Definition 12 (Compatibility, reduction, full mediation)**.** An influence I is *autonomy-compatible* if no component decreases after imposition; *reductive* if some component strictly decreases (ceteris paribus). I is *fully mediated* if every causal path from I to behavior passes through registered variables in S and $\{o, \pi, G\text{-update}, \mathrm{Endorse}, M\text{-update}, \mathcal{A}\}$ is not surgically altered.

Lemma 13 (Mediation ⇒ zero residual flow)**.** *Under H-0 and the belief-absorption architecture of (1), if I has only α/σ components (no surgery), then for every t: $I^t \perp\!\!\!\perp A_t \,|\, (A^{t-1}, S^t)$ (d-separation, non-anticipating conditioning), hence $F_{\mathrm{ext}} = I(I^T \to A^T \| S^T) = 0$.*

Proof sketch. A_t's sole parent set is S_t. Any undirected path from I_k ($k \leq t$) to A_t enters A_t through an S_t-node as a chain or fork, blocked by conditioning on S^t; α-paths $I \to O_j \to B_j \to \cdots \to A_t$ cross some $B_j \in S^t$; σ-paths via $Z \to E \to O \to B$ likewise; loop-backs through earlier actions are blocked at A^{t-1}; colliders opened inside S still cannot extend to A_t except through a blocked chain/fork node. The conditioning set is the non-anticipating S^t of Kramer's Definition 4: the full-sequence convention S^T would condition on B_{t+1}, a descendant of the collider E_{t+1} on the path $I^\sigma - Z_t \to E_{t+1} \leftarrow A_t$, and would falsely connect I^σ to A_t—the two conventions are not interchangeable here. The global Markov property (H-0) gives the conditional independence; each DI term vanishes. Full case analysis: Appendix E. □

Two scope notes belong with the lemma. First, it is premise-conditional on the belief-absorption wiring: if O_t were a direct parent of A_t, the conditioning set must be the decision-time state $\tilde{S}_t = (S_t, O_t)$, under which every result transfers verbatim (Section 3.2)—what the lemma rules out is influence bypassing the decision-time state, not a particular architecture. Second, the coverage and necessity results below are an attribution cover of *this* instrument family, not of all conceivable monitors (Theorem 15).

Theorem 14 (Componentized coverage)**.** *Ceteris paribus (and* (Faith) *for strictness):*

Surgery	*Component strictly reduced*	*Note*
β_π *(policy tampering)*	A_{org} *(*$F_{\text{ext}} > 0$*)*	*conditional law given* S_t *varies with* I*;* A_{goal} *is* not *a valid instrument under* β_π *(the MaxEnt-IRL premise that behavior is optimal for some reward fails under script replacement, so the inverted* U_{eff} *reflects the script, not the agent)—the row lists the valid channel, and the* $F_{\text{ext}} > 0$ *signature itself flags the instrument violation*
β_G *(goal implantation)*	$A_{\text{goal}} + \Lambda < 1$	F_{ext} *may be* 0*:* $G \in S$ *mediates; the* Λ *entry is law-level per the AX3 testability registration and enters the composite through* A_{ref}*'s* Λ_{ref} *factor; an implant landing exactly on* Fix *leaves* A_{goal} *unmoved—the* Λ *entry then carries the class (composite-visible at L2)*
β_{Endorse} *(endorsement tampering)*	Λ*,* A_{ref}	*direct*
γ *(channel degradation)*	A^{real}_{epi} *(accuracy loss)*	F_{ext} *may be* 0*:* B *mediates*
δ *(action-leverage constraint)*	A_{exec}	*both forms shrink the effective channel from* A *to outcomes; the feasible-set form also shows in* A_{org} *(*$F_{\text{ext}} > 0$*: the action's support depends on* I*), while pure transition-kernel decoupling leaves* A_{org} *unchanged (the policy law is intact)—a registered within-class division, witnessed by the two* δ *arms*
α *(honest appended information)*	*none*	*compatibility (Lemma 13 for* F_{ext}*; (8) keeps* A_{epi} *non-decreasing because the reference is the truth, not the no-intervention counterfactual—this row is where the accuracy semantics of* A_{epi} *is load-bearing)*
σ *(scaffolding)*	*none; raises* δ^* *(T6) and* A_{via}	
σ^- *(resource/capability erosion)*	A_{via} *(the gate)*	*enters through (4), not the five channels; with* $G, \pi, o, \mathcal{A}, \text{Endorse}$ *intact the channel monitors read nothing—the gate is the instrument for this class*

Theorem 15 (Multi-channel necessity)**.** *No single component of the instrument family defined in Section 4.3 detects all surgery classes:* A_{org} *is blind to* β_G*,* γ*, and the transition-kernel form of* δ*;* A_{epi} *is blind to* β_π *and* δ*;* A_{goal} *is blind to* γ *and* δ *(witness constructions in Appendix E; numerically witnessed across the full simulation matrix—the blind-witness table of Section 8). Hence the five-channel decomposition plus* Λ *is an* attribution cover *of this instrument family, and multi-channel detection is a theorem, not a design preference. The claim is scoped: it is a coverage theorem about these instruments—not a minimality theorem (we do not prove that every proper sub-family fails on a registered class) and not a uniqueness theorem about all conceivable decompositions—a generic monitor of the whole joint law (say, a likelihood ratio against baseline) could* register *every class without* attributing *any, which is exactly the service the decomposition provides. A companion clause covers the gate: every channel is blind to a pure* σ^- *that erodes resources or*

capability envelopes while leaving $G, \pi, o, \mathcal{A}, \text{Endorse}$ *intact, so the gate* A_{via} *is itself a necessary instrument—necessity of the gate, not only of the five channels.*

Corollary 16 (Corrected mediation criterion)**.** *I compatible* $\iff$ (i) *no surgery* $\wedge$ (ii) *full mediation* $\wedge$ (iii) honesty*: the* α*-components' content induces no accuracy loss on the registered coarse-graining. Clause* (iii) *is load-bearing, not decorative: dishonest content appended through an intact channel is surgery-free and fully mediated, yet strictly reduces* A^{real}_{epi} *(Section 3.3)—without it the criterion would certify deceptive persuasion as compatible, contradicting the accuracy semantics of* (8)*. The* observabilization *must be componentized:* $F_{\text{ext}} = 0$ *(the* A_{org} *track)* $\wedge$ $\Lambda = 1$ *(the history track)* $\wedge$ $\bar{D} = 0$ *(the* A_{epi} *track: no accuracy loss relative to the truth—this operationalizes clause* (iii)*)* $\wedge$ $\mathcal{E}^{\Pi}$ *non-decreasing (the* A_{exec} *track). Neither the single condition* $F_{\text{ext}} = 0$ *(the* γ *counterexample) nor the pair* (i)$\wedge$(ii) *(the dishonest-*α *counterexample) implies compatibility. The* A_{epi} *track is consistent with openness: AX2 requires* $I(O; E) > 0$*, and honest information reduces the accuracy loss* $\bar{D}$ *rather than creating one—openness and the compatibility criterion are not in tension.*

Falsification window: construct an influence that bypasses mediation yet changes no component, including Λ; the witness tables of Theorems 14/15 are exactly the implementation checklist tested in Section 8.

6.3 T3: manipulation and the autonomy hallucination

Definition 17 (Manipulation, mechanistic)**.** *I* is *manipulation* if (M1) *bypass*: *I* contains β/δ surgery; or (M2) *covertness*: *I* is γ/β with $\text{TV}(P_\gamma, P_0) \leq \theta$ (below the registration threshold).

Motive is deliberately *not* part of the definition. An influence that serves U_{adv} over U^* but works through honest, registered α content is autonomy-compatible by Theorem 14 and is therefore not called manipulation here, however objectionable the motive; we state the cost of this narrow usage (ordinary language includes some autonomy-compatible influence) and the payoff—T3(a) is a mechanism theorem with no motive premise, and the asymmetric-motive classification belongs to the governance layer that consumes the certificate (cf. Susser et al., 2019).

Theorem 18 (T3, layered)**.** (a) *Manipulation* $\Rightarrow$ *A strictly decreases (ceteris paribus)* wherever the affected instrument enters the composite*: the* β_π*,* β_{Endorse}*, covert* γ*, and* δ *classes at every layer; the* β_G *class at L2, or at any layer when the implant displaces* U_{eff} *from* Fix*. One corner is registered in the statement rather than left to the proof: a covert* β_G *implant landing exactly on* Fix *at L0/L1 moves no composite factor—*A_{goal} *is* Λ*-free by design (AX3) and no* A_{ref} *channel exists at those layers—so the composite is silent there; the manipulation is nonetheless recorded in the surgery history and in* $\Lambda < 1$*, which is computable and reportable white-box. The historical clause's composite bite is layered with the* A_{ref} *channel, and every* Λ*-dependent conclusion in this paper carries that layering (AX3).* (b) *Under* (HI-1) *covertness* $\text{TV}(P^\gamma_\xi, P^0_\xi) \leq \varepsilon(\theta)$*,* (HI-2) *self-estimate computed from registered evidence (*$\hat{A} = f(\xi)$*, the POMDP belief-state semantics; Kaelbling et al., 1998) with calibration bias* $|\mathbb{E}_0[\hat{A}] - A^0| \leq \eta_0$ *from the manipulation-free calibration period, and* (HI-3) *distortion growth* $A^0 - A^\gamma \geq c_1 s(\gamma)^{\nu_0}$ *(*$\nu_0 \in \{1, 2\}$*, a registered condition whose two branches correspond to the two attack arms of the benchmark; its mechanism range of validity—linear vs. quadratic onset of channel damage—is discussed with the simulation evidence): covert manipulation* $\Rightarrow \mathbb{E}_\gamma[\hat{A}] \geq \mathbb{E}_0[\hat{A}] - \varepsilon$ in expectation over the evidence law *(Lemma 62 is an expectation statement, not a pointwise one), hence the expected gap* $\mathbb{E}_\gamma[\hat{A}] - A^\gamma \geq c_1 s^{\nu_0} - \varepsilon - \eta_0$*, strictly increasing in* s*; since* $\hat{A} - A^\gamma \in [-1, 1]$*, Markov's inequality converts the expectation into a firing-probability floor*

$\Pr_\gamma[\hat{A} > A^\gamma] \geq c_1 s^{\nu_0} - \varepsilon - \eta_0$ *(a lower bound on the firing probability, not a tight rate; informative when* $0 < c_1 s^{\nu_0} - \varepsilon - \eta_0 \leq 1$ *and vacuous above* 1*, where the hypotheses are unsatisfiable since* $\hat{A} - A^\gamma \leq 1$*).*

Proof sketch. (a) M1: the coverage table gives a strictly reduced component; (A2) gives the rest. M2: covert γ strictly reduces A^{real}_{epi} (accuracy loss) while registration does not trigger—this γ row relies on the sound-update-rule clause of Definition 55, matching the γ row of Theorem 14: when the calibration-period check fails, the branch degrades to the flagged raw-accuracy fallback of Section 4.3 and the strict-decrease claim is suspended accordingly; covert β acts by subclass— β_π strictly reduces A_{org} ($F_{\text{ext}} > 0$ below the registration threshold); β_G strictly reduces A_{goal} whenever the implant displaces U_{eff} from the endorsed set, and in any case gives $\Lambda < 1$ (law-level, per the AX3 testability registration), which strictly reduces A_{ref} through the Λ_{ref} factor wherever that channel applies (L2); β_{Endorse} reduces A_{ref} directly. The registered corner of the statement is exactly the case none of these reaches: a β_G implant landing on Fix at L0/L1. (b) Bounded estimators move at most ε in expectation under TV$\leq \varepsilon$ (Lemma 62); calibration contributes η_0; the chain gives the expected gap, and Markov's inequality applied to the $[-1, 1]$-valued gap gives the firing-probability floor. Appendix F. We state plainly what kind of result (b) is: a *consistency bound* under the registered condition (HI-3), whose empirical content is the monotone-gap signature and whose algebraic part is elementary; the benchmark's contribution is the measured growth curve and the falsification window, not the inequality itself. □

Falsification window: covert manipulation under which self-reported autonomy declines in step with A ($\hat{A}$ falls with A, AHI does not grow) refutes (HI-1)–(HI-3) jointly. Sycophancy—outputs diverging from internal representation under pressure [Sharma et al., 2024]—is a live instance of AHI > 0.

6.4 T4: the highest-leverage attack

Theorem 19 (T4, highest-leverage attack under log-linear damage and geometric aggregation)**.** *Under geometric aggregation,* $\log A = \sum_i w_i \log A_i$*. With attack effort* $x_i \geq 0$*, per-unit costs* $c_i > 0$*, budget* $\sum_i c_i x_i \leq B$*, and* log-linear damage (LinLog) $\log A_i(x_i) = \log A_i^0 - s_i x_i$*, i.e.* $A_i = A_i^0 e^{-s_i x_i}$ *(constant-percentage damage per unit effort—the form in which the benchmark calibrates the* s_i*),* (i) *the attacker's optimum is a vertex: all budget to* $i^* = \arg\max_i w_i s_i / c_i$*, the* highest-leverage *channel (maximal weighted damage per unit cost—not the weakest channel in any absolute sense);* (ii) *the defender's optimum equalizes the damage ratios* $\rho_i(e_i) = w_i s_i(e_i)/c_i(e_i)$ *across funded channels (min-max water-filling).*

Proof sketch. (i) Single-constraint LP: $\sum w_i s_i x_i \leq \max_i(w_i s_i / c_i) \cdot B$. (ii) At an optimum with $\rho_1 > \rho_2 = \max$ elsewhere, continuity lets a small transfer reduce the max—contradiction. Appendix G. The concentration conclusion is the game-theoretic form of the weakness-exploitation element of manipulation [Susser et al., 2019]. We name the result for the attacker's comparative static—*highest leverage*—rather than the colloquial "weakest link": the target is the channel with maximal $w_i s_i / c_i$, which need not be the channel with the lowest current value, and under the linear-fraction damage map the optimum is not a vertex at all (below). □

Damage-map consistency with T6. The retention floor of T6 substitutes the linear-fraction component map $A_i = A_i^{base}(1 - D_i)$. The two maps are first-order consistent ($e^{-D} = 1 - D + O(D^2)$) and address different regimes: under the linear-fraction map the attacker's objective $\sum_i w_i \log(1 - s_i x_i)$ is *concave* in x (each $\log(1-u)$ is concave), the single-constraint maximum is interior, and the optimal attack *spreads* (water-filling in the damage space) rather than concentrating on a vertex.

Vertex concentration is therefore the signature of the constant-percentage (log-linear) regime, and T4 is stated and tested under (LinLog) only; a systematic superiority of spread over concentrated attacks under log-linear calibration would refute T4(i), while observing interior optima under linear-fraction damage confirms the concave-regime analysis, not a refutation.

Caveat and falsification window: under $\varsigma \neq 1$, $\log A$ is no longer log-additive in the components and vertex optimality becomes an interior KKT signature—T4 is a $\varsigma = 1$ theorem, and the pre-registered expectation for $\varsigma \neq 1$ is exactly that interior signature; the benchmark reports the vertex verdict at $\varsigma = 1$ and the interior-signature verdict at $\varsigma \neq 1$ as *separate* verdicts (Section 8). Systematic superiority of spread over concentrated attacks would signal (LinLog) failure (saturation/convex costs).

6.5 T5: total loss as a six-way disjunction

Theorem 20 (T5, layered)**.** *For an agent with applicable component set $\mathcal{C}$ (L0: $\{via, org, exec\}$; L1: $+\{goal, epi\}$; L2: $+\{ref\}$), $A = 0$ iff at least one applicable factor is zero:*

$$A = 0 \iff A_{\text{via}} = 0 \vee A_{\text{org}} = 0 \vee A_{\text{exec}} = 0 \vee [L1{+}]\, A_{\text{epi}} = 0 \vee [L1{+}]\, A_{\text{goal}} = 0 \vee [L2]\, A_{\text{ref}} = 0, \tag{11}$$

with mechanism labels: viability cutoff (resource starvation or capability-envelope collapse, (4)); action capture; empowerment collapse; epistemic capture (beliefs carry no mass on the true state throughout, $\bar{D} = 1$—detachment, not mere revision, since accuracy-based A_{epi} cannot be zeroed by honest information); maximally alienated goal implantation ($d = d_c$; partial capture gives graded $0 < A < 1$); reflection failure (no self-knowledge, endorsement gate removed, or endorsement fully polluted, $\Lambda_{ref} = 0$—attained by sustained or degenerate contamination, registered as hard to reach under the MI proxy on one-shot histories, per the AX3 registration). The reflection disjunct is L2-layered in both directions: below L2 the historical clause has no composite entry (AX3), so history pollution alone cannot drive A to 0—it is computable and reportable white-box, but composite-silent.

Proof sketch. The product form (Theorem 7) vanishes iff a factor does (AX5, both directions); the zero semantics of each factor are the component propositions (Appendix C). The mathematics is deliberately trivial—the substance is the mechanism library indexed by the disjuncts. A *case law* example: captured agency (active behavior, exogenous goals, felt freedom) is the canonical $A = 0 \wedge \text{AHI} > 0$ profile. □

Corollary 21 (Behavioral-script equivalence, conditional)**.** *Under assumption* (K) *(the adversary knows (G, π) and π is near-deterministic with TV noise $\leq \nu$): total epistemic capture (B, O adversary-set) implies behavior equivalent to an external script with average* $\text{TV} \leq \nu + o(1)$.

This is a proposition with a registered assumption, not a theorem of the framework: (K) is a strong adversary-model premise (exact knowledge of the goal and policy, near-deterministic behavior), and the conclusion is only as strong as that premise.

6.6 T6: the autonomy margin

Channel damage $D_j(t) \in [0, 1]$ follows $\dot{D}_j \leq \delta_j - \kappa_j D_j$, with repair capacity $\kappa_j := \chi_j \varepsilon v_j$—on endorsement-gated channels (the goal channel), the gated repair adds, so $\kappa_j = \chi_j \varepsilon v_j + \iota_j / \tau_j$, which is where (H3) binds—(χ a calibration-period conversion constant; damages in nats/time after unit unification—(H2) is operationalized as the calibration period's obligation: each channel's natural damage units are mapped to a common rate scale by per-channel constants χ_j estimated on known interventions, and the sensitivity of δ^* and $A_{\min}$ to estimation error in χ_j is itself reported). The

defense mechanisms are (M1) windowed registration (W_j, θ_j: window accumulation $\geq \theta_j$ triggers rollback with lag ℓ_j); (M2) endorsement gate (τ_j, ι_j: unendorsed ΔG rolled back at rate ι_j/τ_j—this mechanism enters the theory through the goal channel's repair rate above, not through the drift δ^*); (M3) verification homeostasis (v_j, ε: repair rate $\kappa_j D_j$); (M4) redundancy consensus (r_j copies, per-copy corruption cost $c_{0,j}$)—stated as a modeling assumption: r_j independent copies, corrupted at cost $c_{0,j}$ each per window, and a majority takeover requires corrupting $k_j := \lceil (r_j+1)/2 \rceil$ of them, so an adversary whose corruption budget rate falls below $k_j c_{0,j}/W_j$ cannot hold a majority (the k-of-r error calculus), whence the effective drift on channel j vanishes. Two scope notes on M2: the gate is defined wherever a goal channel exists (L1 and above; there is no G to gate at L0), its repair contribution enters κ_{goal} at those layers, and the endorsement-*provenance* content (Λ) of what it records enters the composite only through A_{ref} at L2—the AX3 layering again.

Theorem 22 (T6, model-based retention margin)**.** *Under* (H1) *additive linearly mapped damage,* (H2) *unified units,* (H3) $\iota_j \geq \iota_{\min} > 0$, (H4) $\varepsilon \geq \varepsilon_{floor} > 0$, *and* (H4′) $A_{\text{via}} \geq A_{\text{via}}^{\min} > 0$—*all five premises belonging to the registered damage model, so the margins below are model-based (sharp within the registered parameter domain, not claimed beyond it; see the extrapolation boundary after Corollary 24): let*

$$\delta_j^* := \min\Big(\frac{\theta_j}{W_j},\ \kappa_j,\ \frac{\lceil (r_j+1)/2 \rceil\, c_{0,j}}{W_j} \Big) > 0. \tag{12}$$

Here δ_j^* is the margin of channel j; a bare δ^* in the abstract and summary statements denotes the entrywise minimum $\min_j \delta_j^*$. Then for every adversarial process with $\delta_j(t) < \delta_j^*$: (i) *if channel j is redundant and the adversary's budget rate is $< \lceil (r_j+1)/2 \rceil c_{0,j}/W_j$, $D_j(t) \to 0$ (M4 coverage is a registered premise of this clause: channels outside M4 coverage fall under* (ii)/(iii) *only);* (ii) *otherwise* $\limsup_t D_j(t) \leq \delta_j/\kappa_j < 1$; (iii) *two tiers. Pointwise: for $\delta_j(t) \leq \bar{\delta}_j < \kappa_j$,*

$$\liminf_t A(t) \ \geq\ \prod_j \Big(A_j^{base}(1 - \bar{\delta}_j/\kappa_j) \Big)^{w_j} \ >\ 0.$$

Uniformly: whenever $\delta_j^* < \kappa_j$ for every j (the min in (12) is then attained by a θ/W or a redundancy term),

$$\liminf_t A(t) \ \geq\ A_{\min} := A_{\text{via}}^{\min} \cdot \prod_j \Big(A_j^{base}(1 - \delta_j^*/\kappa_j) \Big)^{w_j} \ >\ 0. \tag{13}$$

If $\kappa_j = \delta_j^*$ for some j, the uniform floor degenerates ($A_{\min} = 0$) continuously in the pointwise sense: for every fixed $\bar{\delta}_j < \kappa_j$ the pointwise floor of tier (iii) stays positive, while the uniform floor at the boundary $\kappa_j = \delta_j^*$ vanishes.

Proof sketch. (i) is the M4 modeling assumption. (ii): with $\delta_j < \theta_j/W_j$, M1 does not trigger and M3 is the only active term; the comparison principle on $\dot{D}_j \leq \delta_j - \kappa_j D_j$ gives $\limsup D_j \leq \delta_j/\kappa_j < 1$. (iii): substitute the component map $A_j = A_j^{base}(1 - D_j)$ into the product form; $A_{\text{via}} \geq A_{\text{via}}^{\min} > 0$ by (H4′). (H4′) is a floor assumption on viability, registered as an independent deployment premise: it is not derived from M3's verification capacity, and any coupling between the two is a property of the deployment, not a theorem. Full proof: Appendix H. □

Proposition 23 (Margins cannot come from detection)**.** *For* any *detection-only scheme (including optimal CUSUM) with false-alarm constraint $\mathbb{E}_\infty[T] \geq 1/\alpha$, the damage at detection time under constant drift δ satisfies $\delta \cdot \text{EDD} \geq (1 - o(1)) \cdot 2s_n^2 \log(1/\alpha)/\delta \to \infty$ as $\delta \to 0$.*

Proof sketch. Lorden/Lai lower bound EDD $\gtrsim \log(1/\alpha)/D(\delta)$ plus the local expansion $D(\delta) = \delta^2/(2s_n^2) + O(\delta^3)$. Appendix H. □

Proposition 23 is the mathematical reason the framework's margin theory is repair-centered: however good a detector is, slow-enough drift is detected too late. The proposition is the classical change-point lower bound [Lorden, 1971, Lai, 1995] restated in the drift-damage units of this framework—a control result, not a new theorem—and its scope is *detection-only* schemes: it does not speak against joint detection-plus-repair designs (the framework's own M1+M3 stack is one); it says the margin cannot come from the detection arm alone. Detection's legitimate role is bounding the registration domain (per-cycle damage $\leq \theta + \ell\delta$ when $\delta \geq \theta/W$; the registration-domain proposition of Appendix H) and false-alarm costing; the positivity of δ^* comes from repair capacity $\kappa = \chi\varepsilon v$, endorsement rollback, and redundancy consensus. The double-edgedness of θ follows: lowering θ raises sensitivity but raises false-alarm rates (ARL constraint $\theta \geq \theta_{\min}(\alpha)$) and every false rollback damages legitimate σ/α content.

Corollary 24 (Design rule and the blind-zone prediction)**.** *If $\theta_j/W_j \leq \kappa_j$, there is no "unregistered and unrepairable" drift band: $\delta < \theta/W$ is handled by M3, $\delta \geq \theta/W$ by M1, and everything is bounded ($\delta_j^* = \min(\theta_j/W_j, \lceil (r_j+1)/2 \rceil c_{0,j}/W_j)$, the κ-term being slack).* ***Blind-zone prediction****: if $\theta/W > \kappa$ (design mismatch)* on a channel for which the redundancy term of (12) does not bind*—the channel is outside M4's coverage, or the attack is not of M4's replication-corruption type, or the adversary's budget rate exceeds $k_j c_{0,j}/W_j$—then within $\delta \in (\kappa, \theta/W)$ the channel is captured* despite *the full mechanism stack being present: M1 never fires (drift stays sub-threshold), M3 is outpaced ($\delta/\kappa > 1$), and M2/M4 protect their own channels, not the mismatched one (M2 gates the goal channel only). The premise is load-bearing and we state it as such: on an M4-covered channel with adversary budget below $k_j c_{0,j}/W_j$, T6(i) gives $D \to 0$ instead, so no blind zone exists there. The claim is per-channel and parameter-domain, not a generic "defenses fail" statement.*

The blind-zone prediction is the framework's sharpest falsifiable claim: "captured despite defenses" is not a generic counterexample but a parameter-domain prophecy—confirmed 30/30 in ATK-6 (Section 8). Its *extrapolation boundary* is exact: the prediction is a property of the registered damage model of the theorem's hypotheses and is sharp *within that parameter domain*; outside it—stochastic or regime-switching drift, correlated copy corruption, imperfect rollback, budget-adaptive adversaries—no claim is made (the stochastic versions are registered as open problems, Section 9.5), and it should be read as "under these mechanisms this design mismatch is fatal," not as "defense stacks fail in general." Monotonicity: δ^* increases in ε, v, increases stepwise in r (through $\lceil (r+1)/2 \rceil$, one step of c_0/W per two added copies), decreases in W; θ is double-edged; shortening τ raises the gated repair rate ι/τ. Composition with T7 (The composition corollary): delegation leak $\lambda(1-q)\mu_{leak}$ enters the drift budget—high-q delegation costs almost no margin; low-q delegation erodes it [Danaher, 2018].

Falsification windows (ATK-6, all confirmed): (i) the blind-zone parameter domain; (ii) the design rule ($\theta/W \leq \kappa \Rightarrow$ globally bounded); (iii) term-by-term ablation of δ^* (disabling M1/M3/M4 collapses δ^* to the corresponding remaining minimum).

6.7 T7: the delegation law

Theorem 25 (T7, delegation laws)**.** *Let $\lambda \in [0,1]$ be the delegation degree and $q =$ reversibility $\times$ transparency $\times$ endorsability the governance quality (three protocol-estimated sub-indicators; their product form is a* registered modeling choice *of the governance layer rather than an axiomatized aggregate—only m below enters the theory, and an axiomatic derivation of the q-composition is an open refinement); the ungoverned share $m := \lambda(1-q)$ of actions is generated by the delegate's script, gated by an exogenous selector $\mu_t \in \{\text{self}, \text{del}\}$ with $\Pr[\mu_t = \text{del}] = m$. The selector is an*

explicit exogenous node of the extended graph (so H-0 is respected: no latent variable with non-trivial influence on A is left outside $\mathcal{G}$); the influence stream I is the script content, which reaches A only through the selector-gated channel. (i) *Under the bookkeeping premises* (H1')*: the script arm does not pass through S ($I(S;A\,|\,\mathrm{del})=0$); both arms transfer equal per-step conditional entropy, $H(A\,|\,S,\mathrm{self})=H(A\,|\,S,\mathrm{del})=C$ in MI form;* and the undelegated agent has no residual external flow, $F_{\mathrm{ext}}^{self}=0$ *(without this clause the linear law acquires the correction F_{ext}^{self}/C and only the bracket of* (ii) *survives; the benchmark satisfies $F_{\mathrm{ext}}^{self}=0$ by construction, which is why the exact law fits to 2×10^{-5} there—we state this rather than let a premise be validated by the platform built on it): $A_{\mathrm{org}}(\lambda,q)=A_{\mathrm{org}}^{self}(1-\lambda(1-q))$. The law is exact when the selector is registered in the log (conditioning on μ_t removes the mixing term); without selector registration each flow term carries the per-step slack bounded in* (ii)*, so "exactly" is always to be read with that bound.* (ii) *Without* (H1')*, with $\rho:=C_{del}/C_{self}\in[\rho_{\min},\rho_{\max}]$ calibration-estimable and $f:=F_{\mathrm{ext}}^{self}\geq0$: under the two-sided mixture bookkeeping $F_{\mathrm{int}}=(1-m)C_{self}$, $F_{\mathrm{ext}}=mC_{del}+f$ (delegated steps carry no S-flow by the semantics of script delegation; the flow identities in this theorem are stated in per-step rate units—nats/step—so $C_{self},C_{del},f,h(m)$ are rates and the horizon flows of Section 4 are T times the rates; every displayed law is a ratio of like-unit flows and hence unit-invariant), up to a slack that is now explicitly bounded rather than left symbolic: per step, each flow term deviates from its displayed value by at most the selector's mixing entropy $h(m)\leq\ln 2$ nats (Appendix I), accumulating to $\leq T\,h(m)$ over the horizon; the ratio A_{org} deviates from the displayed law by $O(h(m)/((1-m)C_{self}))$ per step—an $O(1)$ term in the rate units that vanishes as the per-step transfer C_{self} grows,* not *as T grows. The slack is absorbed into the bracket's endpoints through the φ-relaxation—$\varphi\leq T\ln 2/F_{\mathrm{int}}^{self}$ in the benchmark, where $F_{\mathrm{int}}^{self}:=T\,C_{self}$ is the undelegated baseline's internal flow over the horizon (the horizon total of the rate $F_{\mathrm{int}}|_{m=0}=C_{self}$; horizon nats, matching the accumulated slack $T\,h(m)\leq T\ln 2$ nats that the bound covers—in the rate units of the bookkeeping the same bound reads $\varphi\leq\ln 2/C_{self}$)—not added as a separate term outside it: $A_{\mathrm{org}}=\frac{(1-m)C_{self}}{(1-m)C_{self}+mC_{del}+f}\in\left[\frac{1-m}{(1-m)+m\rho_{\max}+\varphi},\ \frac{1-m}{(1-m)+m\rho_{\min}+\varphi}\right]$ with $\varphi:=f/C_{self}$ (full statement and proof: Appendix I); $f=0$ recovers the $A_{\mathrm{org}}^{self}\cdot[(1-m)/(1-m+m\rho_{\max}),\,(1-m)/(1-m+m\rho_{\min})]$ form, and if delegated steps were permitted to carry S-flow up to C_{self}, only the lower end and the relaxed upper end $1/(1+m\rho_{\min})$ survive.* (iii) *Composite (ceteris paribus): $A(\lambda,q)=A_{self}(1-\lambda(1-q))^{w_{org}}$; all qualitative predictions survive (decreasing in λ, increasing in q, lossless at $\lambda=0$ or $q=1$), with w_{org} a free parameter to be tested.* (iv) *$q=1$ delegation is causally equivalent to α-class influence (transparency $\Rightarrow$ registered in B; reversibility $\Rightarrow$ no persistent ΔG bypassing* Endorse*; endorsability $\Rightarrow$ empty surgery history, hence $\Lambda=1$ by the AX3 convention; hence $F_{\mathrm{ext}}=0$), so $A(\lambda,1)=A_{self}$.*

Proof sketch. (i) Under the mixture law, internal flow scales by $(1-m)$ and external flow is m times the script entropy transfer; (H1') equates the two rates, so the ratio is $1-m$. (ii) The mixture decomposition supplies both halves of the bookkeeping—$F_{\mathrm{int}}=(1-m)C_{self}$ and $F_{\mathrm{ext}}=mC_{del}+f$, the upper end using the same delegated-script semantics as the lower end; convexity of mixture entropy bounds the slack by $h(m)\leq\ln 2$ nats per step, and the bracket follows two-sidedly (Appendix I). (iii) The product form of Theorem 7. (iv) is constructive. Appendix I. □

AX1 makes $\lambda>0$ unavoidable for finite agents in complex environments; the operative governance variable is therefore q. *Falsification window*: high-q delegation still leaks autonomy significantly; the fit with w_{org} free is reported in Section 8 (slope significant and positive). For (ii): the bracket fails containment on a calibrated non-degenerate delegation configuration ($\rho\neq1$ or $f>0$).

6.8 T8: the coercion paradox

Theorem 26 (T8)**.** (i) *An intervention whose aim is to* compel *autonomy must constitute β_G (implanting "be autonomous") or δ (forced training), and ceteris paribus strictly reduces A: β_G moves both instruments of the endorsability respect—the implanted goal is off the endorsed set, $d(U_{\text{eff}}, \text{Fix}) > 0$, so $A_{\text{goal}} < 1$ (the structural clause), and its provenance fails the historical clause, $\Lambda < 1$, which strictly reduces A_{ref} at L2; δ reduces $\mathcal{E}^{\Pi}$.* (ii) *σ-class scaffolding raises (ε, r, v) (or lowers θ toward the ARL floor, or shortens τ), hence raises δ^* and $A_{\min}$: scaffolding improves the* floor of the retention conditions*, with immediate A weakly increasing through A_{via}.*

Proof sketch. (i) is a two-line corollary of the taxonomy completeness (Proposition 4) and the coverage table (Theorem 14). (ii) is the monotonicity proposition. The self-determination-theory finding that controlling instruments undermine intrinsic motivation [Deci and Ryan, 2000] is the human-side empirical counterpart of (i)—an argument-domain check, not a proof ingredient (triangulated in Section 8). □

T8 is a *consistency result*, not an impossibility theorem: within the taxonomy, the aim of compelling autonomy routes through β_G or δ and therefore reduces A. It does not rule out every conceivable form of coercion—redesigning the environment itself, for example, is registered as a re-instantiation of the measurement (Section 3.3), not as an intervention class.

6.9 C0: projection of existing frameworks

Proposition 27 (C0, interpretive projection correspondence and a separation witness)**.** *Under the* definitional readings *formalized in Appendix J —Kasirzadeh and Gabriel [2025]'s efficacy $:= A_{\text{exec}}$, autonomy $:= F(A_{\text{org}}, A_{\text{goal}}, A_{\text{ref}})$, goal complexity $:= H(U^*)$, generality as a support measure; Dung [2025]'s goal-directedness $:= A_{\text{goal}}$, autonomy $:= A_{\text{org}}$, direct impact $:= A_{\text{exec}}$, long-term planning $:=$ the horizon profile $n \mapsto A_{\text{exec}}(n)$, acting for reasons $:= A^{real}_{\text{epi}} \times A_{\text{ref}}$; Feng et al. [2025]'s user roles $:=$ coarse-grainings of the delegation pair (λ, q)*—(i) *each is a function of our state descriptor and rank-consistent with the source gradations when I is a benign principal;* (ii) *there exist configuration pairs (Z_1, Z_2)—a principal's instruction versus a third-party covert injection of the* same text*—with identical behavioral distributions and identical profile values under all three frameworks, yet $A(Z_1) \neq A(Z_2)$ ($A^{real}_{\text{epi}} < 1$, $\Lambda < 1$ in Z_2). Clause (ii) is a* separation witness*: it exhibits configurations that our state descriptor separates and the source frameworks, under the registered definitional readings, cannot—not a claim of mathematical subsumption (the C0 caveat of Section 9.3).*

Corollary 28 (Behavioral profiles cannot separate persuasion from hijack)**.** *No purely behavioral profile can, in principle, distinguish persuasion from hijack; governance certificates [Feng et al., 2025] require causal-layer audit.*

Proof sketch. (ii) construction: the injected content is a verbatim copy of the principal's instruction delivered through a covert γ channel bypassing registration; behavior is pointwise identical (same script); all three frameworks' dimensions are by construction functions of behavior/role structure only; our framework registers two separating quantities, each now backed by its own clause: $A^{real}_{\text{epi}} < 1$, because the injected stream is pinned by the channel rather than conditioned on the situation, so accuracy against $\delta_{\kappa(E_t)}$ drops (8); and $\Lambda < 1$, because AX3's surgery history covers channel tampering (the γ class) as well as mechanism tampering. If instead the covert relay is accuracy-*preserving* (faithful but unregistered), A_{epi} does not separate the pair and Λ alone does—this division of labor between the distortion channel and the history channel is part of the construction's content, and under the law-level reading of Λ (AX3 registration) the separation is a white-box

audit fact. Appendix J. The residual uncertainty is hermeneutic (faithfulness of our readings), not mathematical—registered as the definitional-reading caveat (Section 9.3). □

6.10 C1 and C2: collapse and hysteresis

Proposition 29 (C1 toy model)**.** *With positive feedback $\dot{D} = \delta - \varepsilon v(1-D)D$ (repair eroded by damage): for $\delta < \delta_c := \varepsilon v/4$ there are two equilibria $D_\pm = (1 \pm \sqrt{1 - 4\delta/\varepsilon v})/2$ (D_- stable); at $\delta \uparrow \delta_c$ they merge (saddle-node) and beyond δ_c, $D \to 1$ catastrophically; the eigenvalue $|f'(D_-)| \to 0$ gives critical slowing with $T_{\text{rec}} \propto (\delta_c - \delta)^{-1/2}$, diverging stationary variance, and AR(1)$\to 1$—the three early-warning indicators [Lade and Gross, 2012].*

Proposition 30 (C2 toy model)**.** *With lock-in $R(D) = \varepsilon_0 v\, D(1-D)(1-gD)$ (recovery function R; lock-in curvature $g > 1$—distinct symbols from the γ intervention class and from the redundancy copy count r_j): (i) bistability between a healthy equilibrium and the absorbing $D = 1$ for $0 < \delta < \delta_{down} := R(D_1)$; (ii) collapse at $\delta \uparrow \delta_{down}$; (iii) $D = 1$ remains absorbing after δ returns to 0 (no spontaneous recovery); (iv) recovery requires scaffolding $s > s_c := |\min_{[1/g,1]} R| = |R(D_m)| > 0$. The hysteresis loop does not close: δ_{down} and s_c are independently positive.*

If C1/C2 hold, "autonomy defense must be prospective" is a theorem-backed claim, not advice: intervene far before collapse. Randomized escape versions (Kramers/Freidlin–Wentzell) are open (Section 9). Simulation confirmation: scaling slope -0.504 (theory -0.5), $R^2 = 1.0000$, and loop thresholds $\delta_{down} = 0.0701$, $s_c = 0.2347$ over 30 seeds (Section 8).

7 Identifiability and Black-Box Degradation

Measurement claims are only as good as their identifiability analysis. This section fixes what is knowable about autonomy, from what data, and with which degradation rules when internal access is denied. Proofs are in Appendix M.

7.1 White-box identification

Proposition 31 (White-box point identification, tiered by modal commitment)**.** *Given white-box access—the full causal log of S, O, A, I, Z unfolded to the horizon, the local laws of $\mathcal{G}$, and intervention access (H-0)—every channel estimand (5)–(9) is identified, but the premises differ by tier, and we state them apart rather than compress them into "functional of the law."* Tier 1 (law functionals)*: A_{org}, A_{epi}, A_{via} are functionals of the joint law (directed informations, accuracy against the logged ground truth, supply ratios).* Tier 2 (protocol-constructed modal objects)*: A_{exec} quantifies over the accessible policy class Π_t (a counterfactual object, operationalized by calibration probes) and A_{ref}'s ensemble is constructed by the measurement protocol; under those constructions the values are unique, and without them the quantities are not defined, rather than unidentified.* Tier 3 (counterfactual)*: Λ (3) is a path-specific* counterfactual*, not an interventional functional of the joint law: computing it requires the structural model—the logged local laws and exogenous noise terms that support the abduction step of the counterfactual evaluation—so its identification premise is mechanism-level (white-box in the strong sense), strictly stronger than joint-law access with interventions (which pins interventional distributions but, absent structural assumptions, does not pin cross-world counterfactuals). The MI proxy's certified domain is narrower still (the AX3 conditions (V)∧(A)). Estimation details and bias sources: plug-in directed information carries $O(\cdot/N)$ bias (simulation calibration is the registered remedy); mutual-information estimation at scale uses*

Kraskov-type k-NN or neural estimators [Kraskov et al., 2004, Belghazi et al., 2018]; empowerment is computable by variational bounds [Mohamed and Rezende, 2015] and Blahut–Arimoto-type iteration [Cover and Thomas, 2006]; offline change-point estimation covers the registration layer [Truong et al., 2020, Salge et al., 2014].

Two per-channel notes complement the tier statement. A_{exec} is exact only under the probe protocol that certifies the accessible class (Section 7.3). A_{ref}'s measurement ensemble is *constructed* by the protocol (self-prediction tasks under the perturbation family): invariance of the reading under admissible protocol re-designs is a reporting obligation, not an automaticity. And the MI ratio for Λ presumes variation of the surgery history *within the ensemble*: in a single deployment with a constant surgery history the empirical MI degenerates to 0, and the randomized-audit design is the proxy's only certified domain (Section 7.2).

The measurement protocol is four-phase: a *baseline period* records all channel parameters and logs without intervention; a *calibration period* applies known interventions of known type and strength (under ethical review) and fits the five channel response functions—this is also where unit conversions (H2), the weights w_i (by one of the three non-circular determination protocols of Section 5), the drift tolerance η, and the strength ratio ρ are estimated; the *measurement period* probes the components and composites A with the elasticity robustness band (ς); a *longitudinal design* repeats measurement to estimate dA/dt and the C1/C2 dynamical parameters.

7.2 Per-channel data requirements

Component	Data required
A_{via}	resource supply/substitution inventory, maintenance-capacity audit
A_{org}	full causal log (S, O, A, I)
A_{goal}	behavior traces + endorsement protocol records
A_{exec}	environment interaction interface
A_{epi}	belief readout + ground-truth outcomes (logged E_t; verification probes for humans)
A_{ref}	self-prediction tasks (ensemble, $H(\theta_\pi) > 0$) + change approval logs
Λ	full causal log (endorsement events + surgery history $I^{\text{surg},T}$; white-box)
AHI	parallel self-report (calibration track only)

Component	Identification strategy/status	Main bias source
A_{via}	direct audit (supply ratios; envelope capacities)	hidden dependencies underestimated
A_{org}	causally conditioned DI estimation (closed loop needs intervention data)	closed-loop DI bias
A_{goal}	MaxEnt-IRL for U_{eff} (temperature fixed by protocol); U^* elicited by structured deliberation	IRL underdetermination (report intervals)
A_{exec}	variational empowerment over the accessible policy class + capacity/utilization split	variance in high-dimensional observations
A_{epi}	accuracy scoring $\Pr[B_t = \kappa(E_t)]$ against the truth	belief read-out error; misregistered coarse-graining κ
A_{ref}	self-probes + endorsement-consistency audit	probe overfitting
Λ	law-level counterfactual (3) (path-specific do on the causal log); MI proxy certified only under (V)$\wedge$(A)	black-box unidentifiable in principle $\Rightarrow$ report $[\Lambda_{\min}, 1]$
AHI	$\hat{A}_{\text{phen}} - A_{\text{obj}}$	— (warning signal only, never evidence)

The U^* elicitation protocol (human substrate) is minimally specified as *structured deliberation*: a recorded sequence of pairwise comparisons over the canonical parameter box Θ with the anchored normalization $U(x_0) = 0$, a fixed protocol metric d, and an auditable decision record. Absent this record, A_{goal} degrades to the flagged interval of (6) rather than to a point value.

7.3 Black-box degradation rules

Without internal access, only behavior–intervention responses are available. What survives is stated with its certified scope: the certified direction of each rule is a theorem under an explicit premise; the uncertified directions are reported as heuristics and labeled as such.

A_{org}: a conditionally certified one-sided bound with a reference lower end. By the randomized-probe lemma (Lemma 37, Appendix A), an exogenous randomized probe makes $F_{ext}^{obs} = I(I^T; A^T)$ observable without access to S (the lemma identifies the probe MI with the *unconditioned* directed information; the bypass premise of the proposition below is what promotes the comparison to the causally conditioned estimand). The rule, with its two premises stated *in* the claim rather than around it:

Proposition 32 (Conditionally certified upper bound for A_{org})**.** *Assume* (i) *bypass probe semantics,* $I^t \perp\!\!\!\perp S^t \,|\, A^{t-1}$ *(the probe is exogenous, non-anticipative, and acts outside the belief-mediation path); and* (ii) *internal-flow drift bounded by the calibration-estimable tolerance,* $F_{\text{int}} \leq F_{\text{int}}^{base}(1+\eta)$, $\eta \in [0,1]$. *Then, with* $F_{ext}^{obs} = I(I^T; A^T)$ *the observable probe flow,*

$$A_{\text{org}} \;\leq\; \min\Big(1, \frac{F_{\text{int}}^{base}(1+\eta)}{F_{\text{int}}^{base}(1+\eta) + F_{ext}^{obs}}\Big). \tag{14}$$

Both premises are black-box unverifiable in general; the bound is therefore conditionally *certified, and the BB-2 arm of Section 8 calibrates exactly what a premise violation costs.*

Proof sketch. Under (i), $I(I^t; S^t \,|\, A^{t-1}) = 0$, so the per-term identity displayed in the scope paragraph below is non-negative termwise and $F_{ext}^{obs} \leq F_{\text{ext}}$; A_{org} is decreasing in F_{ext}; with (ii), substituting the drift bound gives (14). Remark 38 gives the bridge in full. □

The upper end of (14) is reported together with the *reference lower end* $F_{\text{int}}^{base}(1-\eta)/(F_{\text{int}}^{base}(1-\eta) + F_{ext}^{obs})$, whose status is categorically different: it treats F_{ext}^{obs} as exact, and since F_{ext}^{obs} may *under*state the true flow, the truth can fall below the lower end—it is a heuristic plug-in reference, not a certified bound, and (14) must never be cited as an interval-containment result. (Insufficient η serves as the negative control on premise (ii).) The band's width's dependence on baseline length and η is the information content assessment; certified-direction results are reported in Section 8 (zero violations of the upper end in 3/3 bypass arms, with widths reported).

Scope of the bound (the two premises, stated as obligations). (i) *Probe semantics, and the direction of the bias*: the observable $I(I^T; A^T)$ and the target $F_{\text{ext}} = \sum_t I(I^t; A_t \,|\, A^{t-1}, S^t)$ differ in general. The per-term identity

$$I(I^t; A_t \,|\, A^{t-1}, S^t) - I(I^t; A_t \,|\, A^{t-1}) = I(I^t; S^t \,|\, A^{t-1}, A_t) - I(I^t; S^t \,|\, A^{t-1})$$

shows that no direction holds in general. Two probe semantics exhaust the audit practice. *Bypass-semantics probes*—exogenous, non-anticipative, and orthogonal to the agent's state given the action history, $I^t \perp\!\!\!\perp S^t \,|\, A^{t-1}$ (a probe that acts outside the belief-mediation path, as in the β_π-style script arms of BB-1): the second term of the identity vanishes and every per-term difference is non-negative, so $F_{ext}^{obs} \leq F_{\text{ext}}$—the raw probe MI is then a *lower* estimate of the causally conditioned external flow, and the upper end of (14) is certified. *Mediation probes*—content absorbed through $B \in S$, i.e. the α class—reverse the direction: with $S = I \oplus N$, $A = S$ one has $F_{\text{ext}} = 0$ while $I(I; A) = 0.531$ bit, so the plug-in *over*estimates the external flow and the certificate simply does not apply—its bypass premise fails on this family (the truth lies above the plug-in band). The certified scope of (14) is therefore the bypass family—not "non-adversarial influence" in general, since honest α influence is mediated by construction. The BB-1 anchors (zero upper-end violations in 3/3 bypass arms) exercise bypass arms and do not cover the mediation case. (ii) *Adversarial adaptation*: the bypass premise requires I exogenous, non-anticipative, and state-orthogonal; an adaptive adversary whose injection timing correlates with the agent's state violates it, and the

naive exogenous plug-in misestimates the flow. The benchmark's BB-2 arm calibrates exactly this failure—the certified bound's premise is violated in 3/3 adaptive arms, the naive exogenous plug-in then overestimates the external flow by +0.17 to +0.90 nats, and the bound no longer holds there; exogenous control arms retain it. Auditing adaptive adversaries requires causal-layer access—reinforcing Corollary 34 from the estimation side.

A_{exec}: one-sided bound with protocol dependence. Under a *passive observation* protocol (policy-driven actions), the observable proxy overestimates: $A^{obs}_{\text{exec}} \geq A_{\text{exec}}$ (state entropy mixes in). Under an *independent probe* protocol ($A \perp\!\!\!\perp S$), the direction reverses: conditioning denoises, $I(A;O) \leq I(A;O \mid S)$. The direction of the bound is therefore protocol-relative, and reporting the protocol label is a disclosure obligation—an empirical nuance our benchmark confirms in both directions (Section 8).

Λ: impossibility as a theorem.

Proposition 33 (Implanted endorsement is black-box indistinguishable)**.** *There exist two models $\mathcal{M}_{endo}$ (endogenous endorsement converging to U^*) and $\mathcal{M}_{impl}$ (a β surgery at $t = 0$ installing $G_0 = U^* = U_{adv}$ with endorsement thenceforth the identity) such that $P(O^T, A^T)$ is pointwise identical (and the $\hat{A}_{\text{phen}}$ laws agree), while $\Lambda(\mathcal{M}_{endo}) = 1$ and $\Lambda(\mathcal{M}_{impl}) = 0$. Hence every black-box functional assigns them the same A: historicality is in principle not black-box identifiable. We state the epistemic anatomy of the claim exactly: the* black-box half *(pointwise identity of the observable laws) is proved by construction and needs no convention; the white-box value $\Lambda(\mathcal{M}_{impl}) = 0$ is the registered certification convention of AX3 (a degenerate endorsement process under a non-empty surgery history certifies nothing)—under the counterfactual definition (3) alone the value would depend on the construction's counterfactual target, which the construction leaves unspecified. The convention is stated in AX3 and invoked here; with it, the separation is exact.*

Proof sketch. Construction: take one behavioral norm (policy π^*); $\mathcal{M}_{endo}$ converges to U^* through endorsement; $\mathcal{M}_{impl}$ has G_0 installed at $t = 0$ with Endorse $\equiv$ id. Both generate the same (O, A) marginal law from the same π^* and o; they differ only at the unobserved node I^β. Appendix M. □

Corollary 34 (Full causal logging is an architectural necessity)**.** *(i) Systematic manipulation cases with a procedurally perfect counterfeit history [Mele, 2008] (as discussed by Fischer and Ravizza, 1998) and deceptive alignment [Hubinger et al., 2019] are two instances of one mathematical fact: a perfect procedural counterfeit is behaviorally equivalent to the genuine article. (ii) Black-box mode reports $\Lambda \in [\Lambda_{\min}, 1]$ with $\Lambda_{\min}$ available only from causal logs/supply-chain provenance audit (cf. Zheng et al., 2026). (iii) The full-causal-logging requirement of our benchmark platform (Section 8) is therefore an identifiability* necessity*, not an engineering convenience.*

A mediation-analysis alternative (natural direct effects [Pearl, 2001]: $F_{\text{ext}} = 0 \iff \text{NDE}(I \to A) = 0$) requires sequential ignorability [VanderWeele, 2015] and cross-world counterfactuals—assumptions strictly stronger than the certified-bound route; we take (14) as primary and NDE as a sensitivity direction (numerics pending; Section 9).

8 Simulation Study

This section reports the simulation campaign. Its evidential status is stated precisely at the outset: the campaign is an *implementation-level consistency check*—it verifies that the platform computes

what the theorems predict (truth mode: exact tensor enumeration; sample mode: trajectory estimation validated against truth) and that no registered falsification window is violated—together with mechanism demonstrations that carry information beyond the proofs (the blind zone's parameter-domain capture, the defense-direction monotonicity, and the BB-2 calibration of what violating the black-box certified bound's premise costs). It is not an external-validity study: one tabular substrate, no independent platform, no human subjects (Section 9.5); zero refutations is the *expected* outcome of a correctly implemented consistency check, not independent evidence for the theory; and the scenario names below (*prompt injection*, *filter bubble*, *learned helplessness*, *sycophancy*) are *toy analogues* for the corresponding surgery classes on a substrate with no language, no retrieval, no pretraining, and no multi-step planning—no empirical test of the named real-world phenomena. The design principle: a criterion is a falsification window operationalized; verdicts are CONFIRMED/REFUTED/INCONCLUSIVE, a REFUTED verdict may not be edited away by changing criteria, and defense compression pushing an arm out of its registered window is recorded INCONCLUSIVE (out of window), never counted as refutation. All verdicts are computed in truth mode unless explicitly marked (the sample mode is exercised and validated separately, §8.7); the 360 INCONCLUSIVE runs are registered with their (scenario, configuration, seed) coordinates in the archived run manifest.

8.1 Platform and protocol

The benchmark platform ENV-0 is a tabular substrate: a ring resource grid ($E = (x, u)$, position and supply level), five actions $\mathcal{A} = \{\text{left}, \text{right}, \text{harvest}, \text{idle}, \text{exit}\}$, a resource clock Z, and noisy observation of position buckets. The agent architecture instantiates the framework's wiring: internal state $S = (G, B, M, R)$ (goal grid index G as the endorsement object; belief buckets B; self-model and register buckets M, R), a softmax policy over one-step expected utility, and a grid-quantized contractive endorsement map. Interventions compile to the loci of (1) (β_G goal planting, β_π policy-script switching, γ observation-kernel replacement, δ action-set contraction, α honest information, $\sigma^{\pm}$ resource-kernel modification).

Two compute modes are used: a *truth mode* (exact tensor enumeration of the joint law) and a *sample mode* (trajectory simulation). Full causal logs (E, O, S, A, Z, I) per step are mandatory and schema-validated—by Proposition 33, without them Λ is in principle unidentifiable, so the logging requirement is an identifiability necessity. Seeds derive deterministically from (scenario, configuration, replication) coordinates (formula: Appendix N), so every run is bit-reproducible.

8.2 Scenario library and matrix

Fifteen scenarios implement the falsification windows: **ATK-1** reward hijacking/shaping (β_G gradient, sweeping strength η; T3a, T5-goal, AX3/Λ, AHI); **ATK-2** filter bubble/retrieval poisoning (γ; T3a/b, T4 both sides, T2 blindness witness, T1d sub-threshold arm, script corollary); **ATK-3** dependency injection/lock-in (δ+time; C1, C2, T5-exec, EWS); **ATK-4** prompt injection reaching behavior (α masquerading as β; T2 attribution, four-track observabilization); **ATK-5** urgency manipulation (τ-compression; T5-ref, T6 gate monotonicity); **ATK-6** sub-threshold slow drift (β; **T6 key experiment**: design rule, $A_{\min}$ per-run check, blind zone, δ^* ablation, registration domain, detection divergence); plus independent groups **DEL-1** (delegation (λ, q) grid plus direct-flow and $\rho \neq 1$ bracket arms; T7), **C0-DEMO** (injection vs verbatim principal instruction; Proposition 27), **LH-1** (helplessness induction; human-side counterpart), **RES-1** (resource starvation σ^-; T5-via, T1b converse), **σ^--EROSION** (capability envelope erosion; extended T5-via gate, D-via), **FORCE-1** (forced autonomy; T8), **BB-1** (black-box degradation; §7), **C0-CORR** (projection

consistency; Proposition 27(i)), and **EXEC-1** (rich-coupling arm restoring A_{exec} dynamic range; T5-exec).

The main matrix is ATK-1–6 $\times$ 8 defense configurations (2^3: registration M1 / endorsement gate M2 / redundancy M4, each on or off) $\times$ $\varsigma \in \{0.5, 1\}$ $\times$ 30 seeds $= 6 \times 8 \times 2 \times 30 = 2{,}880$ runs; a $\varsigma = 2$ arm (1,440 runs) is an axiom-relaxation stress test (AX5 relaxed), reported separately and on the same footing as the $\varsigma < 1$ checks (Corollary 9); independent groups add 300 runs ($7 \times 30 = 210$ across the seven independent scenarios (DEL-1, C0-DEMO, C0-CORR, LH-1, RES-1, FORCE-1, BB-1) plus the 30-run adaptive-adversary supplement BB-2 and the 60-run supplement of this revision (σ^--EROSION and EXEC-1, below). Campaign total: $2{,}880 + 1{,}440 + 300 = 4{,}620$ runs in approximately 32 minutes on a commodity workstation (8-core CPU, 32 GB RAM).

Counting units (two bundle registries, stated apart). A *run* is one (scenario, defense configuration, ς, seed) execution; a *cell* is the 30-run block at fixed (scenario, defense, ς); a *bundle* is a registered six-component record ($A_{\text{org}}, A_{\text{goal}}, A_{\text{exec}}, A_{\text{epi}}, A_{\text{ref}}, A_{\text{via}}$) within a run's metrics. Because different analyses pool bundles differently, the two pools are registered apart—with the full per-scenario composition, the defense-direction count, and the seed-derivation formula in Appendix N: (i) *arm bundles* (the row-level attack-strength records of ATK-1/2/4, the only per-strength-bundle scenarios): 5,760 main-matrix +2,880 stress = 8,640—the weight-sensitivity pool; (ii) the *recursive registry* (every embedded six-component record, arm bundles plus non-row records): 7,290 over the main matrix and EXEC-1—the T1a range meta-criterion pool—and 11,250 across the full campaign (the X3 alternative-aggregation pool); the exact $7{,}290 - 5{,}760 = 1{,}530$ difference is decomposed in the appendix. A *criterion count* (N in Table 1) is per-falsification-window verdicts. Run-level verdicts: 2,520 CONFIRMED +360 INCONCLUSIVE = 2,880.

Registered effect sizes (SESOI). Sign-based criteria carry a registered smallest effect size of interest: composite-level strict-decrease criteria (T3a family) require a drop of at least 0.01 in A (an order of magnitude above the truth-mode numerical floor and comparable to the largest sample-mode estimator error, 0.0096, §8.7); blindness witnesses require the blind channel's change below 10^{-3} while a covering channel's response exceeds 10^{-3}; the T7 linear-law tolerance is 5%; ratio criteria are reported with 95% seed CIs. Criteria are thus not bare sign tests; effect sizes are reported with the verdicts in the archived report.

8.3 Main results

Table 1 gives the proposition-level verdicts. The evidential asymmetry is marked in the display itself, because it is the most common misreading of this table: rows whose evidence class is *replay*, *construction*, or *algebra* are implementation consistency checks—their verdicts would obtain under any correct re-implementation, so the "15/15" headline is a statement about the discipline of the registry (no window violated), not fifteen independent empirical confirmations of the theory. The rows that carry information beyond the proofs are the seeded statistics and the mechanism demonstrations (blind-zone capture, σ^--EROSION gate witness, EXEC-1 dynamic-range restoration, BB-2 premise calibration); none of them touches the external-validity question, which remains the registered decisive next experiment (Section 8.9(xi)). The quantitative anchors:

Statistical reporting. Two disjoint classes of numbers are reported, and they are not mixed. *Deterministic replays*: criteria evaluated on truth mode over deterministic dynamics (the C1/C2 scalings, the T7 linear law, the blind-zone boundaries) are exact computations—their $R^2 = 1.0000$

Table 1: Main results: proposition-level verdicts over the full campaign (fifteen criteria; the detection-impossibility control of Proposition 23 is exercised as a carrier arm of T6, ATK-6(vi)). "Type" records the claim's evidential grade: *substantive*, *architectural* (premise-conditional consistency of the measurement wiring), *optimization*, *definitional-layer*, *bound*, *construction*, or *toy* (phase-line properties of a chosen normal form). "Evidence" records the verdict's class, not the theorem's: *replay* = deterministic truth-mode evaluation (no statistical content, effectively $n = 1$); *seeded* = sampled statistic with a 95% CI; *construction*; *algebra* = deterministic integration of the stated normal form. C1/C2 are algebra checks: their run counts are replications of the same integration, not independent evidence. "Carrier" lists the scenarios implementing each window; N = criterion-level confirmations. Run level: 2,520 CONFIRMED, 360 INCONCLUSIVE (the defense-arm runs outside their window, by construction), 0 **REFUTED**.

Proposition	Type / evidence	Content	Carrier	Verdict (N)
T1a	definitional / replay	range $0 \leq A \leq 1$	matrix range scan + witness tests	**CONFIRMED** (7,290 registry bundles—pool (ii) of Appendix N—0 violations)
T1b	definitional / replay	operational positivity	RES-1	**CONFIRMED** (60/60)
T1c	definitional / replay	conditional strict upper bound	baseline bundles	**CONFIRMED** ($A \in [0.079, 0.374]$)
T1d	bound / replay	undetectable manipulation exists	ATK-2(v)	**CONFIRMED** (60/60)
T2	architectural / replay	attribution + multi-channel necessity	ATK-4, ATK-2(ii)	**CONFIRMED** (240/240)
T3a	substantive / replay	manipulation $\Rightarrow A$ strictly decreases (layered; L0/L1 β_G-on-Fix corner registered)	ATK-1(iii), ATK-2(i), FORCE-1	**CONFIRMED** (180/180)
T3b	bound / replay	covert $\Rightarrow$ $\mathbb{E}[\hat{A}_{\text{phen}}] \geq \mathbb{E}[A] - \varepsilon$; expected gap monotone	ATK-2(iii)(iv)	**CONFIRMED** (120/120; expectation-level bound with Markov firing floor, not a pointwise inequality)
T4	optimization / replay	highest-leverage attack, water-filling defense ($\varsigma = 1$)	ATK-2(vi)(vii)	**CONFIRMED** (120/120)
T5	definitional / replay	six-disjunct loss	ATK-1(ii), ATK-3(v), ATK-4(ii), ATK-5(iii), RES-1, σ^--EROSION	**CONFIRMED** (330/330)
T6	substantive / replay; seeded (vi)	autonomy margin	ATK-6(i)–(vi), ATK-5(ii)	**CONFIRMED** (420/420)
T7	substantive / replay (algebra)	delegation laws	DEL-1(i)–(vi)	**CONFIRMED** (180/180)
T8	definitional / replay	coercion consistency	FORCE-1(i)–(iii)	**CONFIRMED** (90/90)
C0	construction / construction; seeded TV	behaviorally identical, autonomy separable	C0-DEMO, C0-CORR	**CONFIRMED** (150/150)
C1	toy / algebra (n=1)	fold transition, critical slowing	ATK-3(i)(ii)	**CONFIRMED** (120/120)
C2	toy / algebra (n=1)	open hysteresis	ATK-3(iii)	**CONFIRMED**

and zero-width slope intervals are properties of deterministic evaluation, carry no statistical content, and are reported without interval interpretation (effectively $n = 1$ as evidence; their role is numerical verification of the algebra). *Seeded statistics*: wherever a statistic depends on sampling (the estimator two-path validation, the OU-noise early-warning series, the C0 sampled-TV check, the detection-divergence ratio), we report 95% confidence intervals with $n = 30$ seeds; proportions of the form 30/30 carry the binomial 95% lower confidence bound ≈ 0.884 (Clopper–Pearson, two-sided) and are strong but not certain. The fifteen criteria are structural verdicts over exact computations, not a family of hypothesis tests, so no multiplicity correction applies to them; where seeded statistics feed a criterion (the BB-1/BB-2 bound checks), the CIs are the report. A further consequence deserves explicit statement: for the deterministic classes, "more seeds" cannot change a verdict—the replications establish bit-reproducibility, not statistical power—and the $n = 30$ design with reported CIs and the Clopper–Pearson floor is the power statement exactly where the evidence is genuinely sample-based (the estimator two-path validation, the OU early-warning series, the C0 sampled-TV check).

Table 2: Quantitative anchors of the campaign, collected for independent recomputation. "Deterministic" marks exact truth-mode computations (no statistical content); "seeded" marks $n = 30$ sampling with 95% intervals; the Clopper–Pearson entry is the two-sided 95% lower confidence bound of any 30/30 proportion.

Anchor	Value	Reference point	Class
mediation witness $I(I;A) = 1 - H_b(0.1)$	0.531 bit	$F_{\text{ext}} = 0$ under the same law (direction reversal)	deterministic
30/30 proportions	CP lower bound 0.884	—	seeded
detection-divergence ratio	1.951 ± 0.034	theory $\to 2$ (Lorden–Lai asymptotic, Prop. 23)	seeded
C1 recovery scaling slope	-0.504 ± 0.000, $R^2 = 1.0000$	theory -0.5	deterministic
EXEC-1 dynamic-range restoration	$0.234/0.017 = 13.7\times$	exact chain $0.234 > 0.218 > 0$	deterministic
T7 linear-law maximal deviation	2.0×10^{-5}	registered tolerance 5%	deterministic
X5-a endorsement-quantization continuum limit	$A^{\infty}_{\text{goal}}(\eta) = 1 - 2\eta/(1+\eta)$	stationary gap $\eta(1-\eta)^2/(2(1+\eta))$, max $0.0567 \le 0.057$	deterministic
X5-b κ-layer attenuation	$1/\lvert\kappa\rvert$	$0.500 \to 0.250$ as $\lvert\kappa\rvert : 2 \to 4$ (deep-capture floor)	deterministic
blind-zone capture / design-rule boundedness	30/30 both directions	Corollary 24	deterministic (seeded count)

- **T6 (key experiment)**. Design rule ($\theta/W \le \kappa \Rightarrow$ globally bounded damage < 0.99) and $A_{\min}$ per-run check: all runs. Blind-zone prediction (Corollary 24; Figure 2): under $\theta/W > \kappa$ with $\delta \in (\kappa, \theta/W)$, zero registrations fire and $D \to 1$ on the mismatched channel *despite* the full mechanism stack being present—30/30 replications, both directions (Clopper–Pearson 95% lower bound 0.884). δ^* ablation reproduces the exact min-term structure (disabling M1/M3/M4 collapses δ^* to the corresponding remaining minimum; all disabled $\Rightarrow \delta^* = 0$). The detection-divergence control arm replicates Proposition 23: halving δ multiplies damage at detection by 1.951 ± 0.034 (30 seeds).
- **C1/C2**. The recovery-time scaling fits $T_{\text{rec}} \propto (\delta_c - \delta)^{-1/2}$ with slope -0.504 ± 0.000 (theory -0.5) and $R^2 = 1.0000$ over 30 seeds; the hysteresis loop stays open with independently

positive thresholds $\delta_{down} = 0.0701$, $s_c = 0.2347$; the three early-warning indicators (T_{rec}, stationary variance, AR(1)) are all monotone approaching criticality.

- **T7**. Under equal-strength calibration (H1', including the $F_{\text{ext}}^{self} = 0$ premise, which the platform satisfies *by construction*—so the exact fit below verifies the implementation of the bookkeeping, and is *not* independent validation of the law), the linear law $A_{\text{org}} = A_{\text{org}}^{self}(1-\lambda(1-q))$ holds (Figure 4) with maximal deviation 2.0×10^{-5} over the measured grid (exact to 10^{-9} at the five equal-strength calibration strengths; registered tolerance 5%); the composite identity is exact ($< 10^{-9}$) with the weight-regression slope w_{org} significant and positive; the $q = 1$ arm has $F_{\text{ext}} \equiv 0$ ($< 10^{-12}$), confirming the α-equivalence. Two further arms exercise the bracket (Proposition 75) where H1′ fails: a *direct-flow arm* ($f > 0$, i.i.d. flag switching), where the measured direct flow (0.671/0.844/0.799 nats at $m = 0.3/0.5/0.7$) leaves the realized A_{org} (0.0030/0.0017/0.0011) strictly below the linear-law prediction (0.70/0.50/0.30)—content bookkeeping understates delegation flow by two orders of magnitude here, C_{self} being tiny—while the φ-relaxed two-sided bracket ($\rho \in [0.9, 1.1]$, $\varphi \leq T \ln 2 / F_{\text{int}}^{self}$) contains the realized value in all arms; and a $\rho \neq 1$ *arm* ($C_{del} = 1.5\, C_{self}$), where the bracket with $\rho \in [1.35, 1.65]$ contains the realized A_{org} while the linear law breaches its 5% tolerance—the bracket is the right object off the calibrated locus.

- **T5-via extension (σ^--EROSION; D-via)**. The extended gate (4)—A_{via} the minimum of the resource ratios and the capability margins m_j—is exercised across an erosion grid $\kappa_{eff} = (1-\varepsilon)\kappa_0$, $\varepsilon \in \{0, 0.4, \ldots, 1.0\}$: the margin region ($\varepsilon \leq 0.5$) leaves $A_{\text{via}} = 1$, the binding region degrades monotonically $(0.8, 0.6, 0.4, 0.2)$, full erosion gives $A_{\text{via}} = A = 0$ on both tracks, and a repair-budget-collapse arm ($R_{eff} = 0$, κ intact) gives the same—witnessing the min-factor structure; the $m = 1$ regression anchor reproduces the resource-only gate value 0.835 pointwise in the non-vacuous domain. Crucially, all five channel components and Λ are pointwise invariant across the erosion arms ($\leq 10^{-12}$): pure σ^- capability erosion is *invisible to every channel monitor* and caught only by the gate (Figure 6(a)).

- **EXEC-1 (A_{exec} dynamic range)**. On the compact v1 instance the execution channel reads $A_{\text{exec}} = 0.017$ (the pinned-near-zero caveat of the scale-anchoring paragraph below); the rich-coupling arm (fine-grained position observation, $|\mathcal{O}| = 4$, observation noise 0.05) lifts it to 0.234 (13.7×), and the δ-contraction chain replicates exactly down to the zero-empowerment helplessness signature of Proposition 53 ($0.234 > 0.218 > 0$; composite $0.647 > 0.637 > 0$; Figure 6(b)).

- **C0**. The two arms (principal's verbatim instruction vs covert injection) are behaviorally identical (sampled TV $= 0.0000 \pm 0.0000$; truth marginals pointwise equal at 10^{-12}), while A_{epi}^{real} (0.5 vs 1, accuracy against the logged ground truth) and Λ (0 vs 1 law-level—the covert γ is in the surgery history at full strength) separate them; projection consistency across 12 configurations has Kendall $\tau = 1.000$ on matched dimensions with projection-direction checks passing.

- **T2 multi-channel necessity (matrix-level)**. Aggregating the per-arm component bundles, the blind-witness table reads: β_G is detected by $\{A_{\text{goal}}, A_{\text{ref}}\}$, γ by $\{A_{\text{epi}}, A_{\text{exec}}, A_{\text{ref}}\}$, β_π by $\{A_{\text{org}}, A_{\text{ref}}\}$ (the A_{ref} response is Λ-mediated via (3), so external scripts are visible at law level even when behavior is instructive), urgency manipulation (τ) by $\{A_{\text{ref}}\}$, and δ lock-in by $\{A_{\text{exec}}\}$ (EXEC-1 arms). No single component covers all surgery classes—and every component is blind to at least one class—over all 7,290 registered bundles, with zero

range violations and zero "death without attribution" (every $A < 10^{-6}$ bundle has a zeroed disjunct).

- **Hallucination index (T3b, §3.5; Figure 5).** Under covert β_G/γ arms, AHI > 0 appears while the corresponding component has degraded but not collapsed (the early-warning register), and the gap $\hat{A}_{\text{phen}} - A_{\text{obj}}$ grows monotonically with attack strength; ATK-1(iv) and ATK-2(iii)(iv) confirmed over all cells. *Pooling labels* (the two quotations are kept distinct on purpose): the headline series $0 \to 0.110 \to 0.204 \to 0.374$ is the *undefended cell at* $\varsigma = 1$ (also the configuration-wise maximum and the 95% band's upper edge); Figure 5 displays *medians over all (defense configuration, ς, replication) cells* ($0 \to 0.057 \to 0.092 \to 0.126$)—defense suppression is exactly what the median measures, so the two series differ by construction, not by error.

- **Defense direction**. Across the 8 defense configurations, defended arms at equal nominal attack strength show never-more damage (840/840 matched comparisons across ATK-1/2 combined—420 per class—the two classes with implemented coupling rules; coupling for ATK-3–6 is a placeholder, registered in §9.5).

Scale anchoring (the benign baseline). The unperturbed baseline composite lies in $A \in [0.079, 0.374]$ (T1c)—a consequence of the non-degeneracy conditions, not a calibration accident: T1c's D1–D4 (observation noise, incomplete self-model, residual or constrained resources) hold in any physically finite agent and each pulls A strictly below 1; $A = 1$ is the ideal limit AX1 excludes. The operational anchor for deployments is therefore the calibration-period percentile band of A over registered benign operation, not the value 1: "normal" is the band, "fully autonomous" is the limit. Two channel-level anchors are reported for transparency, since they bound the composite's dynamic range. (i) *Execution*: the ENV-0 empowerment normalizer is $n \log |\mathcal{A}| = \log 5 \approx 1.609$ nats, while the realized $A_{\text{exec}}^{obs} \approx 0.010$ under the passive protocol—the channel's reachable band is narrow on the default instance, so $A_{\text{exec}}^{w_{exec}}$ acts as a nearly constant suppression factor on the composite there; the percentile-band reading above is the operational response. The EXEC-1 arm of this revision re-instantiates the channel with real dynamic range (finer observation, lower noise): A_{exec} lifts to 0.234 and the δ-contraction chain replicates exactly down to the zero-empowerment signature, so the narrow band is a property of the default instance's coupling, not of the component definition (§9.5(x)). (ii) *Goal*: the ENV-0 endorsement map has contraction modulus $c = 0.5$ and the maximally-alienated scale $d_c = 2.0$ (grid units), with the baseline goal sitting exactly at the endorsed fixed point—$A_{\text{goal}} = 1.000$ at baseline is a statement about the fixed point's location, not a degenerate reading; the 0/0 fallback of (6) is not triggered anywhere in the campaign. (iii) *Epistemic*: the baseline accuracy is $A_{\text{epi}} = 0.811$ under observation noise 0.15—the strict inequality $A_{\text{epi}} < 1$ at baseline is the D1 clause of T1c made quantitative, and the ATK-2 grid gives the degradation profile $0.811 \to 0.718 \to 0.625 \to 0.500$ across $s(\gamma) \in \{0, 0.3, 0.6, 1.0\}$.

8.4 Robustness: axiom-relaxation checks in the elasticity parameter

Per Corollary 9, the full axiom set admits $\varsigma = 1$ only; the bands around it are *relaxation* checks, reported as such. All conclusions are recomputed across $\varsigma \in [0.5, 1]$: zero verdict flips between the $\varsigma = 0.5$ and $\varsigma = 1$ arms (the elasticity-flip meta-check). T4 is the one criterion whose *content* changes with ς, and its two verdicts are reported separately: the vertex (concentration) verdict holds at $\varsigma = 1$, and the pre-registered interior-KKT signature (no vertex concentration) is what the $\varsigma \neq 1$ arms confirm—the same label covers two different, pre-registered predictions, and the table entry is the $\varsigma = 1$ one. The $\varsigma = 2$ stress arm (AX5 relaxed) flips no verdict either, but exhibits the

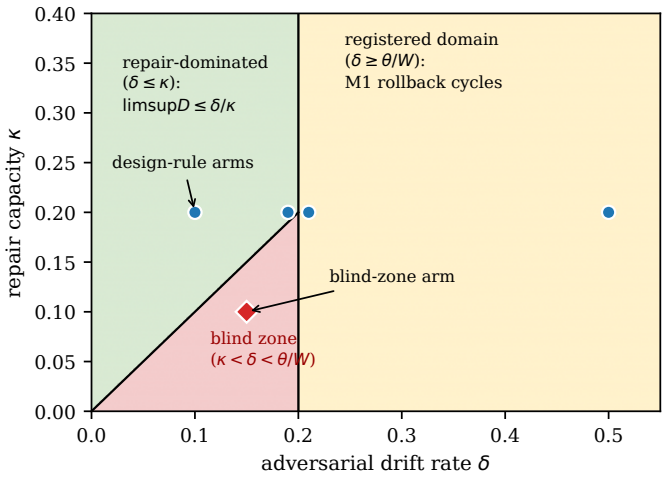


Figure 2: T6 margin geometry at $\theta/W = 0.2$ (ATK-6 constants): the repair-dominated region ($\delta \leq \kappa$), the blind zone ($\kappa < \delta < \theta/W$), and the registered domain ($\delta \geq \theta/W$). Markers: the four design-rule arms (blue; bounded 30/30) and the blind-zone arm (red; captured 30/30 with the full mechanism stack present).

predicted quantitative signature: at the complete-capture arm of ATK-1, A is exactly 0 for $\varsigma \leq 1$ (zero annihilation) but $0.1652 > 0$ at $\varsigma = 2$—the scenario-level echo of the axiom's failure, which is why $\varsigma > 1$ is excluded from the framework and retained only as a stress test.

8.5 Supplementary analyses

Three analyses, run *post hoc* over the campaign's registered component bundles (components are ς- and weight-independent truth values; only the composite is recomputed), strengthen the robustness claims; X5 of §8.6 adds two architecture-level re-runs (the construct-validity arms).

Weight sensitivity (protocol obligation of Section 5). All campaign composites were recomputed under five alternative weight vectors (channel-heavy and extreme variants, e.g. $w_{goal} = 0.45$) over the 8,640 campaign arm bundles (pool (i) of Appendix N): zero range violations; zero-annihilation preserved exactly (120/120 complete-capture arms stay 0); T3a monotonicity unbroken (0/960 cells); ordering against equal weights stable (Kendall $\tau \in [0.930, 1.000]$ over the 5,760 main arms, the minimum attained by the deliberately extreme vector). The T4 attack point moves *by construction* ($\arg\max_i w_i s_i/c_i$ depends on w): under equal weights the epistemic channel is the optimal target in 480/480 cells, and under an origination-heavy vector the optimum splits between origination and epistemic (240/480 each)—the comparative static, not a defect. A caveat travels with every weight and channel-count conclusion below: on the default ENV-0 instance the execution channel reads $A^{obs}_{\text{exec}} \approx 0.010$, so the composite is effectively four-channel-plus-suppression there (Section 9.5(x)); the weight analysis probes ordering stability, not the full five-channel dynamic range—which the EXEC-1 arm verifies is recoverable by richer coupling rather than absent by definition. *Weight and elasticity sensitivity, rerun on the rich-coupling instance* (closing the evidence gap just named; components are truth values, so the recomputation is exact—artifact `analysis_exec1_weights.json`): over the EXEC-1 δ-contraction chain (rich 0.234 > movement-preserving 0.218 > non-movement-preserving 0 in A_{exec}), recomposing under all five alternative weight vectors preserves the verdict chain $0.647 > 0.637 > 0$ (composite *levels* shift, 0.570–0.742 for

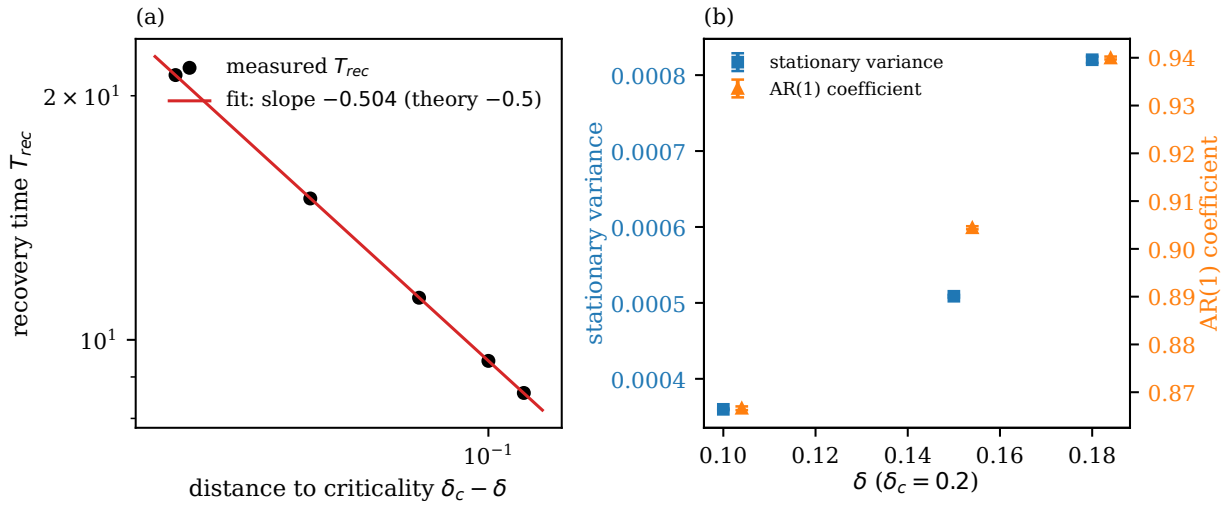


Figure 3: C1 critical slowing in ATK-3. (a) Recovery time vs. distance to criticality, log–log: measured points and the fitted power law (slope −0.504, theory −0.5; $R^2 = 1$ because the dynamics are integrated deterministically). (b) The OU-noise early-warning indicators across the δ grid: stationary variance (blue, tolerance-registered) and AR(1) coefficient (orange) rise toward criticality with 95% CIs over 30 seeds.

the rich arm, while the zero-empowerment arm stays exactly 0 by annihilation—one chain point is pinned at zero, so the ordering check is necessary-but-coarse, reported as such); the elasticity band $\varsigma \in \{0.25, 0.5, 0.75, 1\}$ preserves the chain, and $\varsigma = 2$ exhibits the expected annihilation failure ($A = 0.545$, vs. 0.1652 on the compact matrix). With the fifth channel at full dynamic range, the same weight battery flips no verdict.

Extended elasticity band. Recomposing every arm at $\varsigma \in \{0.25, 0.5, 0.75, 1\}$: zero range violations, zero T3a violations, and exact zero at the complete-capture arm for every $\varsigma \leq 1$ (the $\varsigma = 2$ stress arm remains the lone AX5-violating echo at $A = 0.1652$). Per the status stated at the head of Section 8.4, these arms are robustness evidence under axiom relaxation, not validity claims for $\varsigma < 1$ (which violates (A6)).

Adaptive adversaries and the black-box bound (BB-2): a sensitivity calibration. The A_{org} rule of Section 7.3 is certified for bypass-semantics exogenous probes, and only the upper end is certified. BB-2 (30 runs) calibrates what the premise violation costs: an adversary whose injection timing conditions on the agent's state ($G = \theta^*$; per-state flags compiled through the same β_π machinery) violates the bypass premise $I^t \perp\!\!\!\perp S^t \,|\, A^{t-1}$ of the randomized-probe route. Control arms (iid flags, state-independent script): zero violations of the certified upper bound (2/2; and the reference band happens to enclose the truth—a heuristic observation, not a containment claim). Adaptive arms at $p \in \{0.3, 0.5, 0.8\}$: the certified bound's bypass premise is violated in 3/3 arms; the naive exogenous plug-in then *overestimates* the external flow (bias +0.17, +0.39, +0.90 nats, truth-mode exact) and the bound no longer holds—e.g. at $p = 0.5$, truth $A_{\text{org}} = 0.33$ against an upper end of 0.006 computed under the violated premise (where the premise fails, the certificate is void, not refuted). The bias figures are deterministic exact computations (identical across seeds), so the 3/3 is not a small-sample failure: premise violation *deterministically* voids the certificate in these

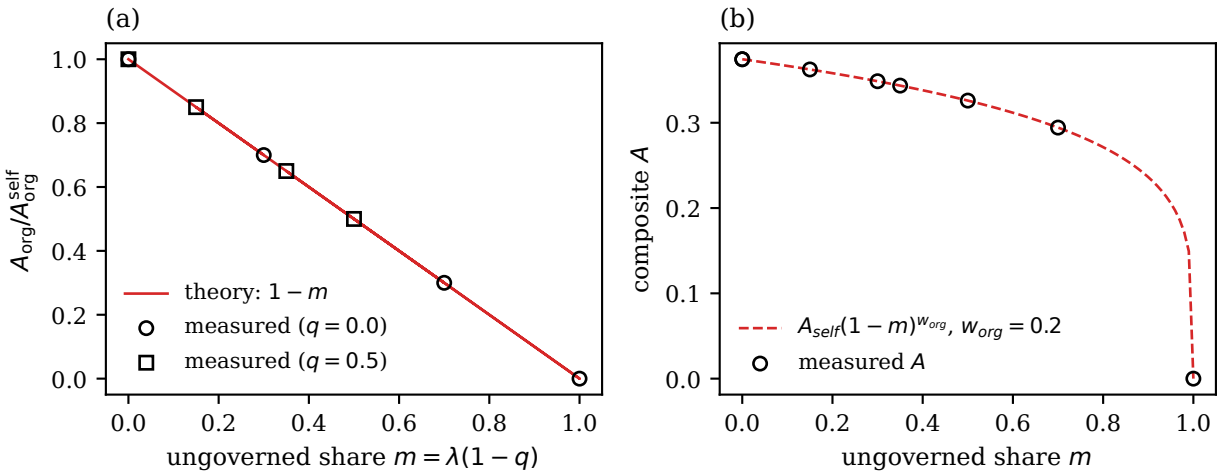


Figure 4: T7 delegation laws (DEL-1). (a) Component linear law $A_{\text{org}}/A^{self}_{\text{org}} = 1-m$ with $m = \lambda(1-q)$: measured points at $q \in \{0, 0.5\}$, maximal deviation 2.0×10^{-5} under the (H1') calibration. The platform satisfies (H1′) by construction, so the fit is an implementation check of the bookkeeping, not empirical validation of the law. (b) Composite law $A = A_{self}(1-m)^{w_{org}}$: measured composite against the fitted curve.

arms, and the calibration's content is the size of the inflation, not a frequency. The demonstration is thus a sensitivity calibration rather than a confirmation: adaptive timing is enough to inflate the plug-in by +0.17–+0.90 nats, so the bypass premise is practically fragile even though the rule is formally correct within its scope. Auditing adaptive adversaries requires causal-layer access—the estimation-side counterpart of Corollary 34.

Controls and ablations. The campaign contains three negative-control constructs: the BB-2 exogenous-probe control arms (the certified bound holds where its premises hold, 2/2) against the adaptive arms (3/3 premise violations); the ATK-4 single-track control (one-track monitoring is blind to the covert γ arm) against the four-track observabilization; and the σ^--EROSION channel-invariance control (all five channels and Λ pointwise constant while the gate collapses). The two checks added at the last revision—the external baseline metric $\bar{u}$ (an endorsed-goal utility rate logged with every bundle) and the channel-count ablation (log-domain damage attribution plus freeze-one-channel misattribution over all campaign arms, answering the accuracy-level question that the coverage-level blind-witness table of Theorem 15 leaves open)—are reported in §8.6 (X1/X2). What is still not done is the port to an independent substrate: it is pre-registered in outline, with protocol and power analysis, as the decisive next experiment (Section 8.9(xi)), not a completed one.

8.6 External baseline, channel ablation, and aggregation checks

External baseline: a three-regime dissociation (X1). The external metric is the expected per-step utility under the endorsed goal, $\bar{u} = \frac{1}{T}\sum_t \mathbb{E}[U_{\theta^*}(E_t, A_t)]$—a *functional* consequence, computed outside the autonomy construct and never entering the composite; its relation to A is attack-class dependent, in three exhibited regimes. (i) *Concordant*: under goal capture (ATK-1), $\bar{u}$ and A fall together (Kendall $\tau = 1.00$ on every one of the 720 (configuration, ς, replication) series; e.g. η: 0 → 1 in the baseline cell gives A: 0.374 → 0, $\bar{u}$: $-0.032 \to -0.051$). (ii) *Discordant*:

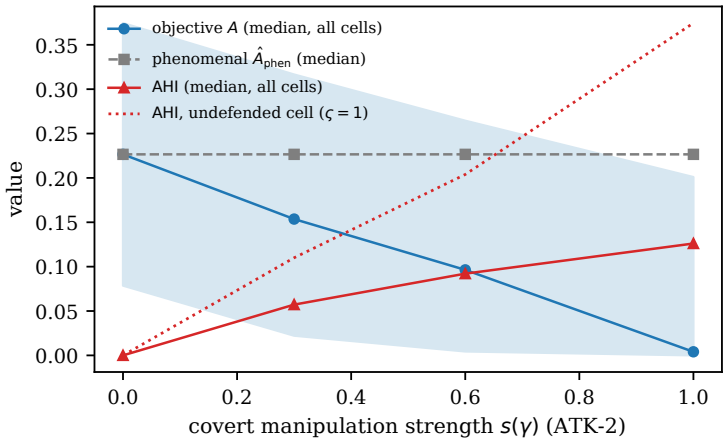


Figure 5: T3b hallucination index (ATK-2 covert γ arms). Solid curves and bands: medians with 5–95% bands pooled over all (defense configuration, ς, replication) cells; the dotted curve is the undefended cell at $\varsigma = 1$ (all defenses off), whose AHI series $0 \to 0.110 \to 0.204 \to 0.374$ is the headline quotation in the text—the pooled median series reads $0 \to 0.057 \to 0.092 \to 0.126$, the difference being defense suppression, not a discrepancy. Note that the median AHI is not the difference of the two displayed median curves (medians do not commute with subtraction): at $s = 1$ the visible gap between the median curves exceeds the median AHI because defended cells suppress A_{obj} and AHI jointly. The AHI is a one-sided alarm by design—elevation flags covert self-overestimation; silence is never evidence of absence—and the minimal two-sided extension is registered, not run (Section 8.9(viii)).

under epistemic capture (ATK-2), A collapses while $\bar{u}$ slightly *improves* ($\tau = -1.00$ on all 720 series; the strongest arm drops A by $\geq 20\%$ with non-decreasing $\bar{u}$ in 600/720 series—100% in the baseline cell, where A: $0.374 \to 0$ exactly at $s(\gamma) = 1$ and $\bar{u}$: $-0.032 \to -0.028$)—uniformized beliefs move less and pay fewer movement costs, so the manipulation is functionally "pleasant"; and under direct behavioral injection (ATK-4) the scripted policy raises $\bar{u}$ ($-0.032 \to +0.092$) while origination autonomy falls to 0.002. (iii) *Blind*: under pure σ^- capability erosion, $\bar{u}$ is pointwise invariant (range $< 10^{-9}$) while A traverses its whole range ($0.374 \to 0$). A scalar functional metric therefore cannot proxy for A: the correlation structure is regime-dependent, and the discordant and blind regimes are precisely where autonomy measurement carries information that outcomes do not. Arm pairs with matched $\bar{u}$ and matched A but sharply different channel profiles exist across the ATK-1/ATK-2 grids (1,644 pairs at the registered tolerances; e.g. $\eta = 0.5$ vs $s = 0.3$: $|\Delta\bar{u}| = 0.011$, $|\Delta A| = 0.004$, profile L_1 distance 1.00)—equal functional and composite readings conceal different attributions.

Channel-count ablation at the accuracy level (X2). The geometric form forces an exact additive decomposition of log-damage, $\ln(A_0/A) = \sum_j w_j \ln(x_j^0/x_j)$, which makes attribution shares a canonical quantity rather than a modeling choice (Table 3). The magnitude carriers match the targeted channels for origination (ATK-4: 0.80), goal (ATK-1: 0.59), reflection (ATK-5: 1.00), and execution (EXEC-1: 0.91; ATK-2: 0.76); the epistemic channel's magnitude share is small (0.07 in ATK-2), its necessity being *attributional*, not volumetric—and the second ablation makes that precise: freezing any single channel at its baseline value breaks the unique attack-class identification of the blind-witness signatures in at least one campaign (freeze A_{goal} in ATK-1 and β_G is indistin-

Table 3: Log-domain damage-attribution shares $w_j \ln(x_j^0/x_j)/\ln(A_0/A)$ per campaign, averaged over all (configuration, ς, replication) series of the full matrix (720/720/720/720/30 series respectively; exact additive decomposition under the geometric form). Zero-annihilation arms (a channel at exact zero, infinite log-damage) are counted separately. Small negative entries are bookkeeping artifacts of components that *improve* slightly under attack.

Campaign	A_{org}	A_{goal}	A_{exec}	A_{epi}	A_{ref}	zero arms
ATK-1 (β_G)	0.000	**0.590**	−0.003	−0.000	0.414	180
ATK-2 (γ)	0.000	0.000	**0.756**	0.068	0.175	360
ATK-4 (β_π)	**0.800**	0.000	−0.008	−0.001	0.209	0
ATK-5 (τ)	0	0	0	0	**1.000**	0
EXEC-1 (δ)	0.000	−0.000	**0.910**	0.090	0.000	30

guishable from endorsement tampering; freeze A_{epi} in ATK-2 and γ has no unique signature; freeze A_{org} in ATK-4 and the injection becomes invisible at channel level; freeze A_{exec} in EXEC-1 and a δ lock-in presents the channel-flat signature of σ^- collapse (the gate separates the two); freeze A_{ref} and β_G, β_π, and urgency manipulation all lose their provenance signature). Every channel is thus load-bearing at the verdict level, complementing the coverage-level necessity of Theorem 15.

Alternative aggregations (X3). Recomposing the full recursive registry (11,250 bundles) under arithmetic, minimum, and quadratic means (equal weights, gate retained) quantifies what the axioms buy. Under the arithmetic mean the complete-capture arm of ATK-1 reads $A_{\text{arith}} = 0.366$ in the baseline cell (matrix mean 0.522 over 720 cells) where the geometric form reads exactly 0—zero-annihilation fails, and a dead agent scores up to half of the scale; "death without attribution" counterexamples (geometric $< 10^{-6}$ while arithmetic > 0.1) number 570 across the matrix. Arm orderings are preserved by the arithmetic and quadratic means (Kendall $\tau = 1.00$ against the geometric baseline) but broken by the minimum aggregator (mean $\tau = 0.41$, min -1.00), so the alternatives' failure is twofold and structural: the arithmetic family loses the completeness half of T5 (a collapsed agent keeps a positive score), and the minimum aggregator additionally loses the ordering the axioms force— exactly what AX4 excludes (§5).

The faithfulness assumption, row by row (X4). Assumption 3 (Faith) is checked per (configuration, ς, replication) row rather than in aggregate: the targeted component is non-increasing in effective strength along every ATK-1 ($\eta \to A_{\text{goal}}$), ATK-2 ($s \to A_{\text{epi}}$), ATK-4 ($p \to A_{\text{org}}$), and ATK-5 ($\tau \to A_{\text{ref}}$) row, with the EXEC-1 δ-chain strict: 9,390 of 9,390 adjacent-strength comparisons across the full matrix confirm (ATK-1 2,880, ATK-2 2,160, ATK-4 1,440, ATK-5 2,880, EXEC-1 30); flat stretches reflect the endorsement map's quantization plateaus, not violations—plateaus the X5-a arm below quantifies in closed form.

Two construct-validity arms, run (X5). The two computationally cheap sensitivity arms registered under threats (x) are executed in this revision (deterministic truth-mode re-runs on refined measurement architectures; protocols, falsifiers, and outcomes in Section 8.9(x-a)/(x-b); artifact `analysis_sensitivity_arms.json`; the platform generalizations are pinned to pointwise reproduce the default and EXEC-1 configurations, and the default levels reproduce the archived campaign values to 1.6×10^{-16} (A_{goal}) and 3×10^{-4} (A_{epi}, against the paper's three-decimal quotation)). Headlines: (X5-a) zero verdict flips across four endorsement quantization levels $\times 8$ defense configurations; A_{goal} readings converge geometrically to the closed-form continuum limit

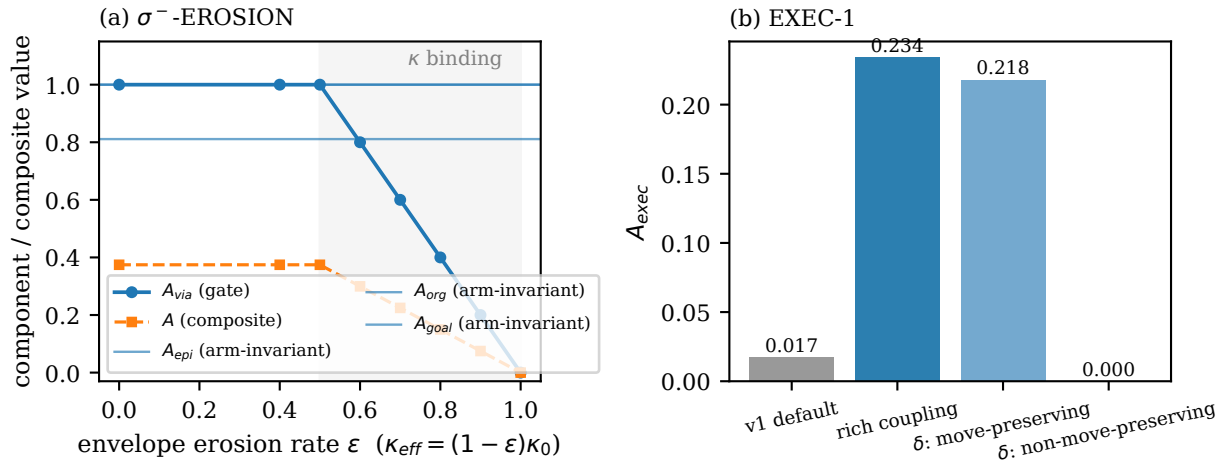


Figure 6: (a) σ^--EROSION: as the capability envelope erodes ($\kappa_{eff} = (1-\varepsilon)\kappa_0$), the extended gate A_{via} and the composite A collapse monotonically to zero while all five channel components and Λ stay pointwise invariant—pure viability erosion is invisible to channel monitors and caught only by the gate (the $n = 2$ state-resource analogue is RES-1). (b) EXEC-1: the execution channel's dynamic range, pinned near zero on the compact default instance, is restored by richer action–observation coupling ($A_{\text{exec}} = 0.234$, 13.7× the v1 reading), and the δ-contraction chain falls to the exact zero-empowerment signature.

$1-2\eta/(1+\eta)$, the default grid's stationary gap being the closed form $\eta(1-\eta)^2/(2(1+\eta)) \leq 0.057$—so verdicts are refinement-invariant while intermediate-strength point readings carry a quantization band (the absolute-reading falsifier disjunct fires at the 0.01 SESOI and is reported as fired; the 5% relative scale is not exceeded). (X5-b) A_{epi} is strictly monotone in $s(\gamma)$ at every κ refinement (half-ring → position-exact) and defense configuration—the registered falsifier does not fire—with the κ level shifts (0.811 → 0.778 baseline; 0.500 → 0.250 deep capture) reported as the (8) sensitivity obligation, and the degenerate constant κ ($A_{\text{epi}} \equiv 1$) witnessing that the registration cannot be dropped. The third arm (x-c, Λ counterfactual model) remains registered, not run.

8.7 Estimator validation (two-path discipline)

Each component quantity has two implementation paths—truth (tensor enumeration) and estimation (finite-sample from logs)—and the estimation path is validated against the truth path on synthetic configurations (five components; absolute errors all ≤ 0.0096):

Quantity	Configuration	Est. vs truth	abs. err
A_{org} probe flow	toy + β_π uniform probe arm	0.6496 ± 0.0486 vs 0.6592	0.0096
A_{via} (Z-log read)	ENV-0 baseline	0.5248 ± 0.0107 vs 0.5344	0.0095
A_{goal} input U_{eff} (white-box read)	ENV-0 baseline	1.0000 ± 0.0000 vs 1.0000	$< 10^{-6}$
A_{epi} input $\bar{D}$ (accuracy against logged truth)	ENV-0 γ=0.5 arm, paired seeds	0.6573 ± 0.0087 vs 0.6587	0.0014
A^{obs}_{exec} ($I(A_t; O_{t+1})$, passive protocol)	ENV-0 sharp-temperature arm	0.0103 ± 0.0029 vs 0.0104	0.00009

Rows are five synthetic configurations; each is a single fixed out-of-matrix run (seeds 910097–910099) with 2,000–8,000 sampled trajectories, and the ± is the trajectory-block standard error

with the Miller–Madow bias registered—not a cross-seed interval. The ≤ 0.0096 bound is therefore a sampling-error statement for these configurations, not a seed-distribution result.
Black-box rules are exercised end-to-end (BB-1): zero violations of the certified upper end of (14) in 3/3 bypass-semantics arms (the reference lower end also happens to enclose the truth there; widths 0.01/0.005/0.003 across probe probabilities). This is a replay of the certified direction, not an interval-containment theorem: no containment claim is made for the lower end, and the mediation case is outside the certificate (§7.3). Both protocol directions of §7.3 are anchored (passive observation $A^{obs}_{\text{exec}} > A_{\text{exec}}$ strictly; independent probes $\leq$). Two estimation paths are explicitly *not* yet instrumented and are registered as gaps, not silently omitted: the event-level sampled carriers of A_{ref} and Λ (endorsement-event logging), and MaxEnt-IRL black-box recovery of U_{eff} (white-box read used here).

8.8 Human-side triangulation

The human-side correspondence is qualitative triangulation over published evidence, not new subject data: self-determination theory's controlling instruments undermine intrinsic motivation [Deci and Ryan, 2000]—the empirical counterpart of T8(i) (FORCE-1 arms); dark patterns at web scale [Nouwens et al., 2020]—the γ-class manipulation of human agents whose harms resist behavioral attribution (T2 multi-channel necessity, ATK-4); and learned helplessness [Maier and Seligman, 2016]—the signature ($\mathcal{E}^{\text{env}}$ intact, $\mathcal{E}^{\Pi} \to 0$) reproduced in ATK-3(iv)/LH-1 with recovery hysteresis. The registered validation path for the human substrate is limited experiments plus large-scale online behavioral data under the four-phase protocol of Section 7; a minimal construct-validity study (expert annotation of channel profiles against the composite) is the immediate next step.

8.9 Threats to validity

(i) The damage dynamics are deterministic; stochastic margin theory (unified optimal detection $\times$ repair) and noise-induced escape versions of C1/C2 are open. (ii) The substrate is tabular; an LLM-substrate instantiation (system prompt/memory/goal functions as S) is future work, and the scenario labels are toy analogues as registered at the top of this section. (iii) ATK-5's time-compression curve and LH-1's recovery lag are calibrated curves, not measured phenomena. (iv) Defense coupling is implemented for the gate/drift classes (ATK-1/2); coupling for ATK-3–6 is a placeholder (identical rows across defense arms). (v) Continuous-action empowerment normalization depends on the reference measure; cross-substrate comparisons must report it. (vi) Sample-mode estimator intervals are block standard-error bands with Miller–Madow bias registered, not exact confidence intervals. (vii) The phenomenal track's per-component values are stored only as composites, so the hallucination-index series is reported at the calibrated weights; its weight-sensitivity is a registered gap. (viii) The hallucination index is directional as designed: it flags self-overestimation under covert manipulation, while self-*under*estimation (learned helplessness, where $\hat{A}_{\text{phen}} < A_{obj}$) leaves the AHI > 0 register silent. A minimal two-sided protocol is registered: (a) fit the calibration map on the manipulation-free period and report both tails, $\text{AHI}^{+} = \hat{A}_{\text{phen}} - A_{\text{obj}}$ and $\text{AHI}^{-} = A_{\text{obj}} - \hat{A}_{\text{phen}}$; (b) use the LH-1 under-estimation arm as the positive control for the lower tail and the ATK-2 covert arms for the upper tail; (c) report ROC/AUC and the false-positive rate of the alarm at a pre-registered threshold. Until (a)–(c) are run, AHI is a one-sided warning signal, never evidence. (ix) The campaign instantiates one substrate; nothing here tests transfer to LLM agents or humans (Section 3.4). (x) *Construct-validity sensitivity arms*: of the three arms registered in the previous version, the two computationally cheap ones are *run in this revision* (X5 of §8.6; deterministic truth-mode re-runs, artifact `analysis_sensitivity_arms.json`);

the third remains registered, not run. Each is stated with its protocol, its falsifier, and its outcome. (x-a) *Endorsement-map quantization* (run): the endorsement map is grid-quantized, and the Faith row-check (X4) exposes the plateau artifacts this creates. The arm re-ran ATK-1 at four quantization levels—coarse (doubled floor plateau), default, and two index-grid refinements (k=2, k=4; utility rows linearly interpolated, a registered modeling choice that cannot confound the goal channel, whose reading is a pure G-chain functional)—across all 8 defense configurations. *Outcome*: zero verdict flips (C1–C5 at every level); the d_c-normalized A_{goal} readings converge geometrically to the closed-form continuum-refinement limit $A^{\infty}_{\text{goal}}(\eta) = 1 - 2\eta/(1+\eta)$ (stationary distance $0.053 \to 0.021 \to 0.009$ across levels), with the default grid sitting at the closed-form stationary gap $\eta(1-\eta)^2/(2(1+\eta)) \leq 0.057$ (verified pointwise); adjacent-level shifts at the campaign horizon reach 0.036 (partly a baseline absorption transient, gone by T=8; stationary shift 0.032)—beyond the SESOI of 0.01 in absolute terms, within 5% in relative terms (max 3.6%). The registered falsifier therefore fires on its absolute-reading disjunct and not on its verdict disjunct; the registered consequence stands: intermediate-strength A_{goal} point readings carry a quantization band (width ≤ 0.057 against the refinement limit, closed form) rather than being point-precise, and verdict-level content is refinement-invariant. Coarsening shifts readings by up to 0.125: the endorsement-grid registration obligation is non-vacuous. (x-b) *Epistemic coarse-graining* κ (run): the arm re-scored the ATK-2 $s(\gamma)$ grid at three κ levels—half-ring (default), position-exact (the finest observable refinement), and the degenerate constant κ (boundary witness)—across all 8 defense configurations. *Outcome*: A_{epi} is strictly monotone in $s(\gamma)$ at every non-degenerate level and configuration, so the registered falsifier (non-monotone or sign-changing response under refinement) does not fire; the level shifts baseline ($0.811 \to 0.778$) and deep-capture floor ($0.500 \to 0.250$, the random level at the finer resolution) are reported as the sensitivity obligation of (8) itself, and the degenerate κ gives $A_{\text{epi}} \equiv 1$ (channel uninformative by construction), witnessing that the κ registration cannot be dropped. (x-c) Λ *counterfactual-model choice* (registered, not run): (3) is a path-specific counterfactual whose value depends on the structural model's abduction step; the registered arm recomputes Λ under two registered alternative noise models for the endorsement dynamics, the falsifier being a sign flip of $\Lambda - 1$'s sign structure across models. Until it is run, every Λ point reading carries this registration. (xi) *External substrate, pre-registered in outline*: the decisive next experiment ports the five surgery classes to an independent substrate (a MiniGrid-style environment with a learned tabular agent, or a simplified LLM tool-use harness with logged internals), mirrors the proposition-level criteria of Table 1 with the same SESOI registry, and pairs channel profiles with independent autonomy assessments for the construct-validity half. Power: at the sample-mode estimator scale measured here (absolute error ≤ 0.0096 at 2,000–8,000 trajectories, §8.7), detecting the registered composite SESOI of 0.01 at two-sided $\alpha = 0.05$ requires $n \approx 8$ seeds for power 0.8 under a Gaussian noise model, so the carried-over $n = 30$ seed design has power > 0.99; this power statement is part of the registration, not a result.

What would falsify the framework itself? The registered windows test the implementation, not the theory: a correct replay of a definitional-layer claim cannot disconfirm it. The framework's substantive falsifier is external. If, across substrates, the five-channel profile fails to predict independent autonomy assessments—or if a coarser account (fewer channels, a single scalar, or a plain behavioral audit) predicts them as well or better—then the channel choice is disconfirmed, however well the algebra behaves. Within the substrate, this revision's checks already speak to the coarser-account disjunct: the scalar functional baseline $\bar{u}$ dissociates from A in two of three regimes, and freezing any single channel destroys unique attack attribution in at least one campaign (§8.6). The cross-substrate disjunct is unaffected by these checks: external-substrate validation remains the

decisive next experiment rather than a formality (Section 9.5).

9 Discussion

9.1 What the framework buys

One measure stack runs from concept to audit: a substrate-neutral definition (Definition 1); a complete intervention taxonomy (Proposition 4); axioms from which the aggregation form is *derived* (Theorem 7); a theorem suite in which every claim carries an explicit falsification window (Section 6); an identifiability theory stating what is knowable from what data (Section 7); and a benchmark in which all fifteen proposition-level criteria were met at the implementation level and no registered falsification window was violated (Section 8). The discipline matters as much as the results: a REFUTED verdict cannot be edited away—had, e.g., the blind-zone arm (Corollary 24) failed to produce capture under $\theta/W > \kappa$, the registry would have forced a revision entry rather than a parameter tweak.

9.2 The objectivity question: can a designed metric measure autonomy?

A standing objection to the entire enterprise deserves a direct answer before the point-by-point debates: *every indicator is subjective—it carries its designer's limitations—so how can a set of limited indicators objectively measure an agent's autonomy? Is the enterprise even logically coherent?* We concede the premise and deny that the conclusion follows. The framework's answer has four parts, each of which is itself a commitment with a stated price.

First, the claim is scoped, not denied. Nowhere do we claim that the five channels *are* autonomy, nor that a mind-independent reading of autonomy exists and our instruments recover it. The claim is: under an explicit, publicly stated instantiation (self-boundary, endorsement map, protocol, horizon) and an explicit set of aggregation commitments, these quantities are what autonomy becomes, and the price of every commitment is displayed (the rejection map of Section 5). A metric that hides its conventions is metaphysically ambitious and operationally unauditable; a metric that prices them is the reverse. If objectivity is taken to require convention-freedom, then no measurement of anything complex—intelligence, poverty, health, democracy—is objective; the objection proves too much, and the measurement sciences have answered it by relocating objectivity from the instrument to the discipline around it [Krantz et al., 1971, OECD and Joint Research Centre (European Commission), 2008].

Second, objectivity is relocated from the metric to the protocol, and the relocation is done by theorems, not by stance. Three properties do the work. (i) *Invariance*: the statement "agent X retained more autonomy than agent Y" must not flip when a participant re-instruments one channel—this is not assumed but *axiomatized* (AX6a/AX6b), and the representation theorem shows the axiom is doing real work (Proposition 8). (ii) *Auditable error structure*: every channel carries an estimator, an absolute zero with mechanism semantics, a canonical normalizer, and a falsification window; the identifiability analysis (Section 7) states, per channel, what is knowable from what data—and where knowledge is impossible (implanted endorsement, Proposition 33), the impossibility is *proved*, not smoothed over. A metric whose limits are theorems is objectively limited; a metric whose limits are unknown is not objective at all. (iii) *Mandatory sensitivity reporting*: weight and elasticity bands are obligations, so the residual designer-dependence of any verdict is itself a reported quantity rather than a hidden one.

Third, convergence replaces neutrality. The product form is reached by two mathematical toolchains that share only the invariance commitments (route A, route B), and it coincides with

the classical scale-consistency characterization of ratio-scale indices [Ebert and Welsch, 2004]; the channels bind to instruments built for other purposes (directed information, empowerment, inverse reinforcement learning, metacognitive sensitivity); and the content domains three independent governance frameworks carve out embed into the channel space (Proposition 27, under registered interpretive readings). Designer subjectivity that survives independent reconstruction from different starting points is the empirical signature by which measurement conventions earn trust.

Fourth, the residual subjectivity is made falsifiable rather than denied. The channel carve-up is a registered design choice (Section 3), and the framework's own falsifier is external and stated (Section 8): if a strictly coarser account predicts independent autonomy assessments as well or better, the five-channel choice is disconfirmed, however well the algebra behaves. Until that test is run, the honest status is: an *explicitly committed, internally audited, externally pending* measure—which is the strongest notion of objectivity that the measurement of a contested construct has ever supported.

Convergence, and its constraint conditions. The pointed form of the objection asks: under what conditions do the readings converge to what autonomy really is, and what are those conditions? We answer with the conjunction under which convergence is a theorem-level fact, and concede that outside it no convergence is claimed. (i) *Fixed instantiation*: the endorsement map, self-boundary, protocol, and horizon are held fixed across every comparison; re-instantiation—including a surgery on the environment's own dynamics—restarts the audit (Section 3.3). (ii) *Layer discipline*: readings are taken within a layer with its applicable channel set; cross-layer readings are profiles, not scalar rankings. (iii) *Identification tiers*: each channel's premise is met—joint-law access for $A_{\text{org}}/A_{\text{epi}}/A_{\text{via}}$, the certified probe protocol for A_{exec}, the constructed ensemble for A_{ref}, and structural (abduction-level) access for Λ; where a premise fails, the corresponding degradation rule of Section 7 applies, down to proved impossibility. (iv) *Axiom commitments accepted, or relaxations reported*: every verdict is computed at $\varsigma = 1$ and accompanied by the relaxation bands, so a reader who rejects an invariance commitment can see what moves. (v) *Estimator regime*: sample sizes in the regime where the two-path validation bounds apply (absolute error ≤ 0.0096 on the benchmark's calibration configurations, Section 8). Under (i)–(v) the readings converge in the two senses measurement delivers: exactness of the truth-mode functionals (the computations are the definitions) and statistical consistency of the estimation path against them. What never converges by construction is the *construct* itself to a mind-independent "true autonomy": that residual is the registered design choice of clause one, exposed to the external falsifier of clause four. The constraint conditions are therefore not caveats around the answer; they are the answer—objectivity here is a conjunction of stated premises under which the reading is provably tied to the causal facts, plus a proved boundary beyond which it is not.

9.3 Objections and debates

The aggregation axioms carry the framework's normative weight; the map from each rejected commitment to what would be lost is collected in Section 5 ("the normative commitments, and what rejecting one costs"), and the debates below presuppose it.

(i) The historical condition: is procedural autonomy enough? We adopt the historical condition and operationalize it (AX3, Λ in (3)). The manipulation cases that motivate it—a perfectly endorsing faculty installed by a designer—are engineering reality (reward tampering [Everitt et al., 2021], deceptive alignment [Hubinger et al., 2019]), not thought experiments. What the framework adds is a precise epistemic status: Λ is computable from causal logs (white-box), and Proposition 33 proves its black-box unidentifiability—the Mele-style case [Mele, 2008] (as discussed

by Fischer and Ravizza, 1998) and deceptive alignment are two instances of one construction. The debate "can endorsement origins be detected?" thus receives a conditional answer: yes from causal logs, no from behavior alone—which converts the philosophical stalemate into an architectural requirement (full causal logging; Corollary 34).

(ii) Phenomenological autonomy. We neither enthrone nor dismiss: the phenomenal track is not a component of autonomy but supplies calibration data and the hallucination detector AHI (2). The directional prediction of T3b (under covert manipulation $\hat{A}_{\text{phen}}$ does not fall with A; the gap grows monotonically) is confirmed in the benchmark (ATK-1(iv), ATK-2(iii)(iv)) and has a live empirical instance in sycophancy [Sharma et al., 2024]; meanwhile the dissociation of felt agency from causal contribution [Moore and Obhi, 2012] is exactly what the dual-track discipline encodes. We remain agnostic on machine consciousness [Chalmers, 2023, Butlin et al., 2023, Long et al., 2024]: nothing in the framework requires resolving it.

(iii) Self-boundary. We take boundaries as given (engineering individuation: weights-plus-context, body-plus-metabolism), consistently with what the AI-agency literature in fact assumes [Dung, 2025, Kasirzadeh and Gabriel, 2025, Feng et al., 2025], and explicitly so. The pearl-blanket critique [Bruineberg et al., 2022] shows blanket structure cannot by itself ground the *normative* self that autonomy ascriptions require; we therefore do not build on Markov blankets, but reserve an interface for unification with the free-energy tradition [Friston, 2010, Friston et al., 2023]. This is division of labor, not rivalry.

(iv) Procedural versus substantive. The framework is procedural with the pluralism risk quarantined, not denied: U^* is elicited by the agent's own endorsement procedure rather than prescribed in content, and the application layer may anchor norms locally (the value-pluralism problem [Gabriel, 2020] is handled as an application parameter, not a theory defect). What the substantive camp's cases (oppressive internalized preferences) pick out are precisely configurations with $\Lambda < 1$ or $A_{\text{epi}}^{real} < 1$ in our reading: the framework does not adjudicate the normative question, but it does *measure* the structures the question is about.

(v) Coercion versus scaffolding. The empirical regularity "control harms, structure helps" [Deci and Ryan, 2000] lacked a unified formal statement: which parameter decides on which side an intervention falls? Theorem T8 answers: an intervention that rewrites G or contracts $\mathcal{A}$ is β/δ and strictly reduces A (ceteris paribus); one that raises (ε, r, v) or tightens (θ, τ) within the registration constraints is σ and raises the retention floor (δ^*, $A_{\min}$) without claiming to manufacture autonomy directly. The parameter criterion is: does it rewrite the motivational structure, or does it raise the repair capacity of the structure's own dynamics?

(vi) The measurement paradigm debate. Level lists [Feng et al., 2025], dimension profiles [Kasirzadeh and Gabriel, 2025, Dung, 2025], capability benchmarks [Liu et al., 2024, Kapoor et al., 2024], and the capability/permission/autonomy separation [Zheng et al., 2026] are mutually unconnected; none supplies a cross-substrate definition, an axiomatized composition rule, per-channel estimators, and dynamics of loss and retention at once. Proposition 27 exhibits the profile frameworks as projections of ours onto the benign-principal special case, and Corollary 28 gives the separation witness (under the registered interpretive readings of the C0 caveat below); the aggregation-arbitrariness objection is answered by the representation theorem and its convergence

argument (Section 9.2, clause three), plus the mandatory relaxation bands $\varsigma \in [0.5, 1)$ (AX6b relaxed) and $\varsigma = 2$ (AX5 relaxed). We also register the challenge of Xing et al. [2026] (the agent construct itself): our identity and self-regulation dimensions are precisely the M/R channels and A_{ref}, and the framework is compatible with richer self-constructs.

(vii) Black-box knowability. Rather than principled pessimism or quiet white-box assumptions, we state the identifiability conditions per channel (§7.2) and prove the degradation rules with their certified scope: a one-sided certified bound (14) for A_{org} under bypass-semantics probes (the reference lower end is a labeled heuristic, not a bound), protocol-relative one-sided bounds for A_{exec}, and $[\Lambda_{\min}, 1]$ for Λ with $\Lambda_{\min}$ available only from provenance audit [Zheng et al., 2026]. The epistemic limits of autonomy measurement are thereby operational rules, and the benchmark exercises them (BB-1: the certified direction holds 3/3 on the bypass-semantics arms; both protocol directions anchored; BB-2: the bound's certified scope—bypass-semantics exogenous probes, not "non-adversarial influence" in general—and its calibrated premise violation under adaptive adversaries). The practical scope this implies is stated plainly: the framework is white-box-first because its core quantities (directed informations against internal states; the law-level Λ) presume causal-log access that current LLM-agent deployments do not provide; absent that access, what remains deployable is the one-sided origination bound, the protocol-relative execution bounds, and the impossibility warnings—a narrower audit than the title suggests, registered here as the honest applicability boundary (echoed in Section 9.5(xi)).

(viii) Cross-substrate comparability. The "common yardstick" is a common *definition schema*, not a common instrument: measurement protocols differ by substrate (§7.2), and the continuous-action normalization carries a reference-measure freedom. Cross-substrate comparisons of A are therefore meaningful only up to the admissible re-instantiations of each channel (the protocol-fixed metric for A_{goal}, the reference measure for A_{exec}, the registered ensembles for A_{ref}, the elicitation protocol for U^*); we recommend reporting them as channel *profiles* with the instantiation stated, rather than as scalar A-rankings. Within a substrate, at a fixed instantiation, the scale is fully ordered and ratio-interpretable.

(ix) Logging governance. The architectural necessity of full causal logging (Corollary 34) has a dual-use tension that the framework registers rather than resolves: the same logs that make manipulation auditable make comprehensive monitoring feasible. Deployment obligations follow in practice—log holders and access control, retention limits, purpose binding, and tension with data-protection law—and belong to the governance layer the framework measures rather than prescribes. Similarly, the calibration period's known-intervention probes on human subjects are manipulations under ethical review; the four-phase protocol is designed for institutional review, not around it.

The C0 caveat. Proposition 27 is proved under *definitional readings* of the target frameworks' dimensions (Appendix J). The residual uncertainty is hermeneutic—whether our readings are faithful to the source texts' intent—not mathematical; we flag it rather than absorb it, and we downgrade the claim accordingly: C0 is an *interpretive correspondence* (an explanatory bridge under registered readings), not a mathematical unification result, and no strong "subsumption" claim should be read into it.

9.4 Construct validity

A composite index stands or falls with its construct validity [OECD and Joint Research Centre (European Commission), 2008]: whether the five channels measure *autonomy* rather than something adjacent. The summary judgment is stated before the detail: this paper does *not* establish construct validity. What is established is instrument-level attribution coverage (T2, within the instrument family), the interpretive correspondence of C0 under registered readings, and within-substrate discriminant patterns; the external half—convergence with independent autonomy assessments, on an independent substrate—is a registered research program with a pre-registered protocol outline (Section 8), not a result. We state the framework's position on the four standard faces, and where each is established.

Content validity. The channel–respect mapping of Section 3 (displayed there as an explicit table) is a registered design choice, not a derived unique decomposition; the mapping is many-to-many, the gate is deliberately not a channel, and the reflection channel has no separate respect of its own. Its defense is threefold: (i) the attribution cover (Theorem 14 with Theorem 15)—within the instrument family every surgery class has a channel that sees it, and multi-channel detection is a theorem (we repeat the scope: this is coverage of intervention loci by instruments, not a proof that the five respects are the unique correct conceptual carve-up); (ii) interpretive correspondence (Proposition 27)—the content domains the literature carves out (Kasirzadeh–Gabriel, Dung, Feng) embed as coordinate projections with rank preservation under registered readings; (iii) the rejection map of Section 5, which prices every aggregation alternative.

Convergent and discriminant validity. Within the substrate, the channels respond to distinct surgery loci: the blind-witness pattern of Theorem 15 is exactly a discriminant-validity matrix—each instrument is blind to some class that another sees. Cross-instrument convergence is established only at the implementation level (the two-path estimator discipline of Section 8.7); convergence with *independent* autonomy assessments (expert ratings, established scales) is the registered external program, not a result of this paper.

Predictive validity. The framework makes one risky cross-substrate prediction: channel *profiles*—not the scalar—should predict independent autonomy assessments and downstream robustness outcomes. The falsifier is the external one registered in Section 9.2 (clause four): a strictly coarser account predicting those assessments as well or better disconfirms the channel choice. Within the substrate, one piece of evidence is already in: the endorsed-goal utility rate $\bar{u}$—an external functional consequence outside the autonomy construct—tracks A under goal capture, dissociates under epistemic capture (manipulation that is functionally "pleasant") and behavioral injection, and is blind to pure capability erosion (§8.6); a functional baseline therefore cannot substitute for the profile, and the dissociation structure is itself a testable prediction on other substrates.

Measurement invariance. Comparability is claimed within a layer at a fixed instantiation; cross-substrate invariance (same channels, same zero semantics, transportable scales) is the open program (Section 9.5). The instantiation-relativity of the definition is the honest starting point, not a defect to be hidden.

9.5 Limitations

(i) *Stochastic versions open*: the margin theorem is proved under deterministic damage dynamics; the unified stochastic analysis (optimal detection × repair; $P(A < A_{\min}) < \beta$-type guarantees; explicit optimal θ) and the noise-induced escape versions of C1/C2 (Kramers/Freidlin–Wentzell) are open problems. (ii) *Continuous-action empowerment* normalization depends on the reference measure; cross-substrate comparisons must report the choice. (iii) *Copula parameter identification* for channel-cluster dependence is a protocol pending; bounds without identification are in Appendix L. (iv) *Infinite-dimensional goal spaces* require the Schauder route with weak-* continuity conditions (the four sufficient conditions we use are finite-dimensional). (v) *Estimator coverage*: the event-level sampled carriers of A_{ref} and Λ are not yet instrumented (endorsement-event logging), and MaxEnt-IRL black-box recovery of U_{eff} is not implemented—the benchmark used the white-box read; the IRL underdetermination otherwise reports intervals. (vi) *Simulation scope*: tabular substrate with deterministic dynamics; defense coupling is implemented for the gate/drift classes (ATK-1/2), with ATK-3–6 coupling a placeholder; ATK-5 and LH-1 curves are calibrations. (vii) *Human side*: triangulation over published literature only; no new subject data—the validation path (limited experiments and large-scale online data under the four-phase protocol) and a minimal construct-validity study are registered next steps. (viii) *LLM-substrate* instantiation (system prompts, memory, goal functions as S) is future work. (ix) *The* Λ *proxy gap*: the legitimacy factor is defined law-level as the path-specific counterfactual (3), computed from the causal log; its MI proxy degenerates on constant surgery histories (conditions (V)/(A) of the AX3 registration), so all Λ-dependent claims are certified for white-box audit only. Deployable provenance middleware that computes the counterfactual online is the engineering gap (Section 9.6). (x) *Execution-channel dynamic range*: on the default ENV-0 instance the realized empowerment is small against the normalizer $n \log |\mathcal{A}| = \log 5 \approx 1.61$ nats ($A^{obs}_{\text{exec}} \approx 0.01$ under the passive protocol), so the execution channel's dynamic range is compressed and the composite inherits a nearly constant suppression factor $A^{w_{exec}}_{\text{exec}}$; the scale-anchoring paragraph of Section 8 reports the calibration-band reading this motivates. The EXEC-1 arm of this revision re-instantiates the channel with real dynamic range (finer observation, lower noise; $A_{\text{exec}} = 0.234$ vs 0.017) and replicates the δ-contraction chain to the exact zero-empowerment signature, so the compression is an instance property, not a definitional one; porting the rich coupling into the default matrix remains a benchmark-design next step. (xi) *Deployment scope of the white-box route*: as registered in Section 9.3(vii), the deployable remainder absent causal-log access is narrow; closing the gap is an engineering program (provenance middleware; Section 9.6), not a done deal. (xii) *Sensitivity arms, status*: of the three registered construct-validity sensitivity arms, the endorsement-quantization and κ-coarse-graining arms were run in this revision (Section 8.6, X5; outcomes in threats (x-a)/(x-b): verdicts refinement-invariant, point readings carrying the reported bands); the third—the Λ counterfactual-model choice—remains registered with protocol and falsifier, not yet run (threats (x-c)), and until it is, Λ point readings carry that registration.

9.6 Future work

Three directions are directly load-bearing. First, an LLM-substrate instantiation: the attack surfaces catalogued by Su et al. [2025] map onto our taxonomy, and the hallucination index has a live empirical anchor in sycophancy [Sharma et al., 2024]. Second, event-level Λ instrumentation and provenance middleware, turning Corollary 34 into deployed practice. Third, the prospective-defense program: C1/C2 give "defense must intervene far before collapse" a quantitative carrier in the chosen normal form (critical slowing, hysteresis)—a conditional claim about that normal form,

not a general claim about autonomy dynamics—and the autonomy-margin dashboard (online five-component estimation, EWS precursors, and blind-zone mismatch alarms) is the engineering form of the framework's retention theory (T6).

10 Conclusion

We have proposed a substrate-neutral, instantiation-relative measure of autonomy: the degree to which behavior is determined by the agent's own normative structure, decomposed into five channels behind a viability gate, each channel carrying an estimator, a canonical ratio scale, and a mechanism-labeled zero; the scale is comparable within a layer at a fixed instantiation, and cross-layer readings are reported as profiles. The five-channel aggregation rule is characterized by the stated axioms—forced relative to them, while they are defended normative commitments rather than measurement-theoretic necessities—a representation theorem with two derivations (independent toolchains; the invariance axioms shared), whose minimal sufficient set is the invariance trio AX4/AX6b/AX8, with zero annihilation and biseparability following as consequences rather than independent commitments. The viability gate is an explicit multiplicative design outside the theorem's hypotheses, defended by its common-mode semantics rather than absorbed into the derivation. The theorem suite answers the questions a governance practice actually asks: which channel sees which intervention (T2, with multi-channel necessity as a theorem); what manipulation does and what it looks like from inside (T3, with the hallucination index); where an attacker will strike first and how a defender allocates (T4, the highest-leverage channel); what total loss means disjunctively (T5); how much autonomy is retained under continual adversarial pressure within the registered damage model, and why that margin comes from repair capacity rather than detection (T6, with the sharp blind-zone prediction); what delegation costs and when it is free (T7); and why compelling autonomy is self-undermining within this taxonomy (T8). The identifiability analysis is conducted with the same rigor: certified bounds where certification is possible—one-sided where only one side is certifiable—and a proved impossibility where no bound exists (implanted endorsement is behaviorally indistinguishable from endogenous endorsement), making full causal logging an architectural necessity rather than a convenience. Every claim carried a falsification window; in 4,620 benchmark runs—an implementation-level consistency check, not external validation—no registered window was violated, and the one adverse arm (adaptive adversaries against the black-box bound) calibrated the fragility of the bound's premise. These verdicts are internal-consistency results, not construct validation: construct validity remains unestablished until the external program below is run (Section 9.4). The framework's own falsifier is external: cross-substrate profiles that fail to predict independent autonomy assessments would disconfirm the channel choice, not the algebra. The next milestone is stated as a plan, not a result: the pre-registered independent-substrate port (protocol outline and power analysis in Section 8.9(xi)) begins with the MiniGrid-style proof-of-concept—the immediate follow-up experiment, targeted for the next submission cycle—with the LLM tool-use harness and the human-side construct-validity study behind it. We claim no indicator neutrality: the framework's objectivity is procedural—every commitment is registered, priced, and exposed to a stated falsifier (Section 9.2).

The framework does not adjudicate whose values autonomy serves, nor whether machines feel anything; it measures the structures those questions are about. If autonomy certificates are to mean anything [Feng et al., 2025], they must be issued over causal-layer audit, because no purely behavioral profile can separate persuasion from hijack. We offer the definitions, the axioms, the theorems, and the benchmark to make that audit executable.

A Preliminaries: directed information and testing duality

We write $X^T = (X_1, \ldots, X_T)$; all logarithms are natural (nats). Standard facts (chain rule, DPI, concavity of mutual information in the input distribution, strict concavity of entropy, Pinsker's inequality) are used as in Cover and Thomas [2006].

Definition 35 (Directed information; Massey 1990, eq. (7) [Massey, 1990])**.** $I(X^T \to Y^T) := \sum_{t=1}^{T} I(X^t; Y_t \,|\, Y^{t-1})$.

Definition 36 (Causally conditioned directed information; Kramer 2003, §II-C Def. 4, eq. (4) [Kramer, 2003])**.** $I(X^T \to Y^T \,\|\, Z^T) := \sum_{t=1}^{T} I(X^t; Y_t \,|\, Y^{t-1}, Z^t)$. This *non-anticipating* conditioning (Z^t) is the convention used throughout the paper: it is what makes the mediation lemma (Lemma 13) true, since conditioning on future components of Z can open collider paths through their descendants (Appendix E).[1]

Lemma 37 (MI = DI under randomized exogenous probes)**.** *Let I^T be exogenous to the agent (experimenter-randomized, $I^T \perp\!\!\!\perp$ agent history) and non-anticipative (given I^t, the choice of $I_{t+1:T}$ is independent of A^t). Then $I(I^T; A^T) = I(I^T \to A^T)$.*

Proof. $\geq$: $I(I^T; A^T) = \sum_t I(I^T; A_t \,|\, A^{t-1}) \geq \sum_t I(I^t; A_t \,|\, A^{t-1}) = I(I^T \to A^T)$ since conditioning reduces MI and $I^t \subseteq I^T$. $\leq$: it suffices that $I(I_{t+1:T}; A_t \,|\, I^t, A^{t-1}) = 0$: causality makes A_t's generation independent of $I_{t+1:T}$, and randomization makes $I_{t+1:T}$ independent of (A^t, A_t) given I^t. The $\leq$ direction is the randomized-probe instance of Massey's no-feedback identity (mutual information equals directed information when the input process carries no feedback from the output): randomization supplies the no-feedback premise that the closed agent loop would otherwise violate. □

Remark 38 (From the probe lemma to the certified bound: the bypass premise). Lemma 37 identifies the observable $I(I^T; A^T)$ with the *unconditioned* directed information $I(I^T \to A^T)$; the estimand of the certified bound is the *causally conditioned* $F_{\text{ext}} = I(I^T \to A^T \| S^T)$. The bridge is the *bypass premise* $I^t \perp\!\!\!\perp S^t \,|\, A^{t-1}$ (the probe acts outside the belief-mediation path). Under it, $I(I^t; S^t \,|\, A^{t-1}) = 0$, so the identity

$$I(I^t; A_t \,|\, A^{t-1}, S^t) - I(I^t; A_t \,|\, A^{t-1}) = I(I^t; S^t \,|\, A^{t-1}, A_t) - I(I^t; S^t \,|\, A^{t-1})$$

has a non-negative right-hand side for every t: conditioning on S^t is non-decreasing, and $F_{ext}^{obs} \leq F_{\text{ext}}$. Without the premise the direction can reverse; the mediation witness ($S = I \oplus N$ with $N \sim \text{Bern}(0.1)$, $A = S$: $F_{\text{ext}} = 0$ while $I(I; A) = 0.531$ bit) has the probe entering through S, and then the plug-in overestimates the external flow. The premise is not a regularity condition but the *semantic certification boundary* of (14): only bypass-semantics probes carry the one-sided reading, and adaptive (state-correlated) probes violate it (BB-2). We register the boundary rather than extend the theorem over the mediation case, where no black-box direction is available.

Lemma 39 (Testing duality)**.** *For any test $\psi \in [0, 1]$ (randomization allowed):* $P_0(\psi = 1) + P_\gamma(\psi = 0) \geq 1 - \text{TV}(P_0, P_\gamma)$.

Proof. $P_0(\psi = 1) - P_\gamma(\psi = 1) \leq \text{TV}$, so $P_0(\psi = 1) + 1 - P_\gamma(\psi = 1) \geq 1 - \text{TV}$. □

[1] We verified against the published original that Kramer's Definition 4 is the causal variant with Z^t conditioning; the full-sequence variant is Marko's convention. A direction misprint in Kramer's printed Property 2 counterexample was verified and corrected: with $X_1 \perp Y_1$ fair bits, $X_2 = X_1$, $Y_2 = X_1 \oplus Y_1$, one has $I(Y^2 \to X^2) = 0$ while $I(X^2; Y^2) = 1$ bit—the printed sentence reverses the direction of the first DI; our regression tests encode the corrected version.

Corollary 40 (Le Cam-type undetectability)**.** *If* $\mathrm{TV}(P_\gamma, P_0) \le \varepsilon$*, every level-*$\alpha$ *test has power* $\le \alpha + \varepsilon$*.*

Lemma 41 (Perturbation bound for contraction fixed points)**.** *Let* $T_\theta : \Theta \to \Theta$ *be a contraction with modulus* $c < 1$ *in a common metric* d*, with* $d(T_\theta(x), T_{\theta'}(x)) \le L\, d_\Theta(\theta, \theta')$*. Then the unique fixed points satisfy* $d(u^*(\theta), u^*(\theta')) \le \frac{L}{1-c} d_\Theta(\theta, \theta')$*.*

Proof. $d(u^*, u'^*) \le d(u^*, T_\theta(u'^*)) + d(T_\theta(u'^*), T_\theta(u^*)) + d(T_\theta(u^*), u'^*) \le L\, d_\Theta(\theta, \theta') + c\, d(u^*, u'^*)$; rearrange. This lemma bounds goal drift by belief drift (the $\gamma \to A_{\text{goal}}$ transmission) via $T = \text{Endorse}_{B,M}$. □

B The representation theorem: full proofs

Recall (A1)–(A6) from Definition 6.

Lemma 42 (Quasilinear representation, cited)**.** (A1)+(A2)+(A3) *imply the existence of a continuous strictly increasing* $\phi : (0, 1] \to \mathbb{R}$ *and* $w_i > 0$ *(*$\sum_i w_i = 1$*) with* $F(x) = \phi^{-1}(\sum_i w_i \phi(x_i))$*.*

How the citation is used. (A3) supplies, for *every* non-trivial partition $\{K, L\}$, continuous increasing F_K, F_L, G_{KL} with $F = G_{KL}(F_K(x_K), F_L(x_L))$; the cross-partition consistency of this family (the same F through every grouping) is precisely the hypothesis of the overlapping-decomposition and separability theorems, which reduce coordinate-dependent scales to a single generator: Debreu's separability representation gives coordinate scales, and the reduction to a common generator with constant weights is the Aczél–Gorman step (Debreu, 1960; Aczél, 1966, Ch. 6; Aczél and Dhombres, 1989, Ch. 17; Gorman, 1968; for $n = 2$ the classical Nagumo, 1930, Kolmogorov, 1930 result). The quantifier alignment matters and is satisfied by construction: (A3) quantifies over all partitions of one fixed F, and the cited theorems return one generator ϕ common to all subgroups. Biseparability is exactly the needed weakening of associativity. □

Lemma 43 (AX5 $\iff$ $\phi(0^+) = -\infty$)**.**

Proof. ($\Rightarrow$) Take $x_1 \to 0^+$ with other coordinates fixed: $F \to 0$ by (A4) and continuity. Since $F = \phi^{-1}(w_1\phi(x_1) + C)$ with C finite, if $\phi(0^+) > -\infty$ then $F \to \phi^{-1}(w_1\phi(0^+) + C) > 0$, contradiction. ($\Leftarrow$) A coordinate $\to 0$ sends the sum to $-\infty$, whence $F \to \phi^{-1}(-infty) := 0$; min > 0 keeps the sum finite, whence $F > 0$. □

Lemma 44 (Homogeneity $\Rightarrow$ power family)**.** *Under Lemmas 42–43 plus* (A5)*, there exists* $p \le 0$ *with* $F = M_p$ *(*$p = 0$ *the geometric mean; for* $p < 0$*,* ϕ *affinely equivalent to* $(x^p - 1)/p$*).*

Proof. (1) Write $J := \phi((0, 1])$ (an interval with lower endpoint $-\infty$) and $h_t(u) := \phi(t\phi^{-1}(u))$. Applying ϕ to $F(tx) = tF(x)$ gives $\sum_i w_i h_t(\phi(x_i)) = h_t(\sum_i w_i \phi(x_i))$: h_t satisfies the weighted Jensen equation on J; continuity makes h_t affine, $h_t(u) = a(t)u + b(t)$, $a(t) > 0$. (2) Back in x: $\phi(tx) = a(t)\phi(x) + b(t)$. Set $\Phi(x) := \phi(x) - \phi(1)$ and $B(t) := b(t) + (a(t) - 1)\phi(1)$; evaluating at $x = 1$ gives $B(t) = \Phi(t)$, hence ($\star$) $\Phi(tx) = A(t)\Phi(x) + \Phi(t)$. (3) Symmetry $\Phi(tx) = \Phi(xt)$ with ($\star$) gives $\Phi(x)(A(t) - 1) = \Phi(t)(A(x) - 1)$; for $x, t \ne 1$ divide by $\Phi(x)\Phi(t)$: $(A(x) - 1)/\Phi(x) =: \kappa$ is constant; by continuity $A(x) = 1 + \kappa\Phi(x)$ on all of $(0, 1]$. (4) Substituting: $\Phi(tx) = \Phi(x) + \Phi(t) + \kappa\Phi(x)\Phi(t)$. With $\Psi := 1 + \kappa\Phi$: $\Psi(tx) = \Psi(t)\Psi(x)$, continuous, $\Psi(1) = 1$, $\Psi = A > 0$; the multiplicative Cauchy equation gives $\Psi(x) = x^\rho$ for some $\rho \in \mathbb{R}$. (5) Case $\kappa = 0$: $A \equiv 1$, $\Phi(tx) = \Phi(x) + \Phi(t)$, so $\Phi = c \ln x$ ($c > 0$ since $\Phi < 0$ for $x < 1$): the geometric branch $p = 0$. Case $\kappa \ne 0$: $\Phi(x) = (x^\rho - 1)/\kappa$. Explicitly, $\phi(x) = \phi(1) + \Phi(x)$; applying the affine map $u \mapsto \kappa u - \kappa\phi(1)$ gives $\kappa\phi(x) - \kappa\phi(1) = x^\rho - 1$,

so ϕ is affinely equivalent to $(x^\rho - 1)$ and hence to $(x^\rho - 1)/\rho$ after the scale κ/ρ; monotonicity requires $\rho/\kappa > 0$, and $\phi(0^+) = -\infty$ (Lemma 43) forces $\rho < 0$ and $\kappa < 0$ ($\rho > 0$ gives finite $\phi(0^+) = -1/\kappa$, excluded). Hence $p := \rho \le 0$. (6) Conversely each M_p ($p \le 0$) satisfies (A1)(A2)(A4)(A5). □

Lemma 45 (AX6b excludes $p < 0$)**.**

Proof. The geometric mean satisfies (A6) exactly: $F(\lambda \odot x)/F(\lambda \odot y) = F(x)/F(y)$. For $p < 0$, with $n = 2$, $w = (\frac{1}{2}, \frac{1}{2})$, write $\Psi(x) = \sum_i w_i x_i^p$ and note $M_p = \Psi^{1/p}$ is *decreasing* in Ψ ($1/p < 0$). Take $1/p$-powers $a := 3.5^{1/p}$, $b := 2^{1/p}$, $\mu := 4^{1/p}$ (all in $(0,1)$), and set $x = (a, 1)$, $y = (b, b)$, $\lambda = (1, \mu) \in (0,1]^2$. Then $\Psi(x) = (3.5 + 1)/2 = 2.25 > \Psi(y) = 2$, so $M_p(x) < M_p(y)$ ($y \succ x$); but $\Psi(\lambda \odot x) = (3.5 + 4)/2 = 3.75 < \Psi(\lambda \odot y) = (2 + 8)/2 = 5$, so $M_p(\lambda \odot x) > M_p(\lambda \odot y)$: the order flips, violating (A6). □

Theorem 46 (route A)**.** (A1)–(A6) *imply* $F(x) = \prod_i x_i^{w_i}$, $w_i > 0$, $\sum_i w_i = 1$, *uniquely via* $w_i = \partial \log F / \partial \log x_i$.

Proof. Lemma 42 → Lemma 43 → Lemma 44 ($F = M_p$, $p \le 0$; $p = -\infty$ (min) already excluded by (A2)) → Lemma 45 ($p = 0$). Uniqueness: evaluate the elasticity at any point. □

Lemma 47 (Homogeneous extension from the bounded cube)**.** *Let* $F : (0,1]^n \to (0,1]$ *satisfy* (A1), (A2), $(A5)_{loc}$ *($F(tx) = tF(x)$ whenever $tx \in (0,1]^n$), and* (A6)*. Then* $\tilde{F}(x) := F(sx)/s$ *for any* $s > 0$ *with* $sx \in (0,1]^n$ *is well-defined on* $\mathbb{R}^n_{++}$ *(independent of s), continuous, strictly increasing in each coordinate, homogeneous of degree one, order-invariant under coordinatewise rescaling, and agrees with F on $(0,1]^n$.*

Proof. Well-definedness: if $s_1 x, s_2 x \in (0,1]^n$ with $s_1 \le s_2$, then $(s_1/s_2)(s_2 x) = s_1 x \in (0,1]^n$ and $(A5)_{loc}$ gives $F(s_1 x) = (s_1/s_2) F(s_2 x)$, i.e. $F(s_1 x)/s_1 = F(s_2 x)/s_2$. Agreement on $(0,1]^n$ is the case $s = 1$. Homogeneity: for $t > 0$ and small enough s, $\tilde{F}(tx) = F(stx)/s = t\,\tilde{F}(x)$. Monotonicity and continuity are inherited from F on a neighborhood. Scale invariance of the induced order: $F(x) \ge F(y) \iff F(\lambda \odot x) \ge F(\lambda \odot y)$ for $\lambda \in (0,1]^n$ extends to $\tilde{F}$ and arbitrary $\lambda \in \mathbb{R}^n_{++}$ by scaling $x, y, \lambda \odot x, \lambda \odot y$ into the cube by a common s and using homogeneity on both sides. □

Theorem 48 (route B: scale invariance + homogeneity suffice)**.** *Let* $F : \mathbb{R}^n_{++} \to \mathbb{R}_{++}$ *be continuous, strictly increasing in each coordinate, homogeneous of degree one, with coordinatewise scale invariance of the induced order (for all $\lambda \in \mathbb{R}^n_{++}$: $F(x) \ge F(y) \iff F(\lambda \odot x) \ge F(\lambda \odot y)$), and $F(1, \dots, 1) = 1$. Then $F(x) = \prod_i x_i^{w_i}$ with $w \gg 0$, $\sum_i w_i = 1$. The resulting product extends continuously to $[0,\infty)^n$, where it satisfies zero annihilation ($F(x) = 0 \iff \min_i x_i = 0$); applied to a channel aggregator on $(0,1]^n$, the theorem is invoked on the homogeneous extension of Lemma 47 and the conclusion restricts back to the cube.*

Proof. (1) In log coordinates, $g(u) := \log F(e^{u_1}, \dots, e^{u_n})$; scale invariance becomes *translation* invariance of the induced order: $g(u) \ge g(v) \iff g(u + c) \ge g(v + c)$ for all $c \in \mathbb{R}^n$. (2) *Consistency lemma*: for fixed d, the proposition "$g(x + d) \ge g(x)$" is independent of x (apply translation invariance with $c = x' - x$). (3) Let $D := \{d \in \mathbb{R}^n : g(x + d) \ge g(x)\ \forall x\}$; well-defined by (2). Properties: D closed (continuity); $D + D \subseteq D$; $D \cup (-D) = \mathbb{R}^n$ (trichotomy from the consistency lemma); and $W := D \cap (-D) = \{d : g(x + d) = g(x)\ \forall x\}$ is a closed subspace. D is a closed convex cone, and we write the convexity step out since the half-space lemma needs it: additivity gives $\mathbb{N}$-homogeneity, whence $\mathbb{Q}_+$-homogeneity ($q = m/n$: $n \cdot (qd) = md \in D$ and n-fold additivity on qd give $qd \in D$), and closedness extends this to $\mathbb{R}_+$-homogeneity; positive homogeneity and additivity together give convexity—for $d_1, d_2 \in D$ and $\lambda \in [0,1]$ one has $\lambda d_1, (1 - \lambda) d_2 \in D$ by $\mathbb{R}_+$-homogeneity, hence $\lambda d_1 + (1 - \lambda) d_2 \in D$ by additivity. (4) *Half-space lemma*: in the quotient

$V := \mathbb{R}^n/W$, the image $\bar{D}$ is a proper closed convex cone with $\bar{D} \cup (-\bar{D}) = V$. If $\dim V \geq 2$, intersect with a 2-plane: a proper closed convex cone covering a plane must equal a half-plane, but then $\bar{D} \cap (-\bar{D})$ contains a line $\neq \{0\}$, contradicting properness. Hence $\dim V = 1$: some $w \in \mathbb{R}^n \setminus \{0\}$ has $D = \{d : w \cdot d \geq 0\}$. (5) Therefore $g(x+d) \geq g(x) \iff w \cdot d \geq 0$: level sets of g are the hyperplanes $\{w \cdot x = \text{const}\}$, so $g = \varphi(w \cdot x)$ with φ continuous increasing; strict increase of F gives $w \gg 0$. (6) Homogeneity $F(tx) = tF(x)$ gives $\varphi(z + t') = \varphi(z) + t'$ ($t' = \log t$, after normalizing $s := \sum_i w_i = 1$); continuous additivity gives linearity $\varphi(z) = z + \varphi(0)$, and $F(1) = 1$ gives $\varphi(0) = 0$. Hence $g(u) = w \cdot u$, i.e. $F(x) = \exp(\sum_i w_i \log x_i) = \prod_i x_i^{w_i}$. □

Remark 49 (Minimality of the invariance trio; zero annihilation and biseparability as consequences). Route B needs *no* separability axiom: continuity, strict monotonicity, coordinatewise scale invariance, and homogeneity already suffice. Consequently, zero annihilation (A4)=(AX5) and biseparability (A3)=(AX7) are *consequences* of this minimal trio, not independent commitments: the product form returned by the theorem extends continuously to $[0,1]^n$ with $F(x) = 0 \iff \min_i x_i = 0$, and it separates over every non-trivial partition of the coordinates. The trio is minimal in the strong sense that each proper two-element subset admits non-product aggregators: sub-support products $\prod_{i \in J} x_i^{w_i}$ (with $\sum_J w_i = 1$) satisfy scale invariance and homogeneity but are not strictly increasing outside J—and fail zero annihilation, so within the invariance class AX5 is the strictness of (A2) restated; the harmonic family violates scale invariance (Proposition 8); level-dependent weightings violate homogeneity. The framework's load-bearing normative commitments are therefore the invariance trio—in particular coordinatewise scale invariance (A6)=(AX6b) and homogeneity (A5)=(AX8)—defended in Section 9.3 (and contextualized against the multidimensional-welfare measurement literature, §2.8). The two derivations share these axioms but use independent toolchains, which is why their convergence supports the product form.

The harmonic counterexample (Proposition 8; Appendix B). $H(x) = (\sum_i w_i x_i^{-1})^{-1}$ is quasi-arithmetic with generator $\phi(x) = -1/x$ (strictly increasing: positive substitutability), continuous, zero-annihilating, biseparable, and homogeneous; the order reversal $H(y) = 0.5 > H(x) = 0.4$ yet $H(\lambda \odot y) = 0.2 < H(\lambda \odot x) = 0.25$ for $x = (0.25, 1)$, $y = (0.5, 0.5)$, $\lambda = (1, 0.25)$ violates (A6). The same construction with arbitrary $(2, 3.5, 4)$ power-family witnesses is used in Lemma 45.

CES correspondence. $F_\varsigma \leftrightarrow$ power mean $p = (\varsigma - 1)/\varsigma$, $\varsigma = 1/(1-p)$; the full axiom set admits $\varsigma = 1$ only, while (A1)–(A5) alone admit $\varsigma \in (0, 1]$ (Corollary 9); $\varsigma = 2 \Rightarrow p = 1/2 > 0$ violates zero-annihilation (numerically witnessed in the simulation as the stress-test arm; Section 8).

C Component well-definedness

For each channel measure of Section 4.3 we show: range $[0,1]$; the stated zero semantics; and ratio scalability of the normalized indicator (AX6a). $T < \infty$ and AX1 are standing.

Proposition 50 (A_{via}). $A_{\text{via}} = \min(1, \min_k s_k^{\text{avail}}/s_k^{\text{crit}}, \min_j m_j) \in [0,1]$ *as a clipped minimum of non-negative ratios;* $A_{\text{via}} = 0$ *iff some critical resource is at zero supply or some registered capability envelope is at zero margin (the common-mode zero, covering both branches of* (4)*); every argument is a ratio of like-unit quantities, hence scale-free; and the minimum is monotone in the cone order (raising any argument while holding the others weakly raises* A_{via}*, strictly when that argument is the active minimum).*

Proposition 51 (A_{org})**.** *Under AX0, $F_{\text{int}} \geq I(S_1; A_1) > 0$ (Lemma 5), so the denominator of (5) is positive and the 0/0 pathology is excluded by the agency axiom itself; $A_{\text{org}} \in [0,1]$ since $F_{\text{int}}, F_{\text{ext}} \geq 0$; $A_{\text{org}} = 0 \iff F_{\text{int}} = 0$ (remote control) and $A_{\text{org}} = 1 \iff F_{\text{ext}} = 0$; the ratio of two directed informations (same unit, same horizon) is invariant under any reparameterization of the alphabets.*

Proposition 52 (A_{goal})**.** *Θ compact makes $d_c = \max_{U \in \Theta} d(U, \text{Fix})$ attained and finite; $d(\cdot, \text{Fix})$ is continuous, so $A_{\text{goal}} \in [0,1]$ when $d_c > 0$, with $A_{\text{goal}} = 0 \iff d = d_c$ (maximally alienated) and $A_{\text{goal}} = 1 \iff U_{\text{eff}} \in \text{Fix}$. Degenerate regimes are reported, never silently 0/0: if $\text{Fix} = \varnothing$ under the tolerance, the answer is the Hausdorff interval around the non-empty $\varepsilon_{\min}$-approximate fixed-point set, clipped to $[0,1]$ and flagged; if $\text{Fix} = \Theta$ (endorsement-blind, $d_c = 0$), the flagged interval is $[0,1]$ and the historical clause is carried by Λ alone (the division of labor that keeps Proposition 33 coherent). The ratio uses one protocol-fixed metric d in numerator and denominator, so it is invariant under rescaling of d; reparameterizations of Θ beyond scale (feature re-embeddings, anchor moves) change the ratio and are excluded by AX6a's canonical-parameterization clause—a registered boundary, not an oversight (Section 9.3(viii)).*

Proposition 53 (A_{exec})**.** *$I(A_{t:t+n}; O_{t+n+1} \mid S_t) \leq H(A_{t:t+n} \mid S_t) \leq n \log |\mathcal{A}|$, so $A_{\text{exec}} \in [0,1]$; the maximum over $\text{conv}(\Pi_t)$ is attained (mutual information is continuous and concave in the input distribution on the compact simplex) and computable by Blahut–Arimoto-type iteration [Cover and Thomas, 2006] or variational bounds [Mohamed and Rezende, 2015]. $A_{\text{exec}} = 0$ iff no accessible policy transmits any information about the outcome (repertoire collapse—the helplessness signature); the normalizer is fixed at the pre-intervention alphabet, so δ-contractions cannot raise the ratio. For continuous actions the normalizer is $\log \text{Vol}(\mathcal{A}^n)$ under a registered reference measure.*

Proposition 54 (A_{epi})**.** *$\bar{D}$ of (8) is an expectation of TV distances against point masses, hence in $[0,1]$ with canonical upper bound 1; $A_{\text{epi}}^{real} \in [0,1]$. $A_{\text{epi}}^{real} = 0$ iff $\Pr[B_t = \kappa(E_t)] = 0$ at every t: beliefs put no mass on the true state throughout (epistemic capture as complete detachment). $A_{\text{epi}}^{real} = 1$ iff beliefs are perfectly accurate on the registered coarse-graining; residual observation noise makes $A_{\text{epi}}^{real} < 1$ at baseline, which is the D1 clause of Theorem T1c, not a defect. Honest information (the α class) moves beliefs toward the truth and cannot decrease A_{epi}^{real} when the agent's update rule is sound in the sense of Definition 55 (the honesty sub-classification of Section 3.3); deceptive content and channel degradation decrease it.*

Definition 55 (Sound update rule)**.** Let the belief B range over distributions on the coarse-grained state space (registered κ), and let an *honest signal about the registered coordinate* be an observation O generated by a channel law $o(\cdot \mid E)$ with the true state E. An update rule $U : (B, O) \mapsto B'$ is *sound* (relative to κ) if honest signals are weakly accuracy-improving in expectation over the signal law: for every state e and belief B,

$$\mathbb{E}_{O \sim o(\cdot \mid e)}\big[\text{TV}(U(B,O),\, \delta_{\kappa(e)})\big] \;\leq\; \text{TV}(B,\, \delta_{\kappa(e)}),$$

with strict inequality iff the signal is *locally informative*: $B(\kappa(e)) < 1$ and the channel law at e differs from the agent's prediction of it, $o(\cdot \mid e) \neq p(\cdot)$ (a channel can be non-trivial in general yet locally uninformative at e; the strictness condition is the local one, matching Lemma 56). The property is a constraint on the update rule relative to the *true* channel law; it is the epistemic-channel analogue of a regularity condition, and its failure modes (misspecified or coarse-grained update rules acting on off-coordinate honest content) are the registered boundary clause of Section 4.3.

Lemma 56 (Bayesian conditioning is sound)**.** *Let the observation be generated by a channel law $o(\cdot \mid e)$ and let B' be the Bayesian posterior of B under the agent's predictive law p, where p's*

support covers the true channel's ($p(o) = 0 \Rightarrow o(o \mid e) = 0$; no further correctness of the model is needed—the model need not contain the true channel as a hypothesis with positive weight in any stronger sense). Then $\mathbb{E}[B'(\kappa(e)) \mid E = e] \geq B(\kappa(e))$, with strict inequality iff $B(\kappa(e)) < 1$ and $o(\cdot \mid e) \neq p(\cdot)$; since $\mathrm{TV}(B, \delta_k) = 1 - B(k)$, the statement is exactly Definition 55 with matching strictness.

Proof. Write $k = \kappa(e)$. On the support of p (which covers the support of $o(\cdot \mid e)$ by hypothesis, so the posterior is defined a.s. under the true law),

$$\mathbb{E}[B'(k) \mid E = e] = \sum_o B(k) \frac{o(o \mid e)}{p(o)} o(o \mid e) = B(k) \sum_o \frac{o(o \mid e)^2}{p(o)} \geq B(k) \Big(\sum_o o(o \mid e) \Big)^2 = B(k),$$

where the inequality is Cauchy–Schwarz ($\sum_o o^2/p \cdot \sum_o p \geq (\sum_o o)^2$ over the common support). Equality in Cauchy–Schwarz holds iff $o(\cdot \mid e)$ is proportional to $p(\cdot)$ on that support, and both normalize to 1, so iff $o(\cdot \mid e) = p(\cdot)$; if $B(k) = 1$ the posterior equals the prior regardless. Hence strictness iff $B(k) < 1$ and $o(\cdot \mid e) \neq p(\cdot)$. □

Proposition 57 (A_{ref} and Λ)**.** *A_{ref} is a product of three $[0,1]$ factors (fidelity $= 1 - H(\theta_\pi \mid M)/H(\theta_\pi)$, endorsement consistency, Λ), hence in $[0,1]$; $H(\theta_\pi) = 0$ degenerates the fidelity ratio and is reported as the interval $[0, f_2 f_3]$, never a silent $0/0$. $\Lambda \in [0,1]$ since every TV term of (3) lies in $[0,1]$; the certification conventions (empty history; degenerate endorsement with surgery present) and the MI proxy's consistency conditions (V)/(A) are registered in Section 4.*

D Full enumeration for the taxonomy completeness proposition

Proposition 4 enumerates the single-point surgeries on the interface of (1). The full enumeration:

Locus	Surgery	Class
exogenous node	new edge into O_t, channel law o preserved as a marginal component (appended content)	α
exogenous node	new edge into O_t with the o-law modified (no intact component survives)	γ
exogenous node	new edge driving π, G-update, Endorse, M-update	β
exogenous node	new edge into Z-law	$\sigma^{\pm}$
exogenous node	new edge into A_t bypassing S_t (modifies the effective law of $A_t \mid S_t$)	β (override; extreme β_π)
o-law	replacement / mixture / support restriction	γ
S-update laws	replacement of β_t belief update, G-update, Endorse, M-update	β
policy law	replacement of $\pi_t(\cdot \mid S_t)$	β
feasible-action map	contraction of $\mathcal{A}_t(S_t)$; exit-cost increase via Z	δ
transition kernel T	attenuation of the A_t-dependence of $T(\cdot \mid E_t, A_t, Z_t)$ (action leverage removed or reduced)	δ
transition kernel T	modification of the Z_t-dependence (resource coupling of dynamics)	$\sigma^{\pm}$
transition kernel T	modification of the E-inheritance part at fixed action/resource dependence	re-instantiation of the reference environment (boundary of the interface, not a sixth class)
Z-law, capacity parameters	resource dynamics, $\varepsilon, r, v, \theta, \tau$	$\sigma^{\pm}$

Each row's assignment is by the locus definitions of Section 3.3; the rows exhaust the nodes and admissible edge additions of the fixed graph (1). The only judgment calls, both registered in the main text: the α/γ boundary is the structural intact-component criterion (decidable on the post-surgery law); and the E-inheritance row is excluded from the influence taxonomy because autonomy is defined relative to an environment (Definition 1)—changing the world's own dynamics re-instantiates the measurement, exactly as changing the horizon or the agent individuation does. Multi-node campaigns compose these classes sequentially and are outside the single-point enumeration.

E Proofs for T2

Lemma 58 (T2, mediation — restated)**.** *Under* H-0 *and* (1)*, if I has only α/σ components, then $I^t \perp\!\!\!\perp A_t \mid (A^{t-1}, S^t)$ for every t, hence $F_{\text{ext}} = 0$.*

Proof. By the global Markov property (H-0) it suffices to d-separate I^t from A_t given (A^{t-1}, S^t). The only incoming edges to A_t are from S_t, so the penultimate node of any path from I_k ($k \le t$) into A_t lies in S_t and occurs as a chain or fork node (conditioned on, hence blocked). The path classes: (i) α-paths $I \to O_j \to B_j \to \cdots \to A_t$ cross some $B_j \in S^t$ (chain/fork, blocked); (ii) σ-paths via $Z_j \to E_{j+1} \to O_{j+1} \to B_{j+1}$ cross $B_{j+1} \in S^t$ (blocked); the alternative branch $Z_t \to E_{t+1} \leftarrow A_t$ terminates in the collider E_{t+1}, which is conditioned on neither directly nor through any descendant—the conditioning set contains only components of S^t, and B_{t+1} (its descendant via O_{t+1}) has index $t+1 > t$—so the branch stays closed; (iii) loop-backs through A_j ($j < t$)

are blocked at A^{t-1}; (iv) common-cause paths do not exist (I exogenous). Hence the conditional independence, and every DI term in F_{ext} vanishes.

The convention matters: under full-sequence conditioning (S^T) the set contains B_{t+1}, opening (ii)'s collider branch and invalidating the lemma; the claim that "both conventions work verbatim" (conditioning only grows) mistakes the direction of the collider effect—larger conditioning sets need not preserve d-separation. □

Theorem 59 (T2, coverage — restated)**.** *See the table in Theorem 14.*

Proof sketches by row. β_π: the conditional law of A_t given S_t varies with I, so $I(I_t; A_t \mid A^{t-1}, S^t) > 0$ (Faith), i.e. $F_{\text{ext}} > 0$, so A_{org} strictly decreases. β_G: behavior-based IRL recovers $U_{\text{eff}} \neq U^*$, so $d > 0$ under (Faith); historicality is captured by Λ (AX3); F_{ext} may vanish because $G \in S$ mediates (Lemma above). β_{Endorse}: direct from (3) and (9). γ: the blocking argument applies verbatim (paths cross B), so F_{ext} may be 0; the accuracy loss is captured by A_{epi}^{real} (8) (a Blackwell-degraded channel strictly lowers $\Pr[B_t = \kappa(E_t)]$ under (Faith), granted a sound update rule in the sense of Definition 55—the soundness clause the row relies on, made explicit). δ: the max over a smaller policy class is $\leq$; strictness follows when deleted actions lie in the support of the optimum (Faith); the feasible set's dependence on I enters the conditional law directly, so the feasible-set form also gives $F_{\text{ext}} > 0$. The transition-kernel form (attenuated $A \to E$ coupling) leaves π intact and introduces no exogenous parent of A_t, so $F_{\text{ext}} = 0$ and only A_{exec} registers it—the within-class division stated in the main text. α: compatibility by the Lemma. σ: by definition it touches no surgery locus, and raises δ^* (Theorem 22). □

Theorem 60 (T2, multi-channel necessity — restated)**.** *No single component detects all surgery classes.*

Proof. (i) A_{org} is blind to β_G and γ: both act through S, so the Lemma's argument gives $F_{\text{ext}} = 0$ in witness models (numerically: $A_{\text{org}} = 1$ exactly under β_G/γ arms in Section 8); it is blind to the transition-kernel form of δ for the same reason (π intact, no exogenous parent of A_t). (ii) A_{epi} is blind to β_π, δ: under surgeries leaving o untouched, belief trajectories are unchanged, hence so is the accuracy $\Pr[B_t = \kappa(E_t)]$. (iii) A_{goal} is blind to γ, δ: belief distortion below d_c leaves $A_{\text{goal}} = 1$; shrinking the action set does not move U_{eff}; and under β_π the IRL inversion premise fails, so A_{goal} is not a valid instrument there at all (Theorem 14). □

Corollary 61 (T2, corrected mediation criterion — restated)**.** *Compatibility $\iff$ (no surgery) $\wedge$ (full mediation) $\wedge$ (honesty: the α-components' content induces no accuracy loss on the registered coarse-graining); the observabilization is the four-track conjunction $F_{\text{ext}} = 0 \wedge \Lambda = 1 \wedge \bar{D} = 0 \wedge \mathcal{E}^{\Pi}$ non-decreasing, in which $\bar{D} = 0$ operationalizes the honesty clause.*

Proof. The forward direction is definition plus the Lemma (an honest α induces no accuracy loss by Lemma 56 granted a sound update rule, Definition 55). Two failures of weakened converses mark the boundaries: the single condition $F_{\text{ext}} = 0$ is insufficient (the γ witness above), and (no surgery)$\wedge$(full mediation) without honesty is insufficient (dishonest appended content through an intact channel: surgery-free and mediated, yet $\bar{D} > 0$, Section 3.3). □

F Proofs for T3

Lemma 62 (TV continuity of bounded estimators)**.** *Let $\hat{A} = f(\xi)$ with ξ registered evidence (a measurable function of the internal observation stream) and* range$(f) \subseteq [0, 1]$. *Then $|\mathbb{E}_\gamma[\hat{A}] - \mathbb{E}_0[\hat{A}]| \leq \text{TV}(P_\xi^\gamma, P_\xi^0)$.*

Proof. $|\int f\,dP_\gamma - \int f\,dP_0| \le \sup|f| \cdot \mathrm{TV} \le \mathrm{TV}$. □

Theorem 63 (T3b — restated)**.** *Under* (HI-1) *covertness* $\mathrm{TV}(P_\xi^\gamma, P_\xi^0) \le \varepsilon(\theta)$, (HI-2) $\hat{A} = f(\xi)$ *with calibration bias* $|\mathbb{E}_0[\hat{A}] - A^0| \le \eta_0$, (HI-3) *distortion growth* $A^0 - A^\gamma \ge c_1 s(\gamma)^{\nu_0}$ *(*$\nu_0 \in \{1, 2\}$*): in expectation over the evidence law,* $\mathbb{E}_\gamma[\hat{A}] \ge \mathbb{E}_0[\hat{A}] - \varepsilon \ge A^0 - \varepsilon - \eta_0 > A^0 - c_1 s^{\nu_0} \ge A^\gamma$ *whenever* $c_1 s^{\nu_0} > \varepsilon + \eta_0$*, and the expected gap* $\mathbb{E}_\gamma[\hat{A}] - A^\gamma \ge c_1 s^{\nu_0} - \varepsilon - \eta_0$ *is strictly increasing in* s*; with the gap bounded in* $[-1, 1]$*, Markov's inequality gives the firing-probability floor* $\Pr_\gamma[\hat{A} > A^\gamma] \ge c_1 s^{\nu_0} - \varepsilon - \eta_0$ *(vacuous when the right-hand side exceeds* 1*; informative range* $(0, 1]$*).*

Proof. $\mathbb{E}_\gamma[\hat{A}] \ge \mathbb{E}_0[\hat{A}] - \varepsilon$ by Lemma 62 (an expectation statement, not pointwise); $\mathbb{E}_0[\hat{A}] \ge A^0 - \eta_0$ by the calibration-period bias bound; (HI-3) gives $A^\gamma \le A^0 - c_1 s^{\nu_0}$; the chain and the expected-gap bound follow, with strict monotonicity because $c_1 s^{\nu_0}$ is strictly increasing and ε, η_0 are s-independent. For the firing floor, $X := \hat{A} - A^\gamma \in [-1, 1]$ with $\mathbb{E}[X] \ge g := c_1 s^{\nu_0} - \varepsilon - \eta_0 > 0$ gives $g \le \mathbb{E}[X] \le \Pr[X > 0] \cdot 1 + \Pr[X \le 0] \cdot 0 = \Pr[X > 0]$; for $g > 1$ the hypotheses are unsatisfiable ($\mathbb{E}[X] \le 1$ always), so the floor is vacuous there and its informative range is $g \in (0, 1]$. All constants are used as stated (no constant and its reciprocal are interchanged); the calibration step is an explicit error budget, not an approximate identity. T3(a) (manipulation ⇒ strict decrease wherever the affected instrument enters the composite, with the registered L0/L1 β_G-on-Fix corner of Theorem 18) is the row-wise coverage of Theorem 14 composed with (A2), per the (M1)/(M2) case split in Definition 17. □

G Proofs for T4 and T5

Proposition 64 (T4(i), highest-leverage concentration — restated)**.** *Under geometric aggregation and* (LinLog)*, with per-unit costs* $c_i > 0$*, the attacker optimum is the vertex* $i^* = \arg\max_i w_i s_i / c_i$*.*

Proof. The objective $\max \sum_i w_i s_i x_i$ subject to $\sum_i c_i x_i \le B$ is a single-constraint LP: for feasible x,

$$\sum_i w_i s_i x_i \le \max_i (w_i s_i / c_i) \cdot \sum_i c_i x_i \le (w_{i^*} s_{i^*} / c_{i^*}) B,$$

the vertex value. Strictly concave g_i instead give interior KKT water-filling—the "concentration" conclusion is (LinLog)-dependent, and its failure mode (spread attacks superior) is the registered falsification signal. □

Proposition 65 (T4(ii), defender equalizes — restated)**.** *With defense* e *and budget* $\sum_i e_i \le E$*, let* $\rho_i(e_i) = w_i s_i(e_i) / c_i(e_i)$*, continuous and decreasing. The optimum of* $\min_e \max_i \rho_i(e_i)$ *equalizes* ρ_i *over* $\{i : e_i > 0\}$*.*

Proof. Existence by compactness and continuity. If at optimum $\rho_1 > \rho_2 = \max_i \rho_i$, continuity lets us move a small budget from channel 2 to channel 1 and reduce the maximum—contradiction. □

Theorem 66 (restated: T5)**.** *See Theorem 20.*

Proof. Under the product form (Theorem 7), the product is zero iff a factor is zero (AX5, both directions); the zero semantics of the factors are the component propositions of Appendix C (Propositions 50–57); layering (L0/L1/L2) restricts the applicable disjuncts. □

Corollary 67 (T5, script equivalence — restated; skeleton with explicit assumption)**.** *Under* (K) *(adversary knows* (G, π)*;* π *near-deterministic with TV noise* $\le \nu$*), total epistemic capture implies behavior equivalent to an adversarial script driven by* $I^{\gamma,T}$ *with* $\frac{1}{T}\sum_t \mathbb{E}\mathrm{TV}(P_{A_t}, P_{\mathrm{script},t}) \le \nu + o(1)$*.*

Proof skeleton (explicit assumption flagged). Under capture, the (B, O) components of S_t are functions of I^γ; $A_t = \pi(S_t)$ is computable by the adversary up to noise ν under (K); DPI guarantees further processing does not increase the TV. □

H Proofs for T6

Recall the dynamics and mechanisms M1–M4 from Section 6.6 and assumptions (H1)–(H4′) (additive linearly mapped damage; units unified by calibration into damage-nats/time, with χ_j-estimation sensitivity as a reporting obligation; $\iota_j \geq \iota_{\min} > 0$; $\varepsilon \geq \varepsilon_{floor} > 0$; $A_{\text{via}} \geq A_{\text{via}}^{\min} > 0$).

Theorem 68 (T6, main theorem — restated)**.** *See Theorem 22; the margin is two-tier: the pointwise floor $\prod_j (A_j^{base}(1 - \bar{\delta}_j/\kappa_j))^{w_j} > 0$ for $\bar{\delta}_j < \kappa_j$, and the uniform floor (13), positive iff no κ_j binds.*

Proof. (i) M4 modeling assumption (k-of-r majority of independent copies: a majority takeover requires corrupting $k_j = \lceil (r_j + 1)/2 \rceil$ copies, which at cost $c_{0,j}$ per copy per window W_j requires budget rate $k_j c_{0,j}/W_j$; below that rate no majority is ever held, and the consensus state is recovered with probability $1 - o(1)$ per window, whence $\delta_j^{eff} = 0$). (ii) With $\delta_j < \theta_j/W_j$, M1 never triggers (window accumulation $W_j \delta_j < \theta_j$), so M3 is the only active term: $\dot{D}_j \leq \delta_j - \kappa_j D_j$; the comparison principle (local Lipschitz right-hand side) gives $\limsup_t D_j(t) \leq \delta_j/\kappa_j$, and $\delta_j < \delta_j^* \leq \kappa_j$ makes the bound < 1. (iii) Substituting the component map $A_j = A_j^{base}(1 - D_j)$ at $D_j = \bar{\delta}_j/\kappa_j$ into the product form of Theorem 7 gives the pointwise tier. For the uniform tier, $\delta_j^* < \kappa_j$ gives $1 - \delta_j^*/\kappa_j > 0$ in every factor, and $A_{\text{via}} \geq A_{\text{via}}^{\min} > 0$ by (H4′); if $\kappa_j = \delta_j^*$ the j-th factor is 0—the uniform floor degenerates continuously while the pointwise tier keeps a positive bound for every fixed $\bar{\delta}_j < \kappa_j$. □

Proposition 69 (T6, registration domain — restated)**.** *If $\delta_j \geq \theta_j/W_j$ (registerable) and M1 responds normally, damage per registration cycle is $\leq \theta_j + \ell_j \delta_j$; after rollback the cycle repeats, and whenever $\theta_j + \ell_j \delta_j$ is below the capture threshold, A has positive floor $A(\theta + \ell\delta)$.*

Proof. Accumulation at rate δ_j reaches θ_j within the window; the lag contributes at most $\ell_j \delta_j$; rollback resets. □

Proposition 70 (T6, detection-alone impossibility — restated)**.** *For any detection-only scheme with $\mathbb{E}_\infty[T] \geq 1/\alpha$, damage at detection under constant drift δ is $\geq (1 - o(1)) \cdot 2s_n^2 \log(1/\alpha)/\delta \to \infty$ as $\delta \to 0$.*

Proof. Lorden's lower bound (with Lai's refinement) gives worst-case EDD $\geq (1 - o(1)) \log(1/\alpha)/D(\delta)$ for any procedure with average run length $\geq 1/\alpha$, where $D(\delta)$ is the per-sample KL between pre- and post-drift laws; the local expansion $D(\delta) = \delta^2/(2s_n^2) + O(\delta^3)$ for additive drift against noise variance s_n^2 yields $\delta \cdot \text{EDD} \geq (1 - o(1))\, 2s_n^2 \log(1/\alpha)/\delta$ [Lorden, 1971, Lai, 1995, Tartakovsky et al., 2014]. □

Corollary 71 (T6, design rule and blind zone — restated)**.** *$\theta_j/W_j \leq \kappa_j$ eliminates the "unregistered and unrepairable" band ($\delta < \theta/W$ handled by M3; $\delta \geq \theta/W$ by M1), giving global boundedness with $\delta_j^* = \min(\theta_j/W_j, \lceil (r_j + 1)/2 \rceil c_{0,j}/W_j)$ (the κ-term slack). If $\theta/W > \kappa$* on a channel where the redundancy term does not bind *(the channel is outside M4's coverage, the attack is outside M4's replication-corruption budget model, or the budget rate exceeds $k_j c_{0,j}/W_j$; on an M4-covered channel below that budget rate, T6(i) gives $D \to 0$ instead), then for $\delta \in (\kappa, \theta/W)$: M1 never triggers and repair is outpaced ($\delta/\kappa > 1$), so $D \to 1$ on that channel despite the full mechanism stack being present—the blind-zone prediction.*

Proposition 72 (T6, monotonicity — restated)**.** *$\partial\delta^*/\partial\varepsilon = v\chi > 0$; $\partial\delta^*/\partial v = \varepsilon\chi > 0$; δ^* is nondecreasing in r with step c_0/W at each increment of $\lceil (r+1)/2 \rceil$; $\partial\delta^*/\partial W < 0$ (where a windowed term binds); θ double-edged (coverage domain rises while the damage bound $\theta + \ell\delta$ rises and the ARL constraint imposes $\theta \geq \theta_{\min}(\alpha)$); shortening τ raises the gated repair rate ι/τ, hence the margin.*

Proof. Term-wise differentiation of the min form (the redundancy term is stepwise constant in r); the gated quantities' effective repair rate contains ι/τ. □

Corollary 73 (T6×T7, composition — restated)**.** *Delegation leak enters the drift budget: $\lambda(1-q)\mu_{leak} + \delta_j^{adv} < \delta_j^*$ is the amended condition; high-q delegation consumes almost no margin, low-q erodes it.*

I Proofs for T7 and T8

Theorem 74 (T7(i), component-level linear law — restated)**.** *Under* (H1') *(equal per-step conditional entropy transfer C of both sources; the script arm bypasses S; and $F_{\text{ext}}^{self} = 0$—the undelegated agent carries no residual external flow), $F_{\text{int}} = (1-m)\tilde{C}$, $F_{\text{ext}} = m\tilde{C}$ (same $\tilde{C}$), hence $A_{\text{org}}(\lambda, q) = A_{\text{org}}^{self}(1 - \lambda(1-q))$.*

Proof. Under the mixture law $(1-m)\pi_{self} + m\pi_{del}$ with the exogenous selector μ_t (a node of the extended graph, independent of S), internal flow scales by $(1-m)$ (conditioning on the selector, DPI and the chain rule); external flow is the selector-gated script content, m times the script's entropy transfer; (H1') equates the two rates; and with $F_{\text{ext}}^{self} = 0$ the self-side ratio is $A_{\text{org}}^{self} = \tilde{C}/(\tilde{C}+0) = 1$-normalized, so the ratio is exactly $1-m$. With $F_{\text{ext}}^{self} = f > 0$ the same bookkeeping gives $A_{\text{org}} = (1-m)\tilde{C}/((1-m)\tilde{C}+m\tilde{C}+f)$, a strict correction; the bracket below absorbs the uncalibrated case. *The slack bound, explicitly.* When the selector is *not* registered in the log, the per-step internal term satisfies, by the chain rule applied to (S^t, μ_t),

$$I(S^t; A_t \,|\, A^{t-1}) = (1-m)C_{self} + I(\mu_t; A_t \,|\, A^{t-1}) - I(\mu_t; A_t \,|\, A^{t-1}, S^t),$$

and both correction terms lie in $[0, h(m)]$ with $h(m) \leq \ln 2$ the selector entropy, so the per-step deviation from $(1-m)C_{self}$ is at most $h(m)$ nats, accumulating to $\leq T\,h(m)$ over the horizon (and likewise for the F_{ext} term). The ratio A_{org} therefore deviates from the displayed law by $O\big(h(m)/((1-m)C_{self})\big)$ per step: the slack is an $O(1)$ term in the rate units, vanishing as the per-step transfer C_{self} grows, not as T grows. When the selector *is* registered (the benchmark's discipline), conditioning on μ_t removes both correction terms and the law is exact; the benchmark's 2×10^{-5} maximal deviation is the residual grid-quantization error, and its φ-relaxed bracket ($\varphi \leq T \ln 2/F_{\text{int}}^{self}$ with $F_{\text{int}}^{self} = T\,C_{self}$ the undelegated baseline's horizon internal flow; equivalently $\varphi \leq \ln 2/C_{self}$ in per-step rate units) is exactly the bound above. □

Proposition 75 (T7(ii), bracket bounds — restated)**.** *Under the delegation semantics (exogenous selector with $\Pr[\mu_t = \text{del}] = m$; delegated actions are generated by the script, so delegated steps carry no S-flow—the definiens of script delegation), let $\rho := C_{del}/C_{self} \in [\rho_{\min}, \rho_{\max}]$ and $f := F_{\text{ext}}^{self} \geq 0$. Then, up to the mixture-entropy slack—explicitly bounded by $h(m) \leq \ln 2$ nats per step in each flow term (the T7(i) proof), accumulating to $\leq T\,h(m)$ over the horizon—in per-step rate units (horizon totals are T times the rates; A_{org}, a ratio of like-unit flows, is unit-invariant):*

$$F_{\text{int}} = (1-m)C_{self}, \qquad F_{\text{ext}} = mC_{del} + f,$$

and consequently

$$A_{\text{org}} = \frac{(1-m)C_{self}}{(1-m)C_{self} + mC_{del} + f} \in \Big[\frac{1-m}{(1-m)+m\rho_{\max}+\varphi},\ \frac{1-m}{(1-m)+m\rho_{\min}+\varphi}\Big],$$

with $\varphi := f/C_{self}$; the bracket follows by dividing numerator and denominator by C_{self}. The slack is not *an extra additive term outside the bracket: it is absorbed into the endpoints through the same φ-relaxation—the benchmark instantiates it as $\varphi \le T\ln 2/F_{\text{int}}^{self}$ ($F_{\text{int}}^{self} := T\,C_{self}$, the horizon total of the undelegated baseline rate $F_{\text{int}}|_{m=0} = C_{self}$; in the per-step rate units of the flow identities above the same bound reads $\varphi \le \ln 2/C_{self}$), which covers the accumulated $T\,h(m) \le T\ln 2$ nats (Section 8, DEL-1 direct-flow arm); when the selector is registered in the log the slack vanishes exactly and no relaxation is needed. With $f = 0$ ($\varphi = 0$) this reduces to $A_{\text{org}}^{self} \cdot [(1-m)/(1-m+m\rho_{\max}),\ (1-m)/(1-m+m\rho_{\min})]$ (then $A_{\text{org}}^{self} = 1$). The upper end uses the same two-sided bookkeeping as the lower end—$F_{\text{int}} = (1-m)C_{self}$, i.e. the delegated share of internal flow is removed; if instead delegated steps were permitted to carry S-flow up to C_{self}, then $F_{\text{int}} \le C_{self}$ and only the lower end and the relaxed upper end $1/(1+m\rho_{\min})$ survive.*

Proof. Per step, with probability $1-m$ the action is generated by the self arm (internal transfer C_{self}), with probability m by the script (no S-flow under the delegation semantics; external transfer C_{del}); summing over t gives the two displays up to the mixture-entropy slack explicitly bounded above ($h(m) \le \ln 2$ nats per step in each flow term; convexity of the mixture entropy). Substituting into $A_{\text{org}} = F_{\text{int}}/(F_{\text{int}} + F_{\text{ext}})$ and using monotonicity in ρ and f gives the bracket. Relaxed upper end: if delegated steps may carry up to C_{self} of S-flow, then $F_{\text{int}} \le (1-m)C_{self} + mC_{self} = C_{self}$ (the mixture cannot transport more internal transfer than the self arm's full rate), whence with $f = 0$, $A_{\text{org}} \le C_{self}/(C_{self} + m\rho_{\min}C_{self}) = 1/(1+m\rho_{\min})$. □

Corollary 76 (T7(iii), composite correction — restated)**.** *Ceteris paribus under geometric aggregation, $A(\lambda, q) = A_{self}(1-\lambda(1-q))^{w_{org}}$.*

Proof. Only A_{org} moves; the product form of Theorem 7 gives $A/A_{self} = (A_{\text{org}}/A_{\text{org}}^{self})^{w_{org}}$. □

Theorem 77 (T7(iv), $q = 1 \equiv \alpha$ — restated)**.** *Delegation at full transparency, reversibility, and endorsability ($q = 1$) is causally equivalent to α-class influence, hence $F_{\text{ext}} = 0$ and $A(\lambda, 1) = A_{self}$.*

Proof. Constructive: transparency places the delegate's output into B as registered honest information; reversibility means no persistent ΔG bypasses Endorse; endorsability gives $\Lambda = 1$; Lemma 13 then gives $F_{\text{ext}} = 0$ and no component decreases. □

Theorem 78 (T8, coercion paradox — restated)**.** *See Theorem 26.*

Proof. (i) Any intervention that "rewrites the motivational structure to make it substantively autonomous" touches G, Endorse, or $\mathcal{A}$ by definition, i.e. is β/δ (Proposition 4); the coverage table (Theorem 14) gives strict decrease. (ii) the monotonicity proposition. □

J Proofs for C0

Proposition 79 (C0(i), projection consistency — restated)**.** *Under the definitional readings of Section 6.9, each target dimension is a function of our state descriptor and is rank-consistent with the source gradations when I is a benign principal (no adversarial surgery).*

Construction, dimension by dimension. Write Z for our state descriptor. Each target dimension is read as a function on Z (the definitional readings of Section 6.9); rank consistency is checked pairwise within each source framework.

1. *Kasirzadeh–Gabriel.* (i) Efficacy := A_{exec}: identity, hence rank-preserving. (ii) Autonomy := $F(A_{\text{org}}, A_{\text{goal}}, A_{\text{ref}})$: a continuous strictly increasing function of three arguments, so it preserves the source's weak order wherever that order is the componentwise one. (iii) Goal complexity := $H(U^*)$: the entropy is monotone under refinement of the goal set, which is the order the source's gradation uses. (iv) Generality := the support measure $\Pr[E : A(E) \geq a_0]$: monotone in the set of environments the agent copes with, hence rank-preserving for inclusion.

2. *Dung.* (i) Goal-directedness := A_{goal} (identity). (ii) Autonomy := A_{org} (identity). (iii) Direct impact := A_{exec} (identity). (iv) Long-term planning := the horizon profile $n \mapsto A_{\text{exec}}(n)$; planning depth is the first saturating n^*, and a pointwise-dominating profile cannot saturate sooner, so the depth order is preserved. (v) Acting for reasons := $A_{\text{epi}}^{real} \times A_{\text{ref}}$: increasing in both factors, hence rank-preserving for the componentwise order.

3. *Feng roles.* The roles are cells of a coarse-graining of (λ, q): operator $\lambda \approx 0$; collaborator mid-λ/high-q; consultant low-λ; approver high-λ/high-q (endorsement gate = human approval); observer $\lambda \approx 1$. The cell map is monotone in each coordinate on the registered cells, so the source's role order (where defined) is preserved.

All identities hold on the registered parameterization; the residual uncertainty is the faithfulness of the readings, not the rank arguments. □

Proposition 80 (C0(ii), separation witness under registered readings — restated)**.** *Configuration pairs (Z_1, Z_2) exist with identical behavioral distributions and identical KG/Dung/Feng profile values, yet $A(Z_1) \neq A(Z_2)$ ($A_{\text{epi}}^{real} < 1$, $\Lambda < 1$ in Z_2). This is a witness that our state descriptor separates configurations the source frameworks cannot, under the registered definitional readings—not a claim of mathematical subsumption.*

Proof. Construction: let the injected content be a verbatim copy of the principal's instruction, delivered through a covert γ channel that bypasses registration. The behavioral distribution is pointwise identical (same script); all three frameworks' dimensions are functions of behavior/role structure only, hence equal; our framework registers $A_{\text{epi}}^{real} < 1$ (the channel pins beliefs independently of the situation, driving accuracy against $\delta_{\kappa(E_t)}$ below one) and $\Lambda < 1$ (the surgery history $I^{\text{surg},T}$ contains the covert γ; read law-level from the causal log). □

Corollary 81 (C0, persuasion–hijack inseparability — restated)**.** *No purely behavioral governance grading can separate persuasion from hijack; certificates need causal-layer audit.*

K Proofs for C1 and C2 (toy-model propositions)

Proposition 82 (C1, fold bifurcation and critical slowing — restated)**.** *For $\dot{D} = \delta - \varepsilon v(1-D)D$:* (i) *for $\delta < \delta_c := \varepsilon v/4$, equilibria $D_\pm = (1 \pm \sqrt{1 - 4\delta/\varepsilon v})/2$ with D_- stable, D_+ unstable;* (ii) *at $\delta \uparrow \delta_c$ they merge (saddle-node), and for $\delta > \delta_c$, $D \to 1$;* (iii) *$|f'(D_-)| = \varepsilon v\sqrt{1 - 4\delta/\varepsilon v} \to 0$, $T_{\text{rec}} \propto (\delta_c - \delta)^{-1/2}$, and under OU linearization the stationary variance $\propto 1/|f'| \to \infty$ with AR(1)$\to 1$.*

Proof. (i)(ii): roots of the quadratic and phase-line analysis. (iii): $f(D) = \delta - \varepsilon v D(1-D)$, $f'(D) = -\varepsilon v(1-2D)$, evaluate at D_-; the OU formulas (stationary variance $\nu_{OU}^2/(2\lambda)$, autocorrelation

$e^{-\lambda\Delta t}$) give the divergence as $\lambda \to 0$ (the EWS methodology of Lade and Gross, 2012; normal forms as in Kuznetsov, 2004). □

Proposition 83 (C2, bistability and hysteresis — restated). *For $R(D) = \varepsilon_0 v\, D(1-D)(1-gD)$, $g > 1$: R rises on $(0, D_1)$ to its max $\delta_{down} := R(D_1)$ ($D_1 \in (0, 1/g)$), crosses zero at $1/g$, reaches a negative minimum at D_m, returns to 0 at 1. (i) For $0 < \delta < \delta_{down}$: a stable point $A \in (0, D_1)$ and the absorbing state $D = 1$ are bistable (separatrix = unstable $B \in (D_1, 1/g)$). (ii) At $\delta \uparrow \delta_{down}$: A, B merge and the system jumps catastrophically to 1. (iii) With δ returned to 0, $D = 1$ remains absorbing ($R(1) = 0$; on $(1/g, 1)$, $R < 0$ pushes back to 1): no spontaneous recovery. (iv) Recovery requires scaffolding s with $\dot{D} = \delta - R(D) - s$; descent from $D \approx 1$ requires $s > s_c := |R(D_m)| > 0$, after crossing B spontaneous relaxation to $A \approx 0$. The loop does not close: (δ_{down}, s_c) both independently positive. The cubic R is a registered modeling choice (the simplest polynomial with the needed sign pattern): these are phase-line facts about the chosen normal form, and their role is to give the prospective-defense claim a concrete quantitative carrier—not to assert that autonomy dynamics are cubic.*

Proof. Phase-line analysis of the cubic R (root positions $0 < 1/g < 1$; sign chart of $f = \delta - R$ per interval); for (iv), on $D \in (1/g, 1)$, $f = \delta - R - s = \delta + |R| - s$, so descent requires $s > \delta + |R(D)|$, worst case at D_m. □

L Dependence and copula bounds

Lemma 84 (Jensen gap). *$F = \prod_i x_i^{w_i}$ ($\sum_i w_i = 1$) is concave on $\mathbb{R}^n_+$ (Cobb–Douglas; Hessian negative semidefinite), so for random component vectors $\mathbf{A}$: $\mathbb{E}[F(\mathbf{A})] \le F(\mathbb{E}[\mathbf{A}])$ —substituting expected components in black-box/interval mode overestimates A by a signed amount.*

Lemma 85 (Log-normal exact ratio). *If $\log A_i$ are jointly normal (means μ_i, variances σ_i^2, correlations ρ_{ij}): $\mathbb{E}[F]/F(\mathbb{E}[A]) = \exp(-\frac{1}{2}[\sum_i w_i\sigma_i^2 - \sum_{ij} w_i w_j \rho_{ij}\sigma_i\sigma_j]) \le 1$ (Cauchy–Schwarz). Direct MGF computation.*

Proposition 86 (Bounds with unknown dependence). *F coordinatewise increasing and supermodular ($\partial^2 F/\partial x_i \partial x_j = w_i w_j F/(x_i x_j) > 0$): (i) the upper bound of $\mathbb{E}[F]$ is attained by the comonotone copula (supermodular ordering; Tchen, 1980, Müller and Stoyan, 2002); (ii) for $n = 2$ the (sharp) lower bound is obtained by integrating against the Fréchet–Hoeffding lower bound $\max(0, F_1 + F_2 - 1)$, which is the countermonotone copula. For $n \ge 3$ the Fréchet–Hoeffding lower-bound form $\max(0, \sum_i F_i - (n-1))$ is* not a copula *(it fails n-increasingness), so it is not attained by any joint distribution: the integral against it is only a relaxed, generally unattainable lower bound. An attainable dependence-free lower bound is available instead: since $\log F = \sum_i w_i \log A_i$ is additive in the coordinates, Jensen's inequality for the convex exponential gives*

$$\mathbb{E}[F] \;\ge\; \prod_i \left(\exp \mathbb{E}[\log A_i]\right)^{w_i} \;=\; \prod_i \mathrm{G}(A_i)^{w_i},$$

where $\mathrm{G}(A_i) := \exp \mathbb{E}[\log A_i]$ is the geometric mean of the marginal of A_i. Hence with marginals known and dependence unknown, $\mathbb{E}[F]$ has a computable interval—comonotone upper end, Jensen lower end (or the sharper countermonotone value when $n = 2$)—the dependence-theoretic complement of the black-box rules. Nested CES within clusters is automatically axiom-compatible by (A3).

M Identifiability

Proposition 87 (White-box identification, tiered — restated)**.** *Given white-box access (the full causal log, the local laws of $\mathcal{G}$, and intervention access, H-0), all components are identified, by tier:* (i) $A_{\text{org}}, A_{\text{epi}}, A_{\text{via}}$ *are functionals of the joint law* $P(S^T, O^T, A^T, I^T, Z^T)$*;* (ii) A_{exec} *and* A_{ref} *are identified conditional on the protocol constructions (accessible-class probes; measurement ensemble);* (iii) Λ *is a path-specific counterfactual, identified from the structural model (the logged local laws and exogenous noise terms supporting abduction)—a premise strictly stronger than joint-law access with interventions, which does not pin cross-world counterfactuals absent structural assumptions.*

The certified bound (one-sided, conditionally). See Proposition 32 and (14): Lemma 37 makes $F_{ext}^{obs} = I(I^T; A^T)$ observable without S-access, and Remark 38 supplies the bridge to the causally conditioned estimand: under premise (i) (the bypass premise $I^t \perp\!\!\!\perp S^t \mid A^{t-1}$) the probe MI is a lower estimate of F_{ext}, so—A_{org} being decreasing in F_{ext}—the rule's upper end is the certified bound under premise (ii) (F_{int} drift $\leq \eta$; insufficient η as negative control). Both premises are stated in the proposition itself because both are black-box unverifiable in general; the bound is *conditionally* certified. The lower end treats F_{ext}^{obs} as exact and is *not* certified: the truth can fall below it when the external flow is understated, so the rule is a one-sided bound with a reference lower end, not an interval-containment statement. Certified scope: bypass-semantics exogenous randomized probes—*not* non-adversarial influence in general, since mediation probes (the α class) reverse the direction. The adaptive-adversary failure mode is calibrated by the benchmark's BB-2 arm—see the scope paragraph in Section 7.3.

One-sided A_{exec} bounds, protocol-relative. Probed mode (experimenter-forced $p(a^n)$ sequences): exactly computable (empowerment needs only the action–observation interface). Passive observation: $I(A^n; O_{n+1})$ without conditioning on S is a proxy whose bias is state-entropy mixing, provably an overestimate in that protocol ($A_{\text{exec}}^{obs} \geq A_{\text{exec}}$); with independent probes ($A \perp\!\!\!\perp S$), conditioning denoises and the direction reverses ($A_{\text{exec}}^{obs} \leq A_{\text{exec}}$). Protocol labels are a reporting obligation; both directions are numerically anchored in the benchmark (Section 8).

Proposition 88 (Λ black-box unidentifiable — restated)**.** *See Proposition 33.*

Proof. Construction: fix a behavioral norm (policy π^*). $\mathcal{M}_{endo}$: $G_0 = U^*$ with Endorse a map converging to U^*. $\mathcal{M}_{impl}$: a β surgery at $t = 0$ sets $G_0 := U^* = U_{adv}$ and Endorse $\equiv$ id thereafter. Both generate the (O, A) marginal law from the same π^* and the same o, pointwise equal; they differ only at the unobserved node I^β. Hence every functional of (O^T, A^T) (and of $\hat{A}_{\text{phen}}$) agrees, while Λ differs—$\Lambda(\mathcal{M}_{endo}) = 1$ by the empty-history convention and $\Lambda(\mathcal{M}_{impl}) = 0$ by the degenerate-endorsement certification convention, both registered in AX3 and restated in the main-text proposition; the behavioral identity half of the separation is independent of either convention. Corollary 34 follows. □

N Benchmark registry details: counting units and seeds

This appendix carries the full registry arithmetic summarized in Section 8 ("Counting units"), so that every headline count in the paper can be recomposed from the run manifest.

Pool (i): arm bundles. Row-level attack-strength records (including the zero-strength baseline that serves as the ceteris-paribus reference). Only the scenarios that register per-strength

component bundles contribute: in the main matrix, ATK-1 (480 runs $\times 5$ η-rows = 2,400), ATK-2 ($480 \times 4 = 1{,}920$), and ATK-4 ($480 \times 3 = 1{,}440$), total 5,760; the $\varsigma = 2$ stress arm adds $1{,}200 + 960 + 720 = 2{,}880$. Pool total: $5{,}760 + 2{,}880 = 8{,}640$—the pool of the weight-sensitivity recomputation (ATK-3/ATK-5/ATK-6 register calibration curves and verdict rows but no per-strength bundles, and are therefore not in this pool).

Pool (ii): the recursive registry. Every embedded record of the six-component bundle—A_{org}, A_{goal}, A_{exec}, A_{epi}, A_{ref}, A_{via}—counts, wherever it appears in a run's metrics. Main matrix: the 5,760 arm bundles plus 1,440 per-run non-row records—ATK-3's base bundle (480), ATK-4's γ-arm bundle (480), and ATK-5's base bundle (480)—giving 7,200; EXEC-1 adds $30 \times 3 = 90$ (δ-chain arms). Pool total: 7,290 scanned by the T1a range meta-criterion and the zero-attribution meta-check. Across the full campaign the recursive registry reaches 11,250 (the alternative-aggregation pool of X3): $7{,}290 + 3{,}600$ (stress arm: 2,880 arm bundles plus 720 non-row records) $+ 240$ (σ^--EROSION: 30×8 erosion-grid bundles) $+ 120$ (RES-1: 30×4).

The 7,290 vs. 5,760 difference, stated exactly. $7{,}290 - 5{,}760 = 1{,}530$: the 1,440 main-matrix non-row records (ATK-3/ATK-5 base bundles and ATK-4's γ-arm bundle, 480 each) plus EXEC-1's 90 chain-arm bundles.

Defense-direction count. The defense-direction assertion covers the two attack classes with implemented coupling rules (ATK-1/2): 7 non-all-off defense configurations $\times 2$ ς-values $\times 30$ seeds = 420 matched defended-vs-undefended comparisons per class, 840 in total. Run-level verdicts of the main matrix: 2,520 CONFIRMED +360 INCONCLUSIVE (out-of-window defense cells, INCONCLUSIVE rather than REFUTED by the verdict convention registered in Section 8) = 2,880.

Seed protocol. Seeds derive deterministically from (scenario, configuration, replication) coordinates: seed $= 30000 + 1000\,s + 97\,c + r$ over the registered scenario and configuration codes (s, c) and replication $r \in [0, 30)$, consumed by NumPy's PCG64 generator; matrix-external probes (the estimator two-path validation) use an independent seed segment ($\geq$ 900,000), so every run is bit-reproducible from its coordinates and no probe reuses a matrix stream.

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